1.0.21

EOS-ANALYSIS-2026-05

Determination of SSE form factor parameters for EOS v1.1

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Overview

This report was generated with EOS v1.0.21 from the analysis file `analysis.yaml`. It documents the determination of the `SSE` form-factor coefficients for 26 posteriors, grouped by transition type. For each posterior it lists the local p values of the individual constraints (and the global p value where the fit is over-constrained), together with the posterior-predicted form factors overlaid on the input constraints.

Pseudoscalar to pseudoscalar ($P \to P$)

Posterior `B-to-D`

This posterior varies 5 `SSE` coefficients. It is constrained by:

- `B->D::f_++f_0@FNAL+MILC:2015B`
- `B->D::f_++f_0@HPQCD:2015A`

Note. No (lattice or LCSR) tensor form-factor data are available for this transition. The tensor coefficients are therefore not varied and retain their database default values; the tensor form factors below are not determined by this fit.

Goodness of fit

The table lists the local p value of each constraint together with the global p value and total χ^2 of the best-fit point.

Constraint	local p value
`B->D::f_++f_0@FNAL+MILC:2015B`	86.3%
`B->D::f_++f_0@HPQCD:2015A`	74.2%
total (χ^2 / d.o.f. = 5.96 / 7)	**54.5%**

Posterior-predicted form factors

The bands show the posterior-predicted form factors (`SSE`); the points show the constraints entering the fit.

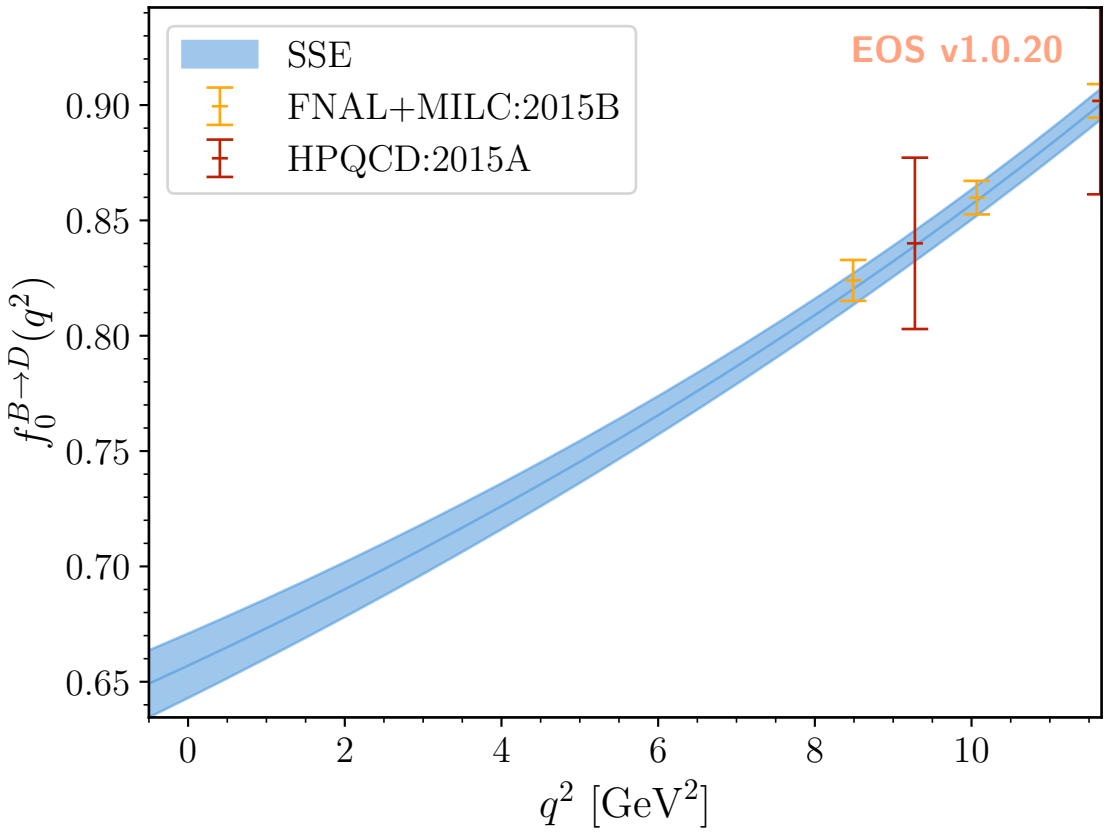

Figure 1: Posterior-predicted form factor for `B-to-D` (`B-to-D-f0`).

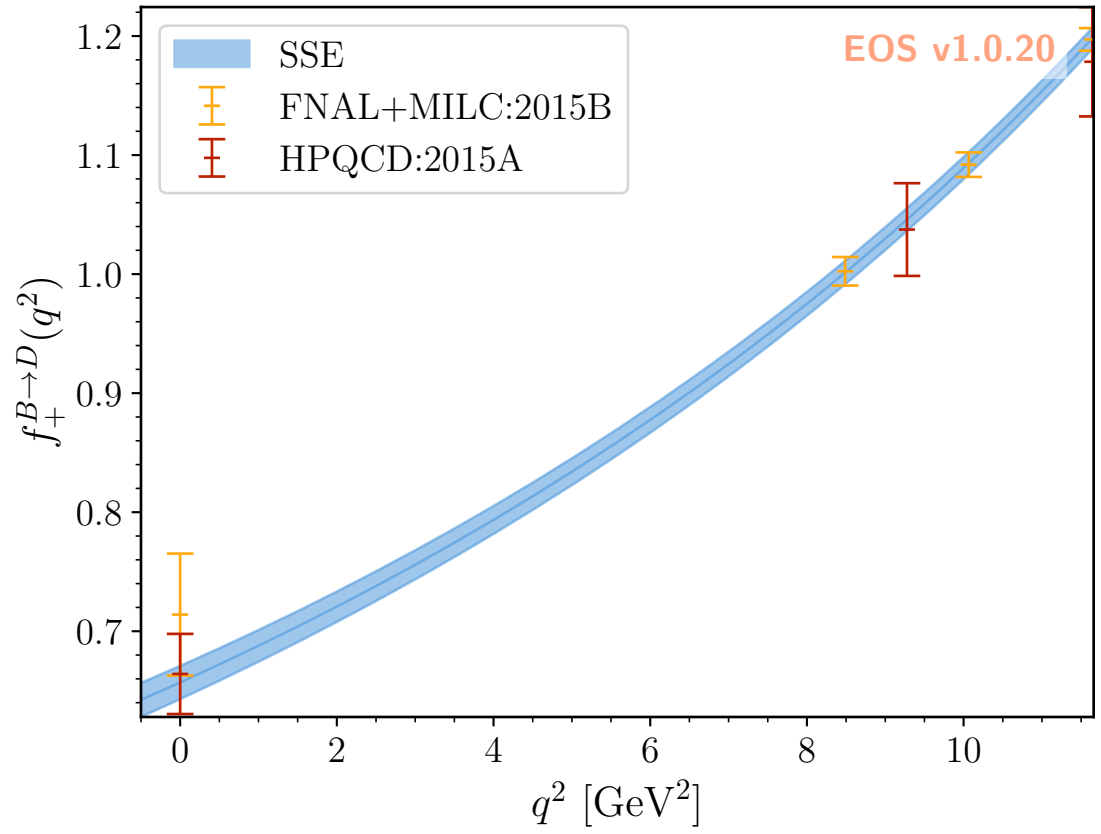

Figure 2: Posterior-predicted form factor for `B-to-D` (`B-to-D-fp`).

Posterior B-to-K

This posterior varies 8 SSE coefficients. It is constrained by:

- B->K::f_0+f_++f_T@FNAL+MILC:2015D
- B->K::f_0+f_++f_T@HPQCD:2013A
- B->K::f_0+f_++f_T@HPQCD:2022A

Goodness of fit

The table lists the local p value of each constraint together with the global p value and total χ^2 of the best-fit point.

Constraint	local p value
B->K::f_0+f_++f_T@FNAL+MILC:2015D	100.0%
B->K::f_0+f_++f_T@HPQCD:2013A	75.9%
B->K::f_0+f_++f_T@HPQCD:2022A	98.5%
total (χ^2 / d.o.f. = 8.56 / 18)	**96.9%**

Posterior-predicted form factors

The bands show the posterior-predicted form factors (SSE); the points show the constraints entering the fit.

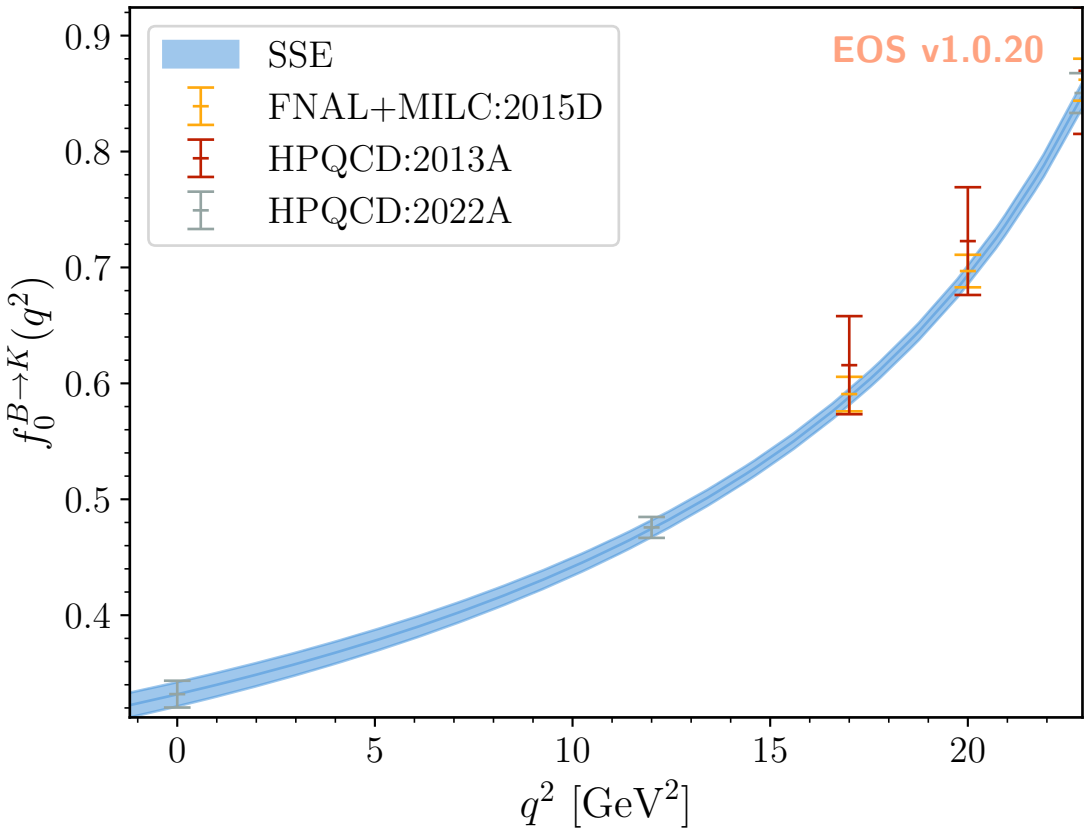

Figure 3: Posterior-predicted form factor for `B-to-K` (`B-to-K-f0`).

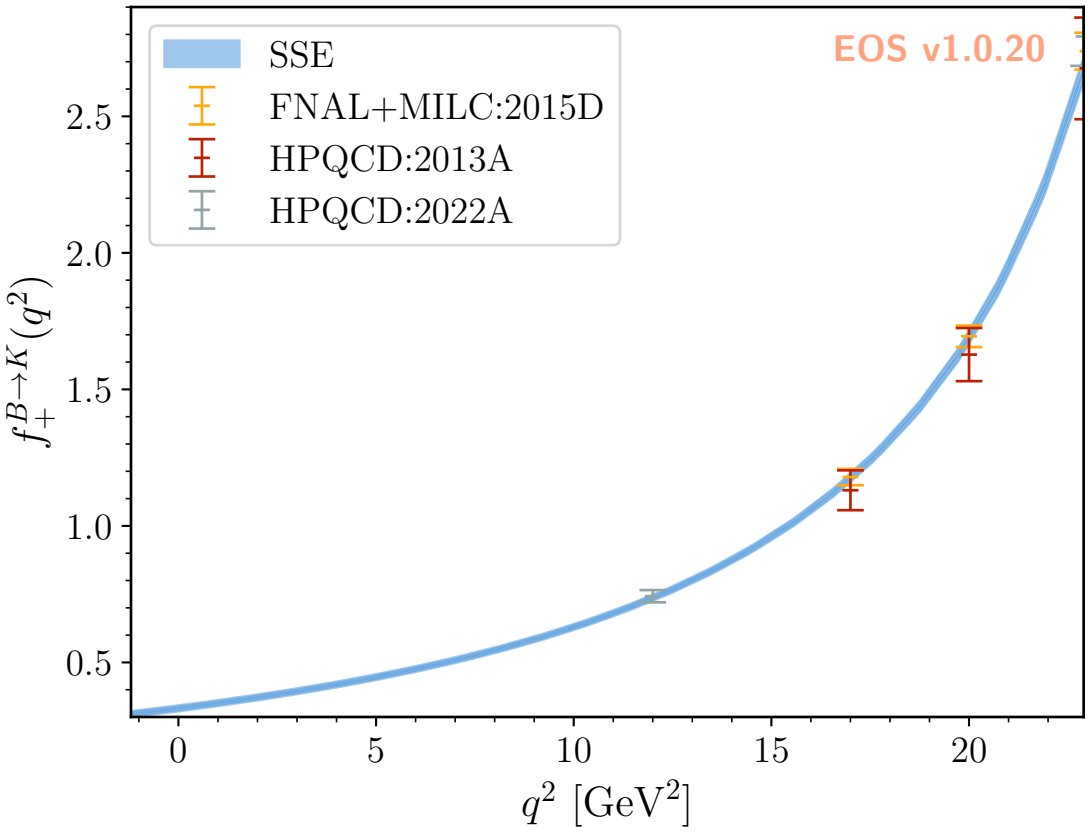

Figure 4: Posterior-predicted form factor for `B-to-K` (`B-to-K-fp`).

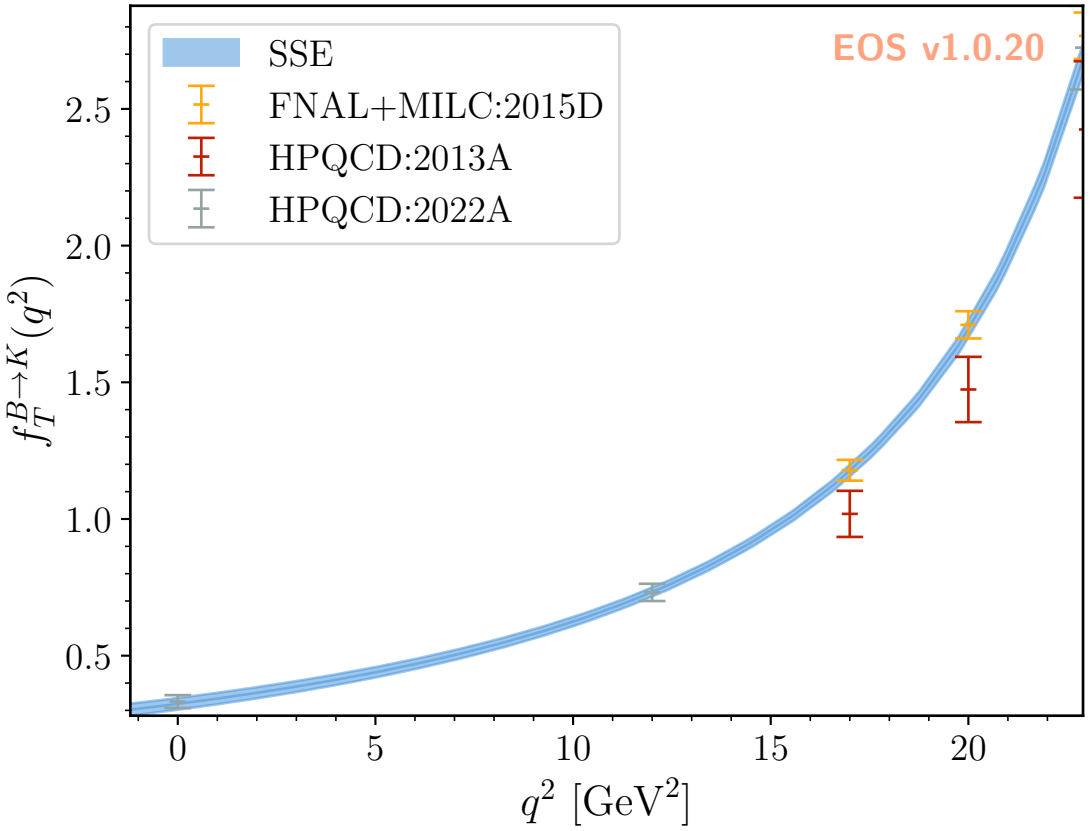

Figure 5: Posterior-predicted form factor for `B-to-K` (`B-to-K-fT`).

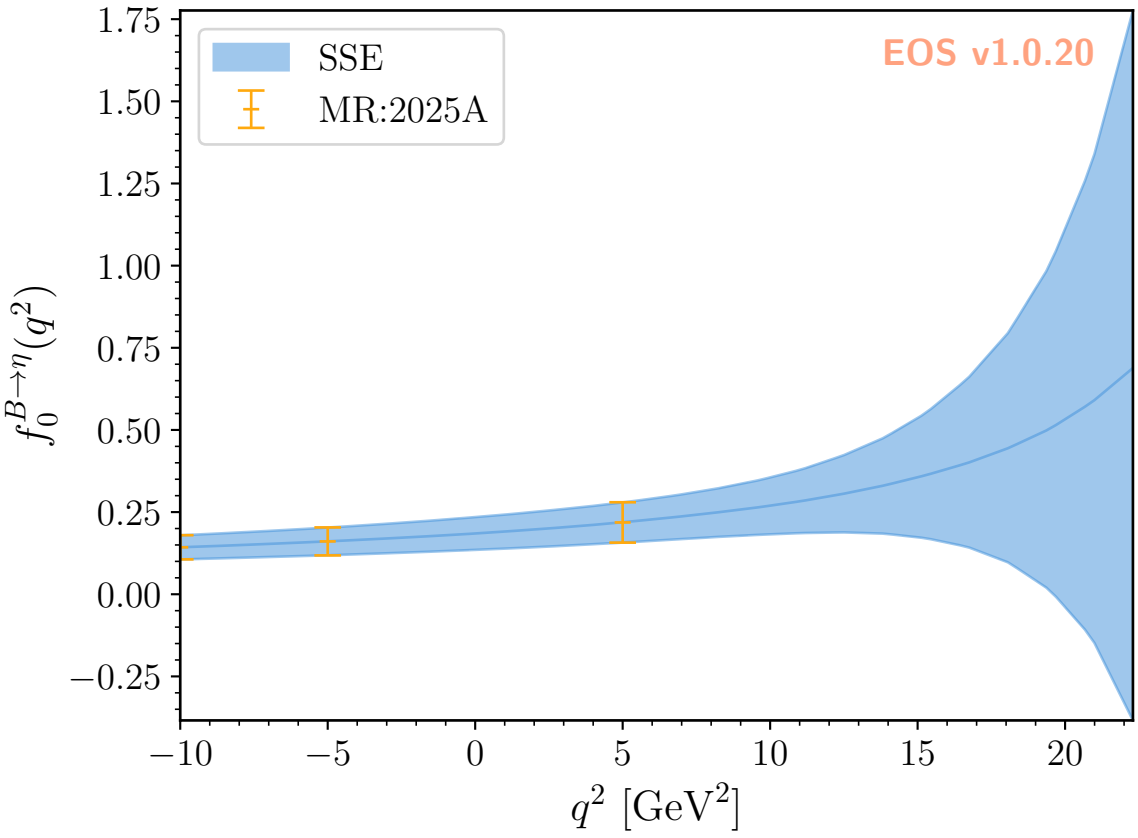

Figure 6: Posterior-predicted form factor for `B-to-eta` (`B-to-eta-f0`).

Posterior `B-to-eta`

This posterior varies 8 `SSE` coefficients. It is constrained by:

- `B->eta::f_0+f_++f_T@MR:2025A`

Goodness of fit

The table lists the local p value of each constraint together with the global p value and total χ^2 of the best-fit point.

Constraint	local p value
`B->eta::f_0+f_++f_T@MR:2025A`	100.0%
total (χ^2 / d.o.f. = 0.00 / 3)	**100.0%**

Posterior-predicted form factors

The bands show the posterior-predicted form factors (`SSE`); the points show the constraints entering the fit.

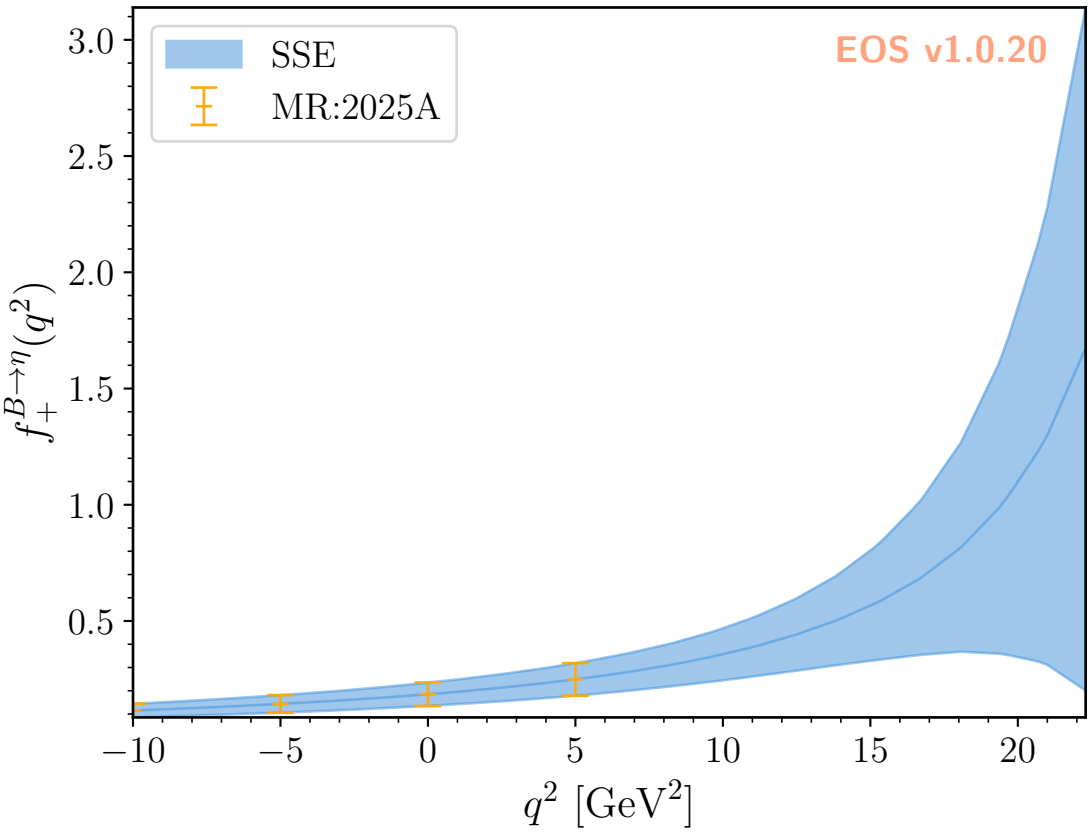

Figure 7: Posterior-predicted form factor for `B-to-eta` (`B-to-eta-fp`).

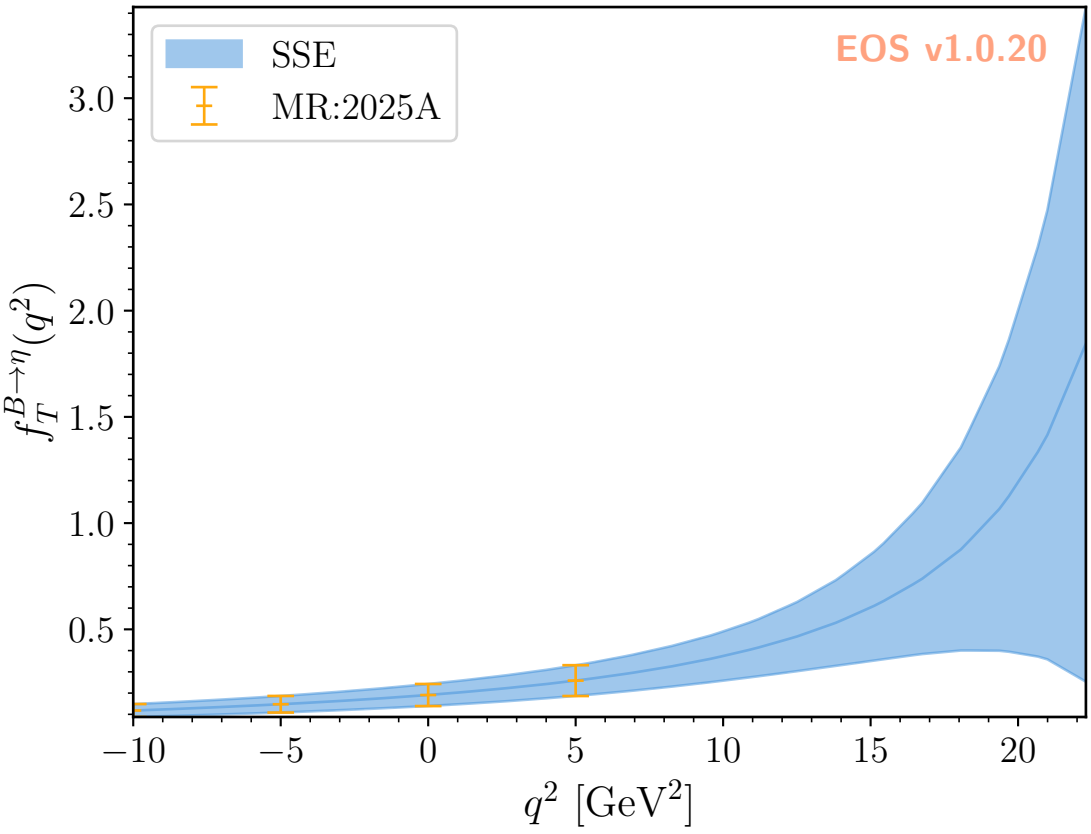

Figure 8: Posterior-predicted form factor for `B-to-eta` (`B-to-eta-fT`).

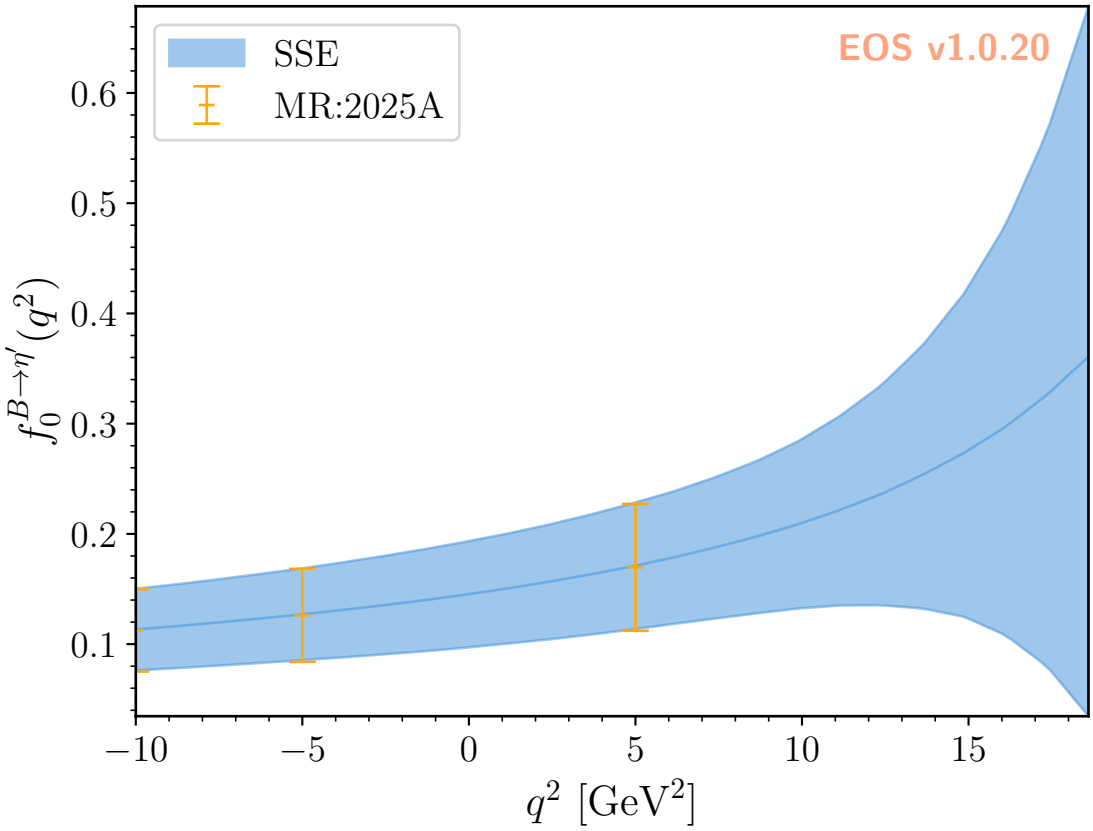

Figure 9: Posterior-predicted form factor for `B-to-etaprime` (`B-to-etaprime-f0`).

Posterior `B-to-etaprime`

This posterior varies 8 `SSE` coefficients. It is constrained by:

- `B->eta_prime::f_0+f_++f_T@MR:2025A`

Goodness of fit

The table lists the local p value of each constraint together with the global p value and total χ^2 of the best-fit point.

Constraint	local p value
`B->eta_prime::f_0+f_++f_T@MR:2025A`	100.0%
total (χ^2 / d.o.f. = 0.00 / 3)	**100.0%**

Posterior-predicted form factors

The bands show the posterior-predicted form factors (`SSE`); the points show the constraints entering the fit.

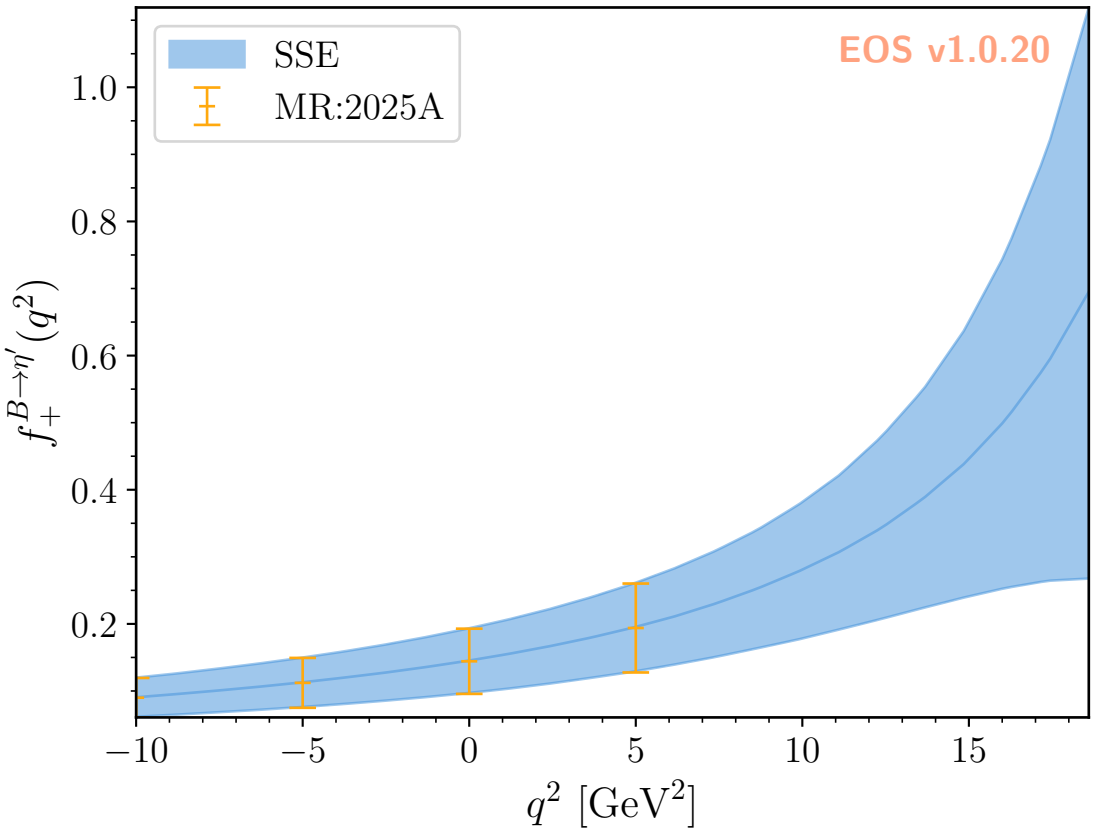

Figure 10: Posterior-predicted form factor for `B-to-etaprime` (`B-to-etaprime-fp`).

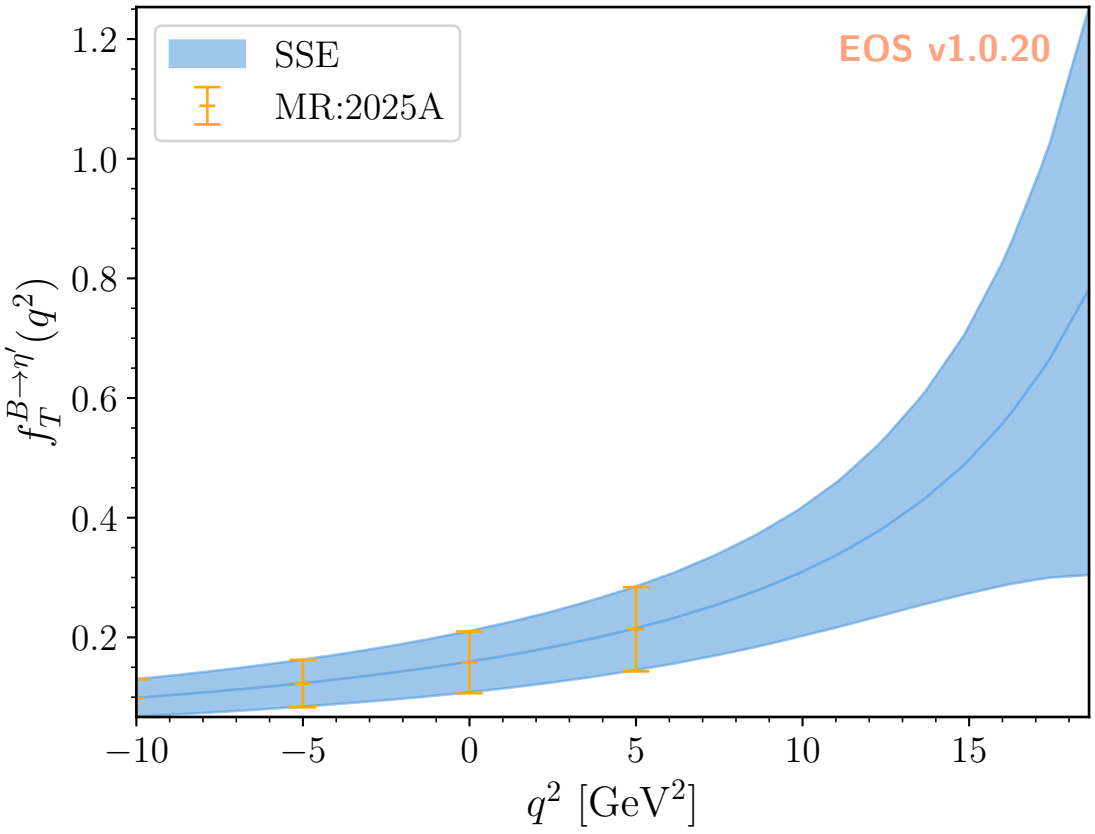

Figure 11: Posterior-predicted form factor for `B-to-etaprime` (`B-to-etaprime-fT`).

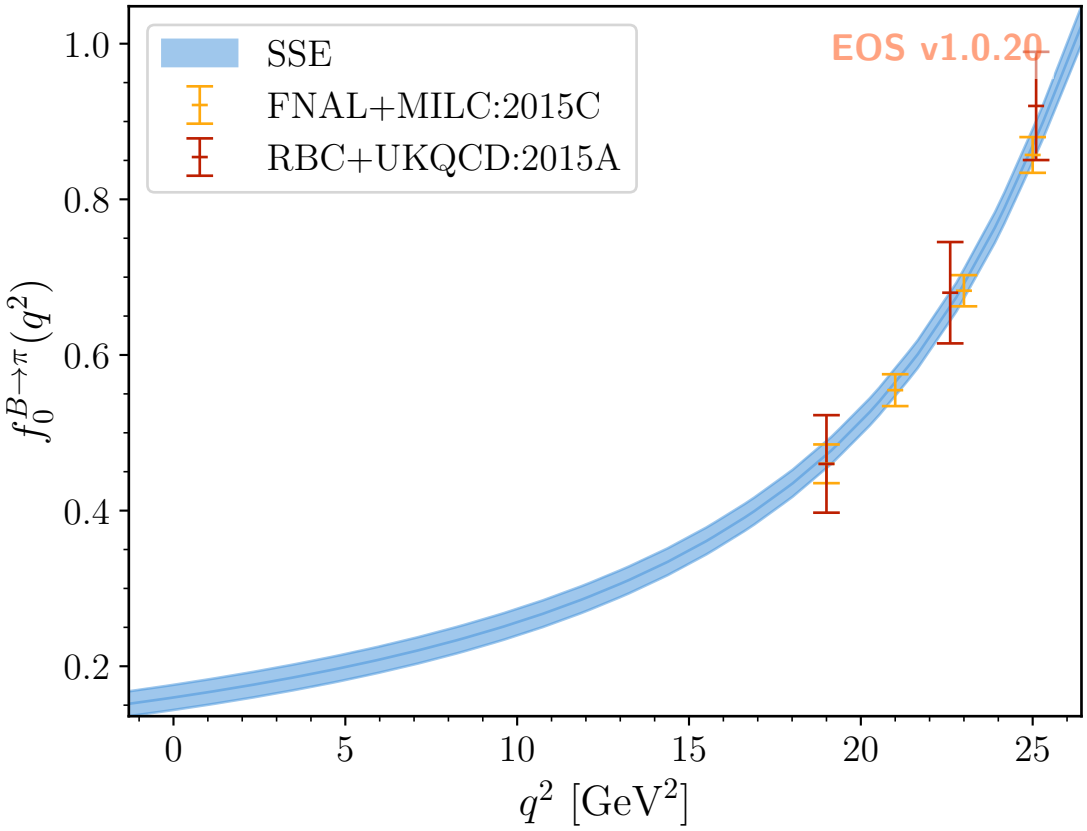

Figure 12: Posterior-predicted form factor for `B-to-pi` (`B-to-pi-f0`).

Posterior `B-to-pi`

This posterior varies 8 `SSE` coefficients. It is constrained by:

- `B->pi::f_++f_0+f_T@FNAL+MILC:2015C`
- `B->pi::f_++f_0@RBC+UKQCD:2015A`

Goodness of fit

The table lists the local p value of each constraint together with the global p value and total χ^2 of the best-fit point.

Constraint	local p value
`B->pi::f_++f_0+f_T@FNAL+MILC:2015C`	99.7%
`B->pi::f_++f_0@RBC+UKQCD:2015A`	57.7%
total (χ^2 / d.o.f. = 7.06 / 9)	**63.1%**

Posterior-predicted form factors

The bands show the posterior-predicted form factors (`SSE`); the points show the constraints entering the fit.

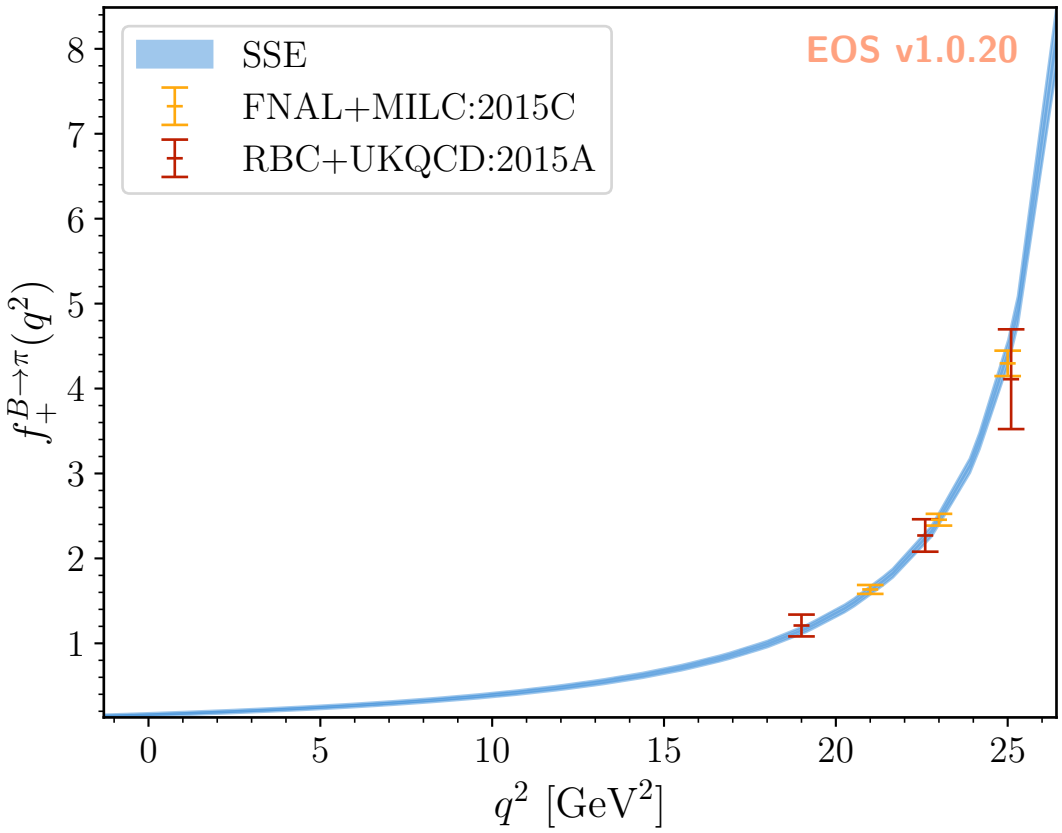

Figure 13: Posterior-predicted form factor for `B-to-pi` (`B-to-pi-fp`).

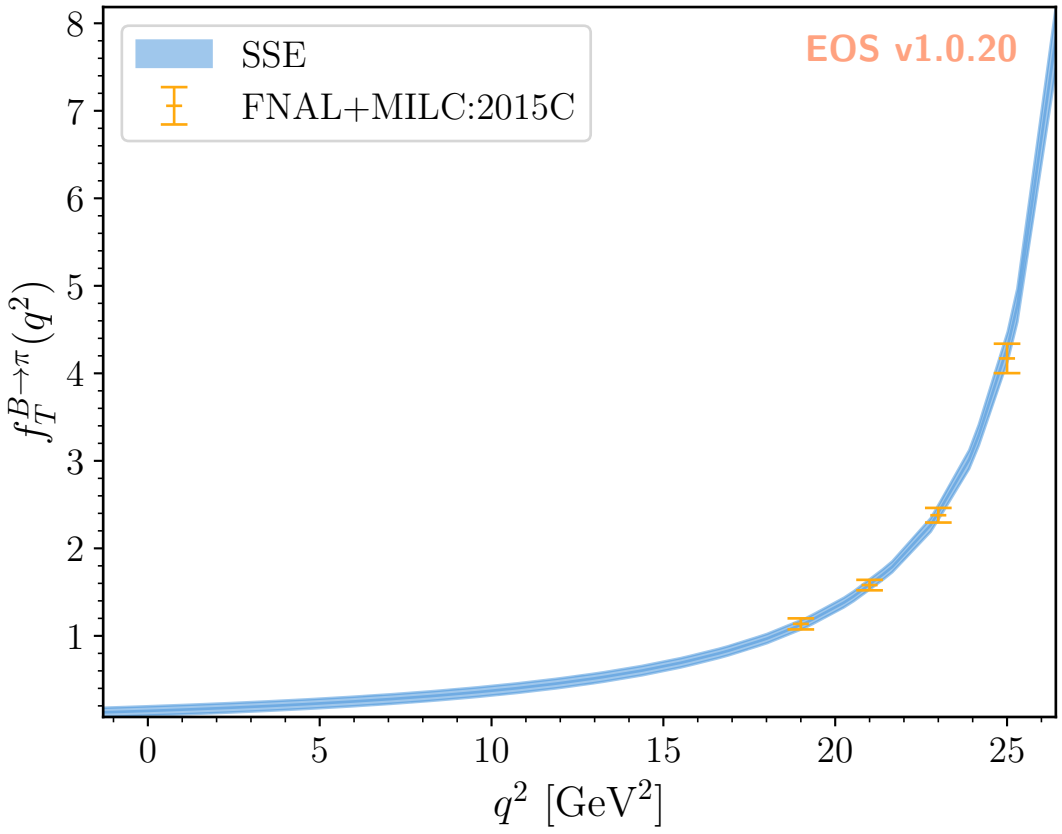

Figure 14: Posterior-predicted form factor for `B-to-pi` (`B-to-pi-fT`).

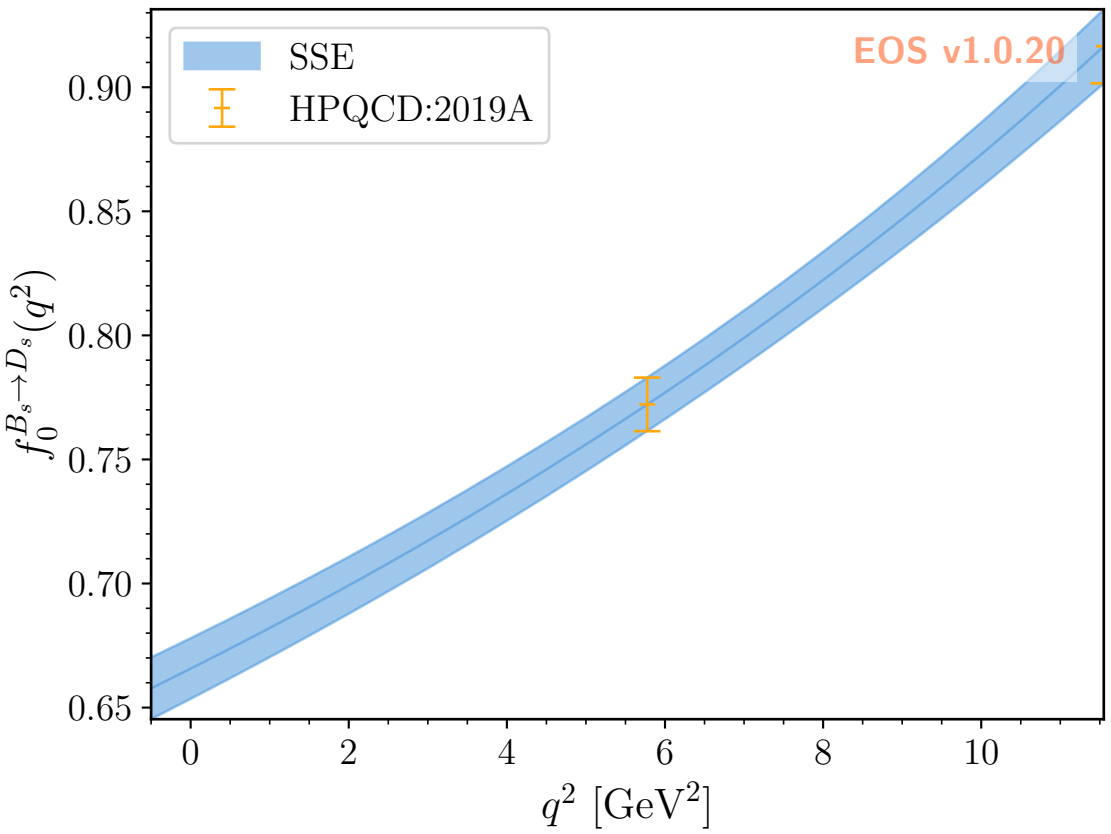

Figure 15: Posterior-predicted form factor for `Bs-to-Ds` (`Bs-to-Ds-f0`).

Posterior `Bs-to-Ds`

This posterior varies 5 `SSE` coefficients. It is constrained by:

- `B_s->D_s::f_++f_0@HPQCD:2019A`

  Note. No (lattice or LCSR) tensor form-factor data are available for this transition. The tensor coefficients are therefore not varied and retain their database default values; the tensor form factors below are not determined by this fit.

Goodness of fit

The table lists the local p value of each constraint. The fit is saturated (the number of data points equals the number of coefficients, d.o.f. $= 0$), so no global p value is defined.

Constraint	local p value
`B_s->D_s::f_++f_0@HPQCD:2019A`	100.0%

Posterior-predicted form factors

The bands show the posterior-predicted form factors (`SSE`); the points show the constraints entering the fit.

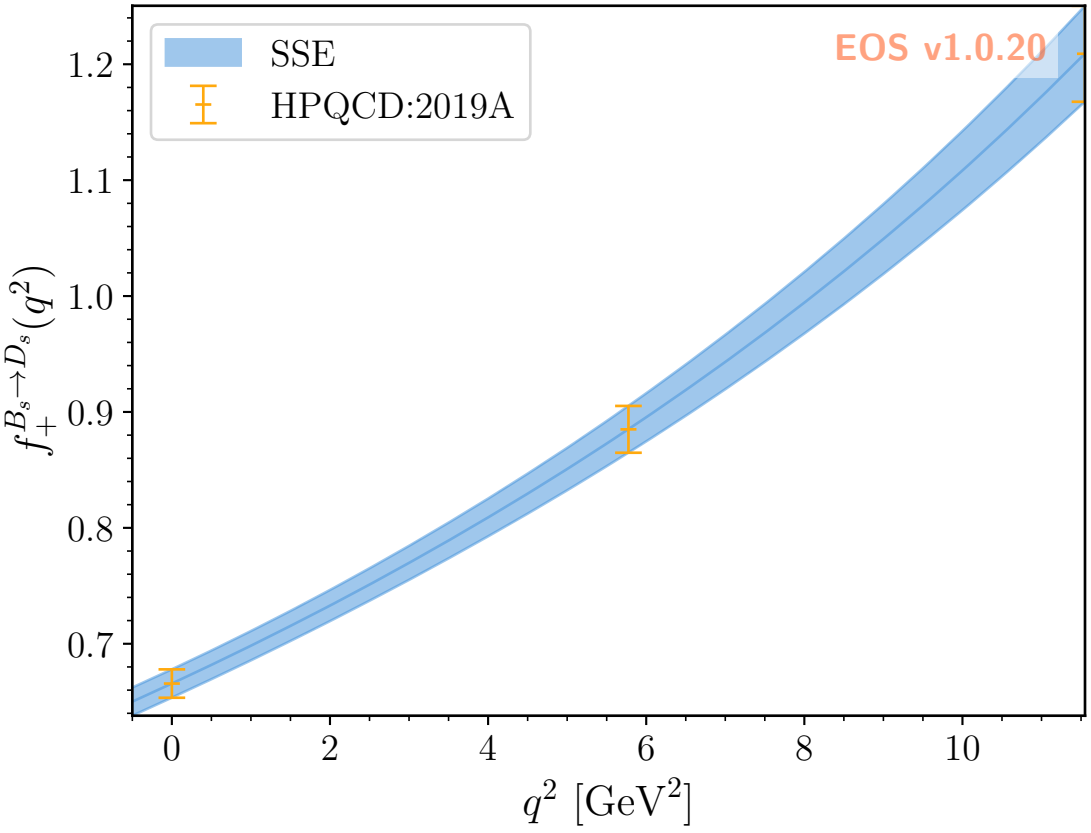

Figure 16: Posterior-predicted form factor for `Bs-to-Ds` (`Bs-to-Ds-fp`).

Posterior Bs-to-K

This posterior varies 8 SSE coefficients. It is constrained by:

- B_s->K::f_++f_0@FNAL+MILC:2019A;rescale-factor=11.311
- B_s->K::f_++f_0@HPQCD:2014A;rescale-factor=11.311
- B_s->K::f_++f_0@RBC+UKQCD:2023A;rescale-factor=11.311
- B_s->K::FormFactors[f_+,f_0,f_T]@GvDV:2020A

Note. The lattice inputs for this transition are mutually inconsistent. Their covariances have been rescaled by the PDG factor $\mathcal{S}^2 = 11.311$ (scale factor $\mathcal{S} = 3.36$), as indicated by the ;rescale-factor option on the constraints above.

Goodness of fit

The table lists the local p value of each constraint together with the global p value and total χ^2 of the best-fit point.

Constraint	local p value
B_s->K::FormFactors[f_+,f_0,f_T]@GvDV:2020A	100.0%
B_s->K::f_++f_0@FNAL+MILC:2019A	7.5%
B_s->K::f_++f_0@HPQCD:2014A	99.4%
B_s->K::f_++f_0@RBC+UKQCD:2023A	54.4%
total (χ^2 / d.o.f. = 19.69 / 25)	**76.3%**

Posterior-predicted form factors

The bands show the posterior-predicted form factors (SSE); the points show the constraints entering the fit.

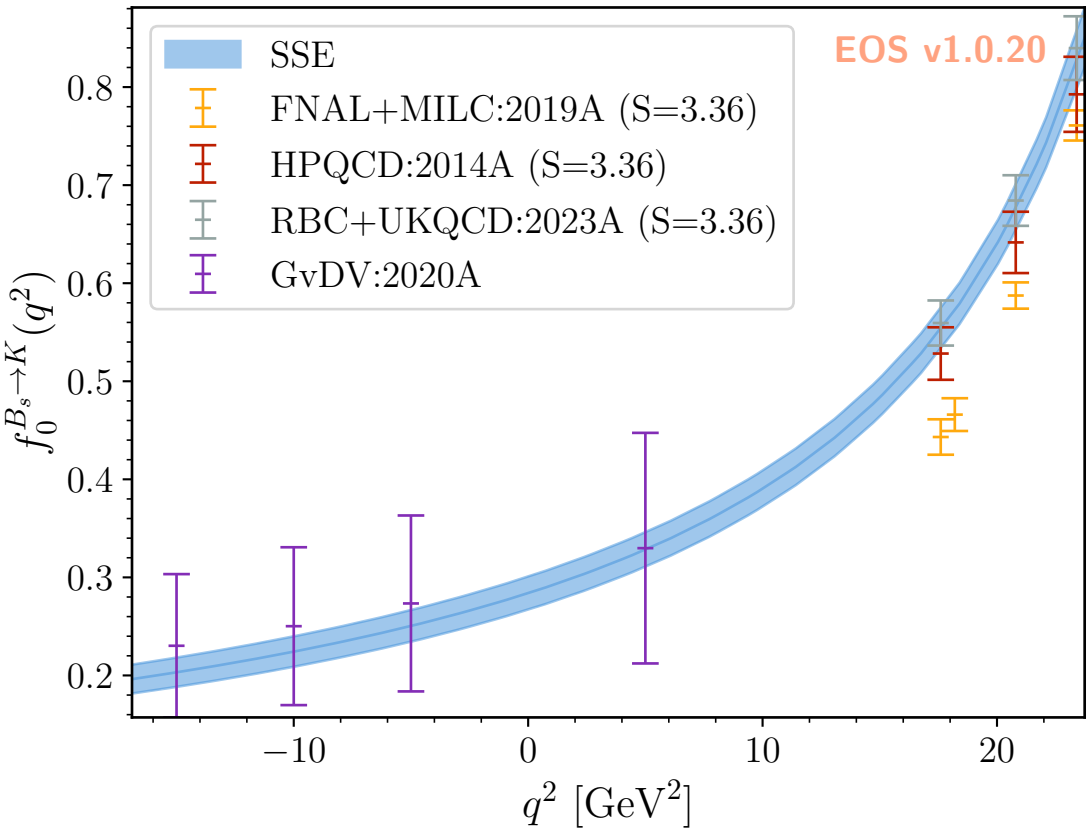

Figure 17: Posterior-predicted form factor for `Bs-to-K` (`Bs-to-K-f0`).

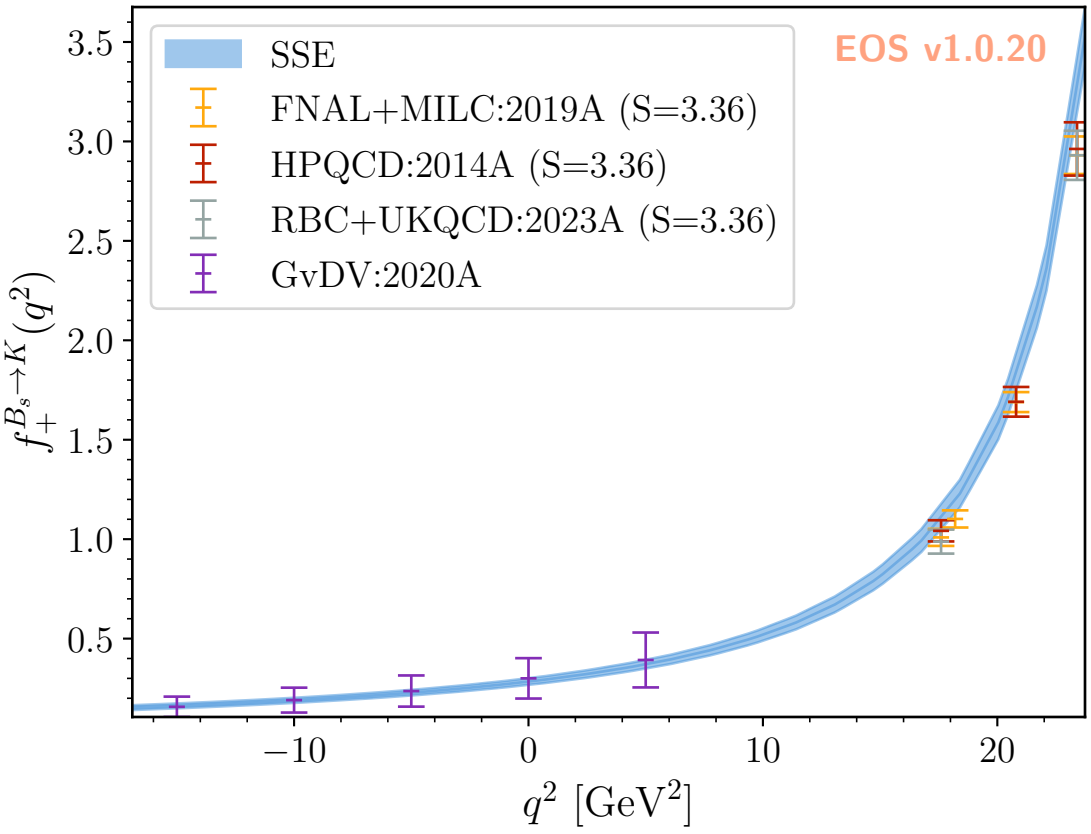

Figure 18: Posterior-predicted form factor for `Bs-to-K` (`Bs-to-K-fp`).

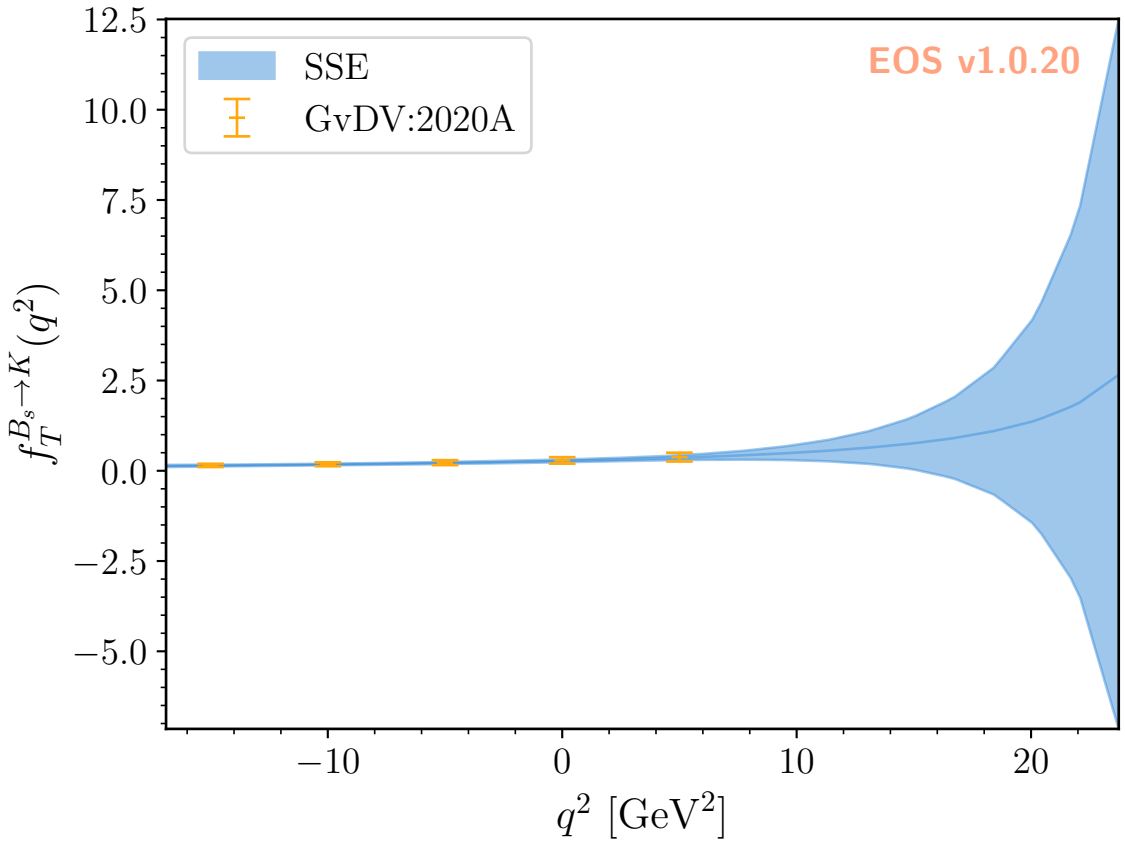

Figure 19: Posterior-predicted form factor for `Bs-to-K` (`Bs-to-K-fT`).

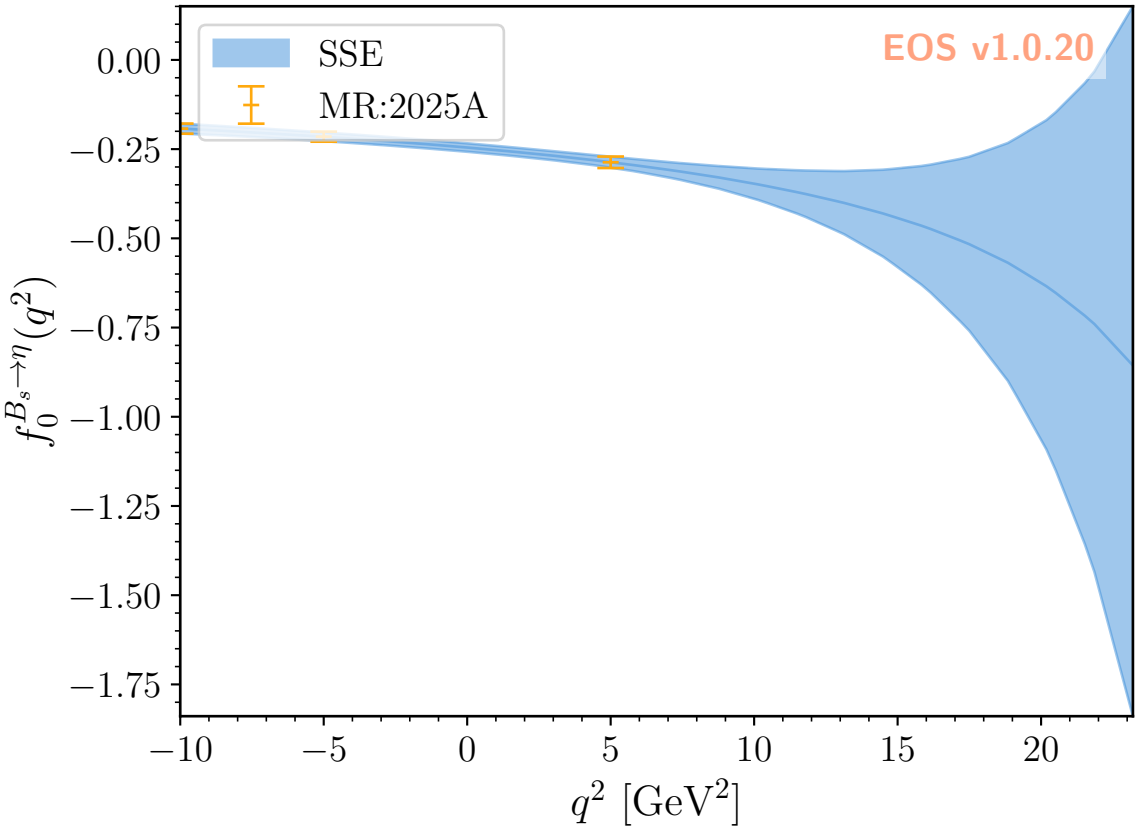

Figure 20: Posterior-predicted form factor for `Bs-to-eta` (`Bs-to-eta-f0`).

Posterior `Bs-to-eta`

This posterior varies 8 `SSE` coefficients. It is constrained by:

- `B_s->eta::f_0+f_++f_T@MR:2025A`

Goodness of fit

The table lists the local p value of each constraint together with the global p value and total χ^2 of the best-fit point.

Constraint	local p value
`B_s->eta::f_0+f_++f_T@MR:2025A`	100.0%
total (χ^2 / d.o.f. = 0.00 / 3)	**100.0%**

Posterior-predicted form factors

The bands show the posterior-predicted form factors (`SSE`); the points show the constraints entering the fit.

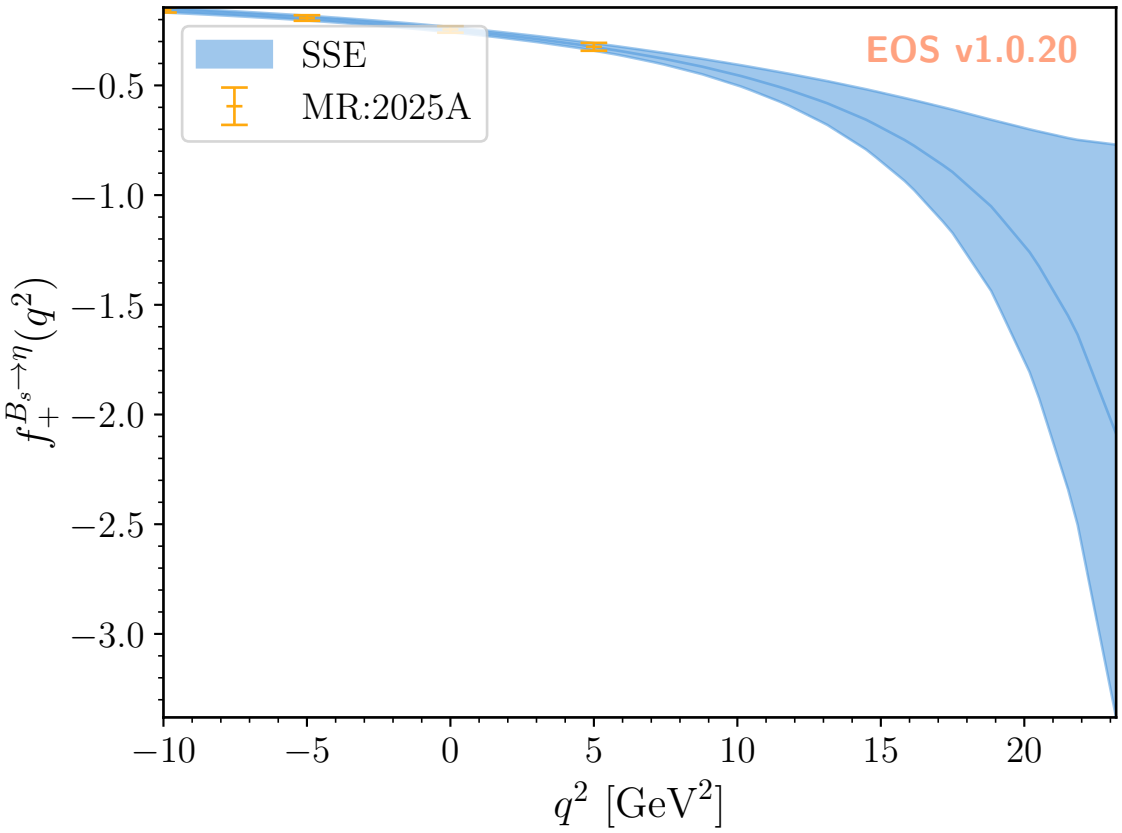

Figure 21: Posterior-predicted form factor for `Bs-to-eta` (`Bs-to-eta-fp`).

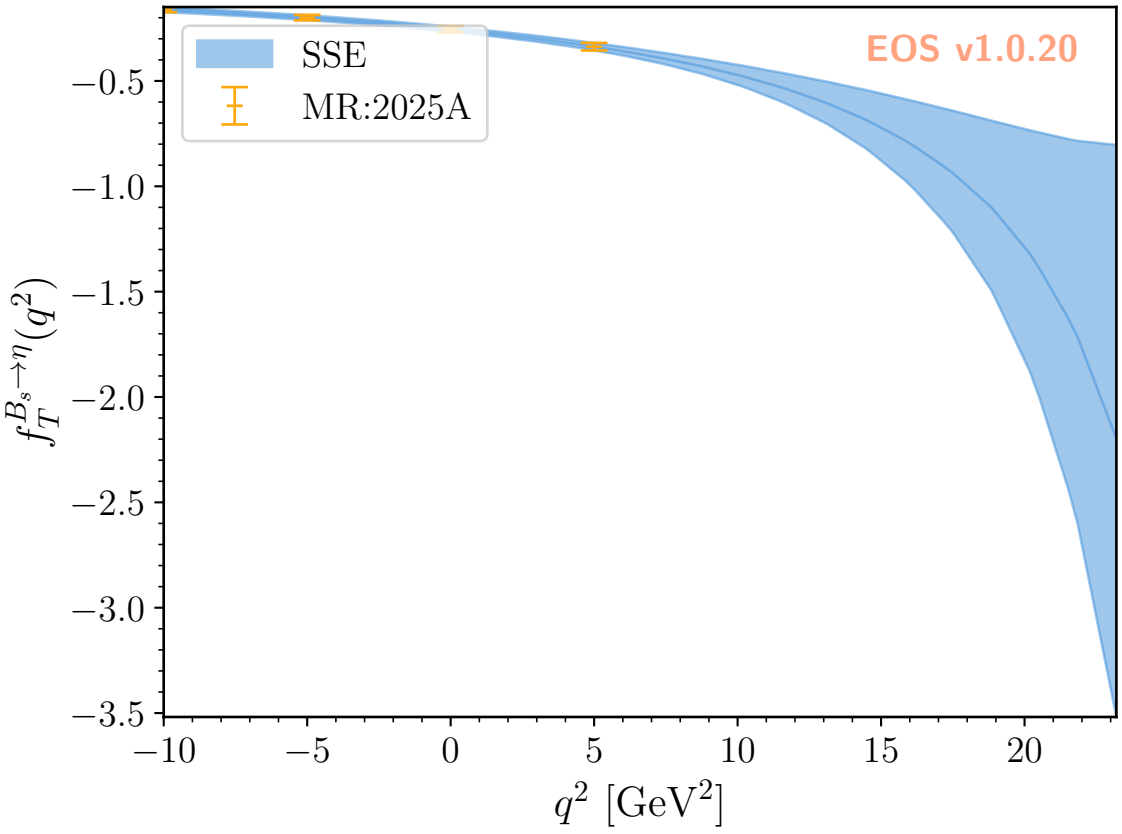

Figure 22: Posterior-predicted form factor for `Bs-to-eta` (`Bs-to-eta-fT`).

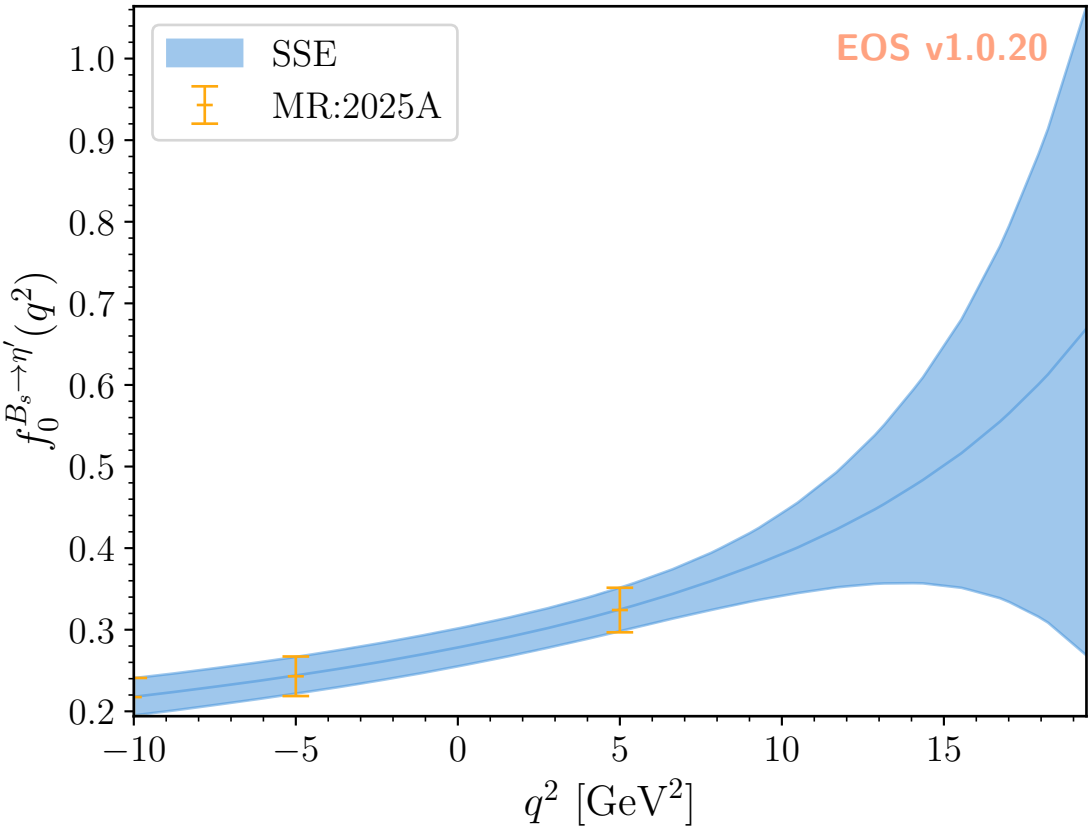

Figure 23: Posterior-predicted form factor for `Bs-to-etaprime` (`Bs-to-etaprime-f0`).

Posterior `Bs-to-etaprime`

This posterior varies 8 `SSE` coefficients. It is constrained by:

- `B_s->eta_prime::f_0+f_++f_T@MR:2025A`

Goodness of fit

The table lists the local p value of each constraint together with the global p value and total χ^2 of the best-fit point.

Constraint	local p value
`B_s->eta_prime::f_0+f_++f_T@MR:2025A`	100.0%
total (χ^2 / d.o.f. = 0.00 / 3)	**100.0%**

Posterior-predicted form factors

The bands show the posterior-predicted form factors (`SSE`); the points show the constraints entering the fit.

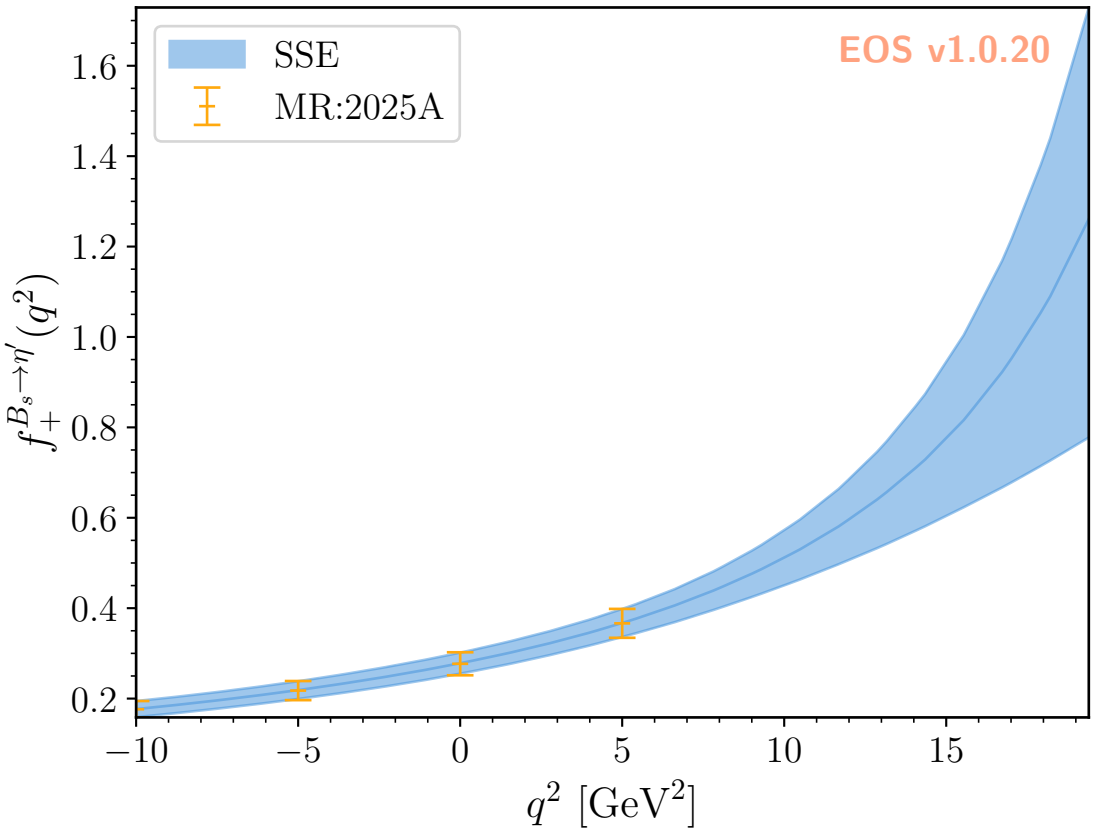

Figure 24: Posterior-predicted form factor for `Bs-to-etaprime` (`Bs-to-etaprime-fp`).

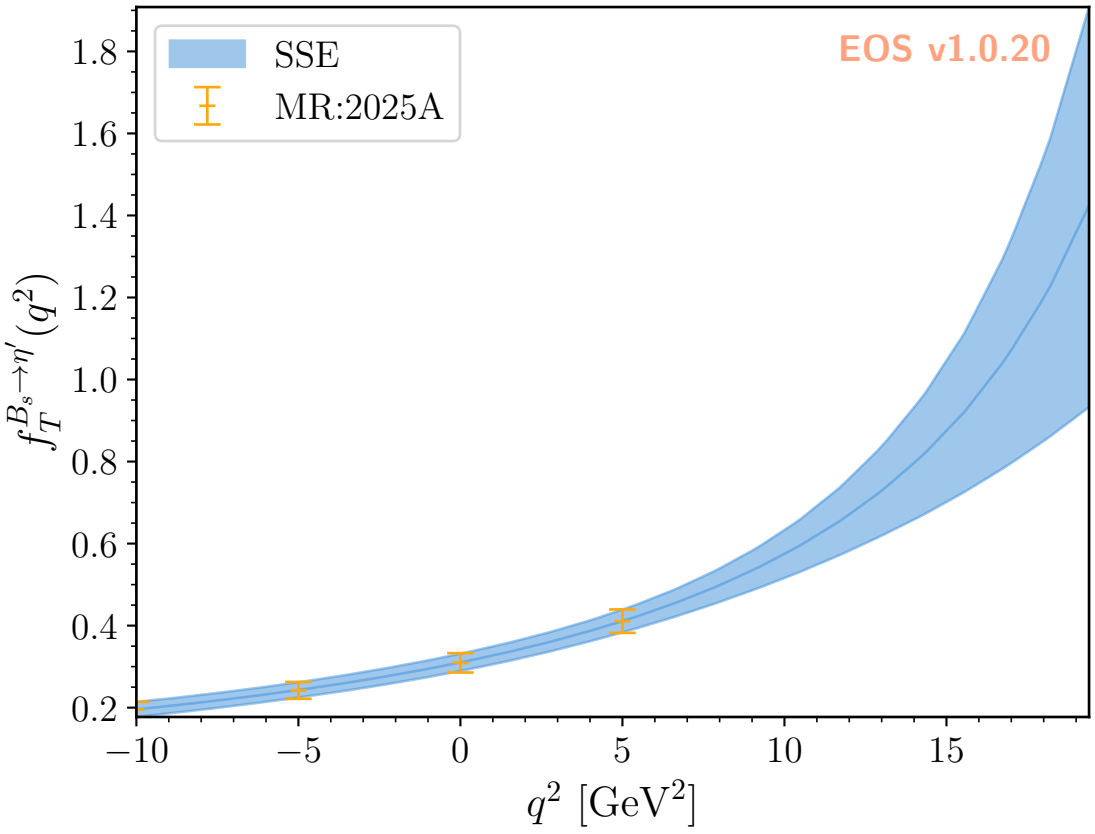

Figure 25: Posterior-predicted form factor for `Bs-to-etaprime` (`Bs-to-etaprime-fT`).

Posterior D-to-K

This posterior varies 8 SSE coefficients. It is constrained by:

- D->K::f_++f_0@ETM:2017B;rescale-factor=3.241
- D->K::f_T@ETM:2018A;rescale-factor=3.241
- D->K::f_0+f_++f_T@HPQCD:2022A;rescale-factor=3.241
- D->K::f_++f_0@FNALMILC:2022A;rescale-factor=3.241

Note. The lattice inputs for this transition are mutually inconsistent. Their covariances have been rescaled by the PDG factor $\mathcal{S}^2 = 3.241$ (scale factor $\mathcal{S} = 1.80$), as indicated by the ;rescale-factor option on the constraints above.

Goodness of fit

The table lists the local p value of each constraint together with the global p value and total χ^2 of the best-fit point.

Constraint	local p value
D->K::f_++f_0@ETM:2017B	8.8%
D->K::f_++f_0@FNALMILC:2022A	99.8%
D->K::f_0+f_++f_T@HPQCD:2022A	53.7%
D->K::f_T@ETM:2018A	8.3%
total (χ^2 / d.o.f. = 21.00 / 13)	**7.3%**

Posterior-predicted form factors

The bands show the posterior-predicted form factors (SSE); the points show the constraints entering the fit.

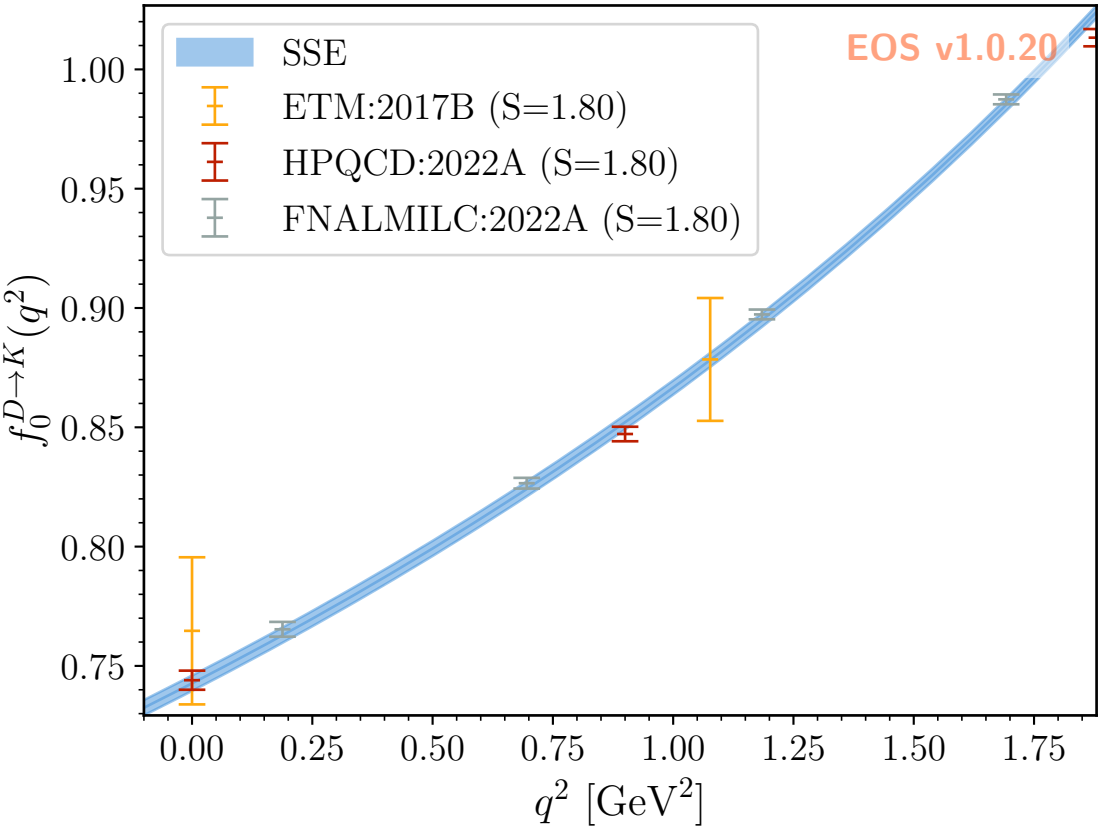

Figure 26: Posterior-predicted form factor for `D-to-K` (`D-to-K-f0`).

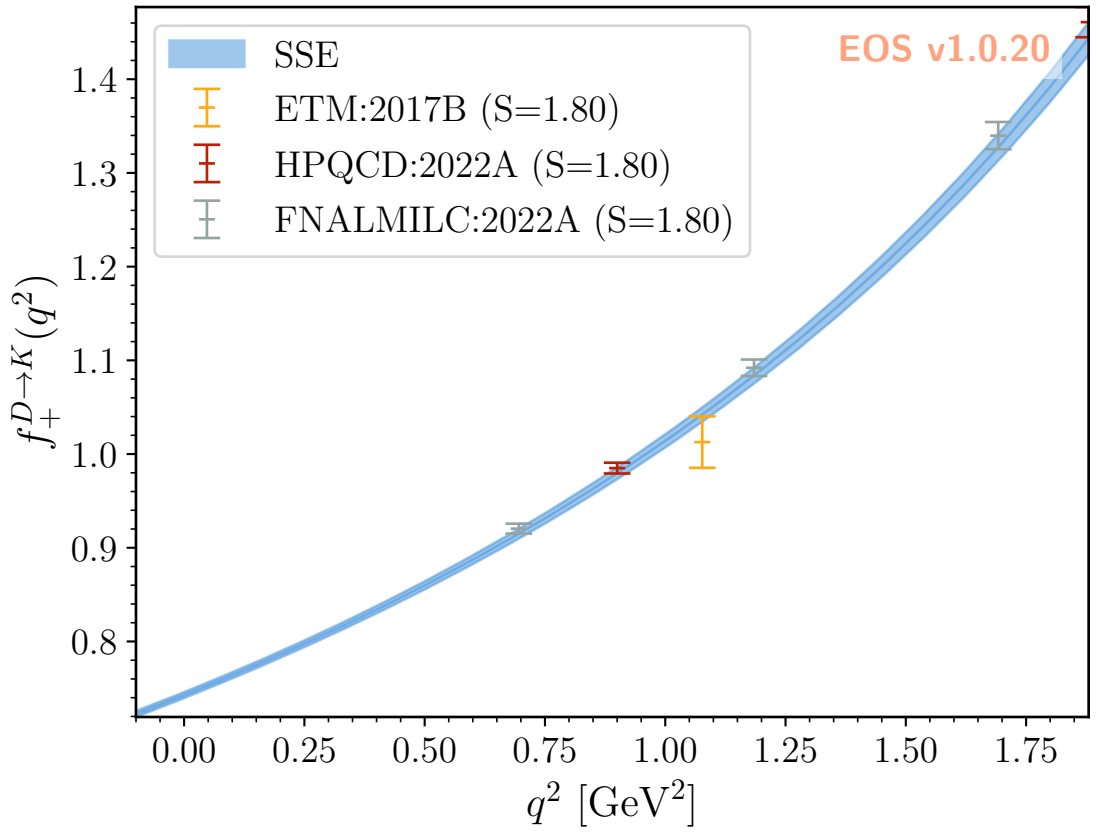

Figure 27: Posterior-predicted form factor for `D-to-K` (`D-to-K-fp`).

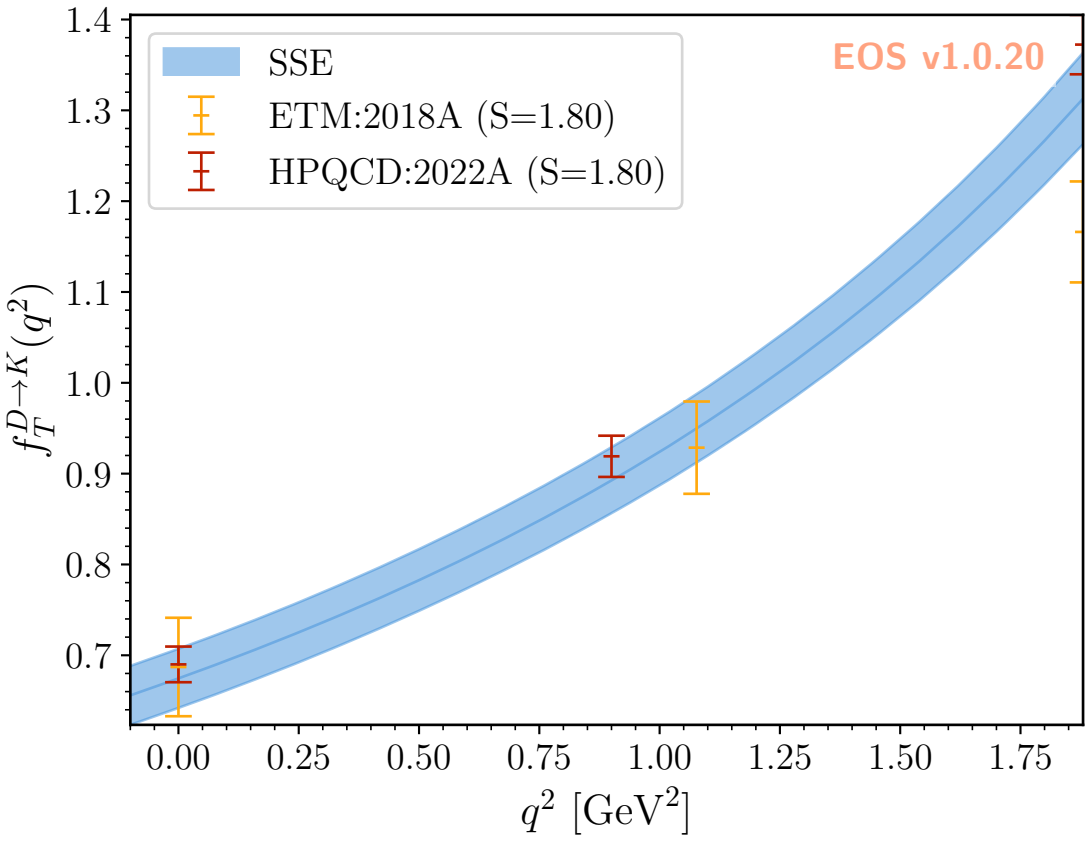

Figure 28: Posterior-predicted form factor for `D-to-K` (`D-to-K-fT`).

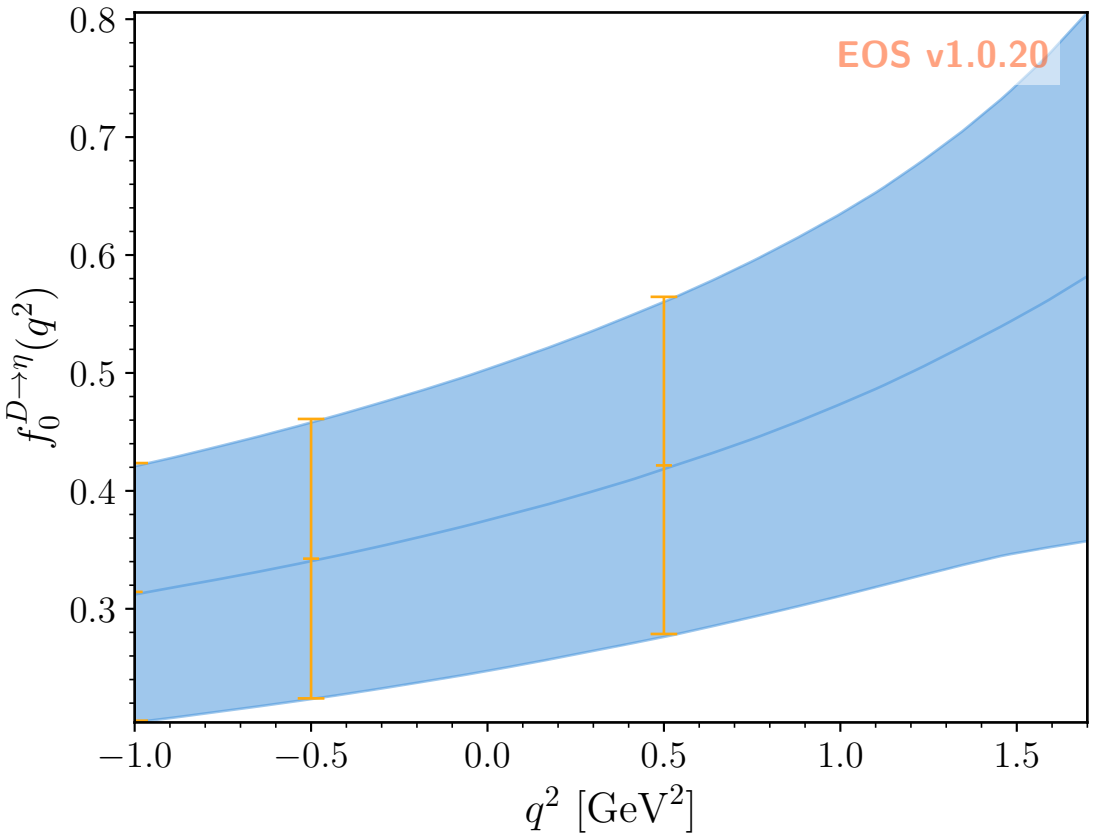

Figure 29: Posterior-predicted form factor for `D-to-eta` (`D-to-eta-f0`).

Posterior `D-to-eta`

This posterior varies 8 `SSE` coefficients. It is constrained by:

- `D->eta::f_0+f_++f_T@MR:2025A`

Goodness of fit

The table lists the local p value of each constraint together with the global p value and total χ^2 of the best-fit point.

Constraint	local p value
`D->eta::f_0+f_++f_T@MR:2025A`	100.0%
total (χ^2 / d.o.f. = 0.02 / 3)	**99.9%**

Posterior-predicted form factors

The bands show the posterior-predicted form factors (`SSE`); the points show the constraints entering the fit.

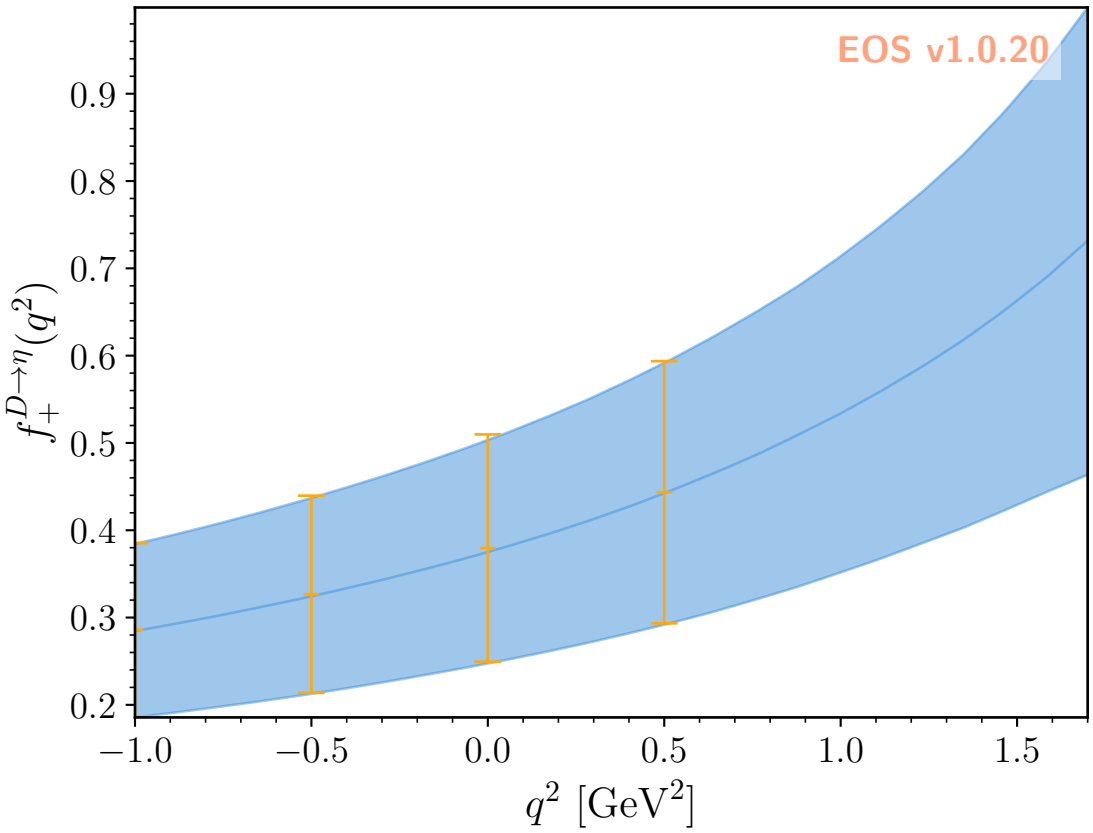

Figure 30: Posterior-predicted form factor for `D-to-eta` (`D-to-eta-fp`).

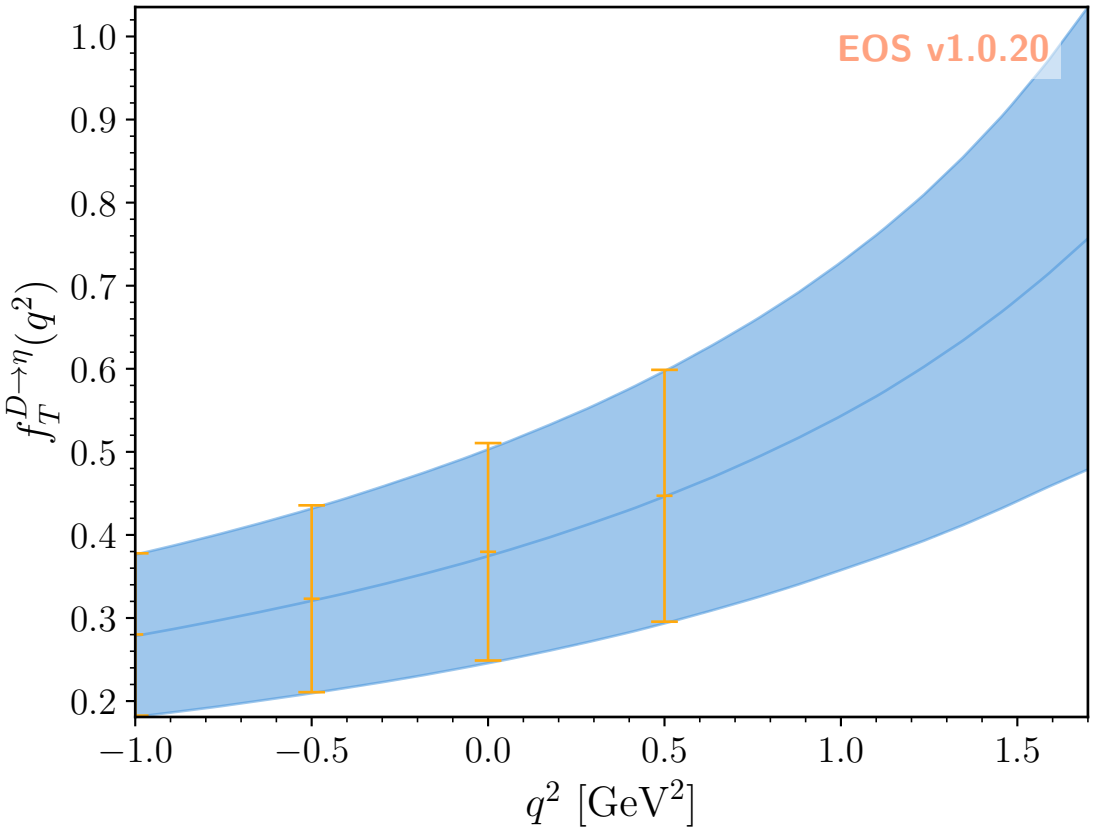

Figure 31: Posterior-predicted form factor for `D-to-eta` (`D-to-eta-fT`).

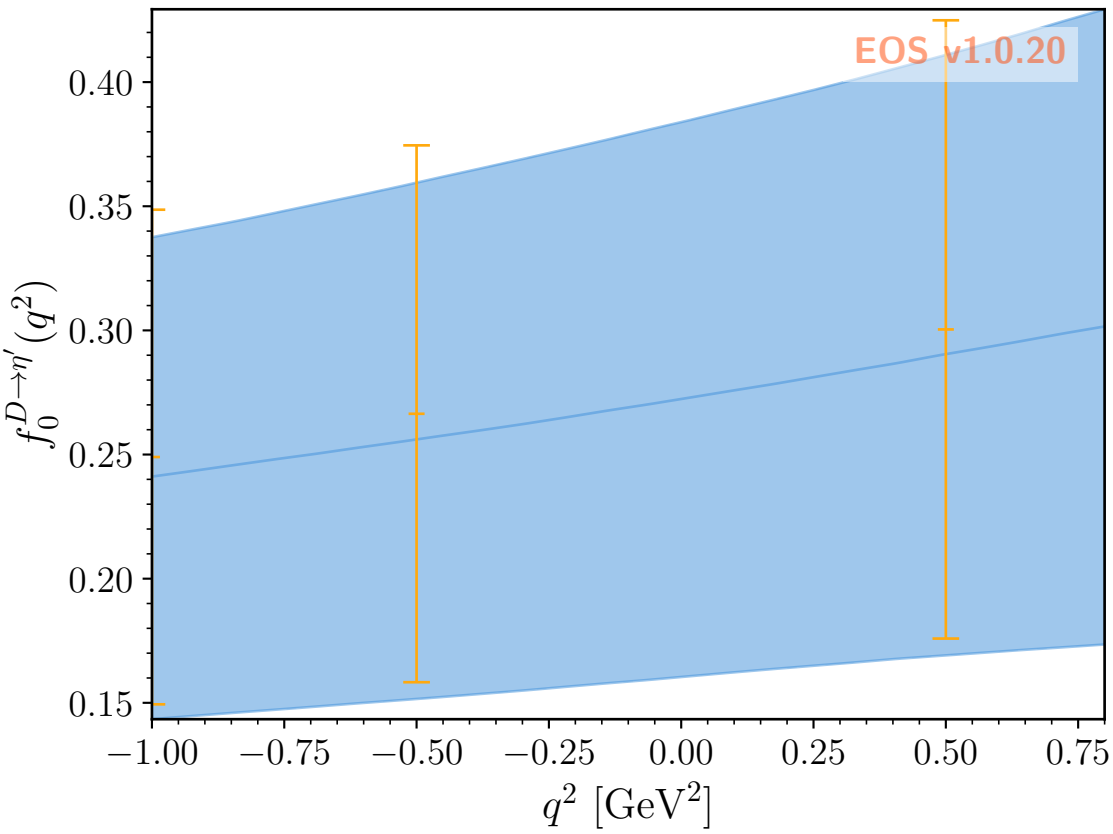

Figure 32: Posterior-predicted form factor for `D-to-etaprime` (`D-to-etaprime-f0`).

Posterior `D-to-etaprime`

This posterior varies 8 `SSE` coefficients. It is constrained by:

- `D->eta_prime::f_0+f_++f_T@MR:2025A`

Goodness of fit

The table lists the local p value of each constraint together with the global p value and total χ^2 of the best-fit point.

Constraint	local p value
`D->eta_prime::f_0+f_++f_T@MR:2025A`	100.0%
total (χ^2 / d.o.f. = 0.02 / 3)	**99.9%**

Posterior-predicted form factors

The bands show the posterior-predicted form factors (`SSE`); the points show the constraints entering the fit.

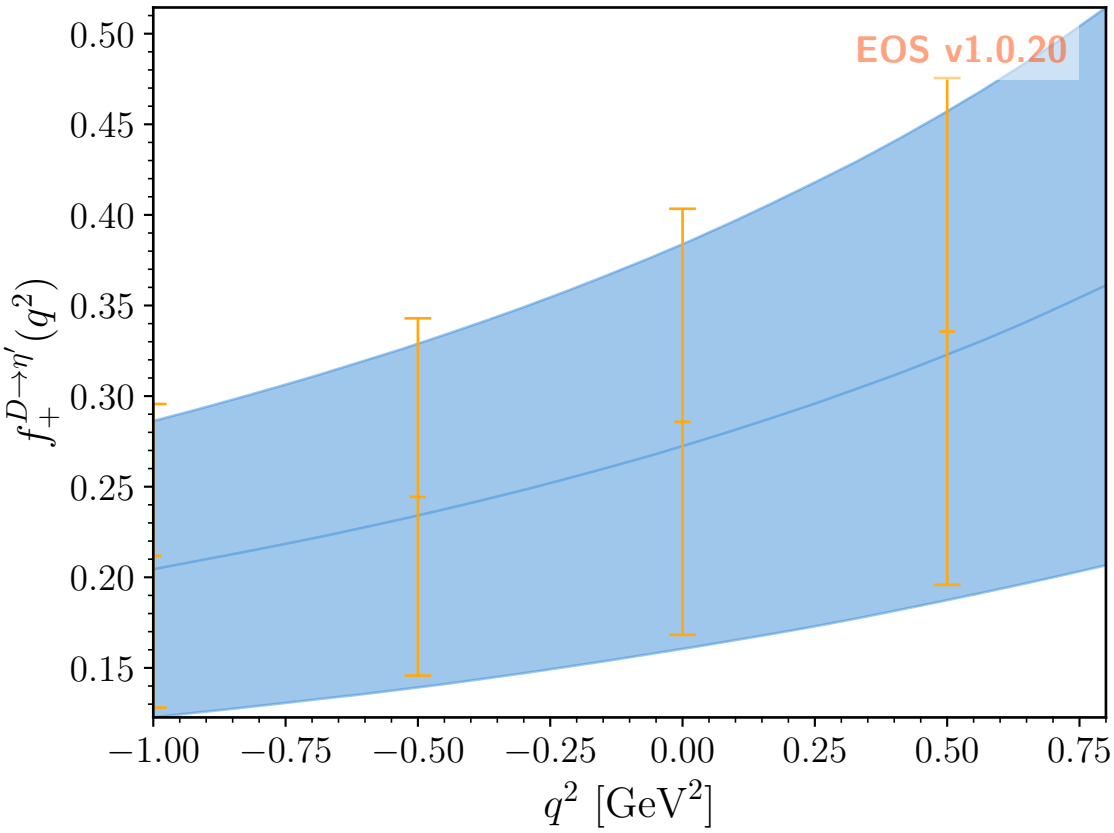

Figure 33: Posterior-predicted form factor for `D-to-etaprime` (`D-to-etaprime-fp`).

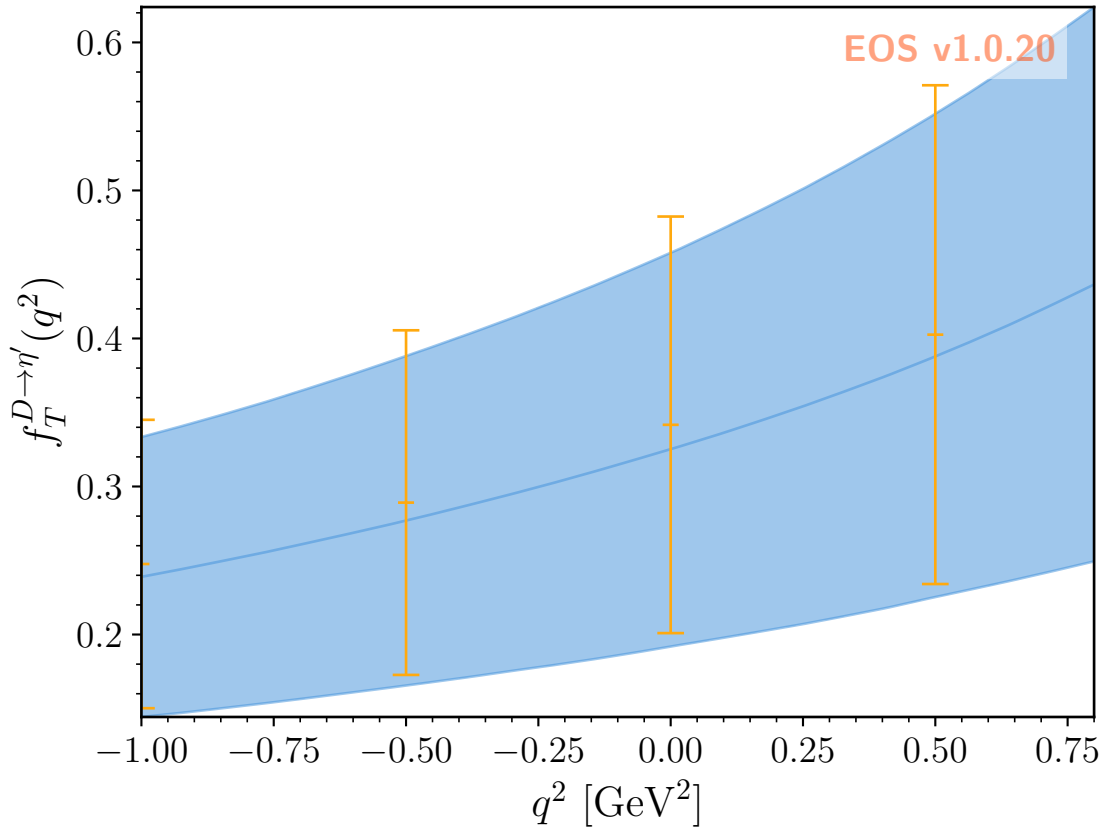

Figure 34: Posterior-predicted form factor for `D-to-etaprime` (`D-to-etaprime-fT`).

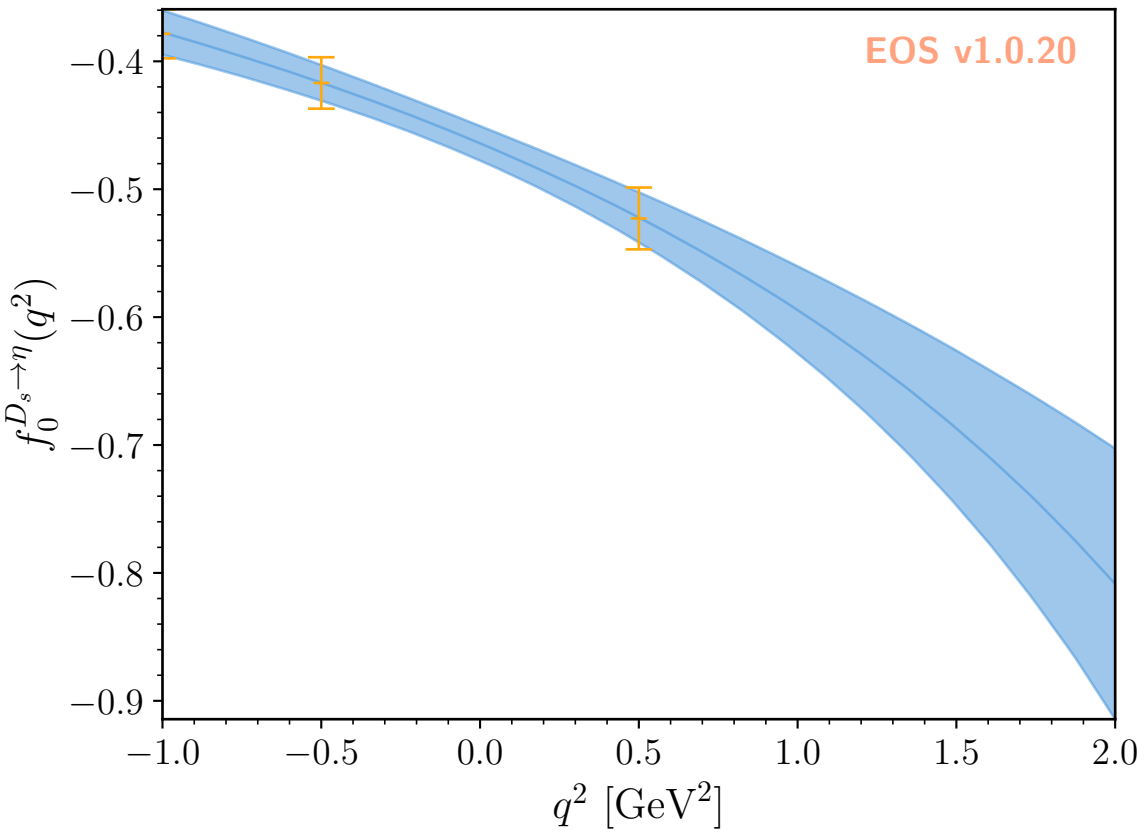

Figure 35: Posterior-predicted form factor for `Ds-to-eta` (`Ds-to-eta-f0`).

Posterior `Ds-to-eta`

This posterior varies 8 `SSE` coefficients. It is constrained by:

- `D_s->eta::f_0+f_++f_T@MR:2025A`

Goodness of fit

The table lists the local p value of each constraint together with the global p value and total χ^2 of the best-fit point.

Constraint	local p value
`D_s->eta::f_0+f_++f_T@MR:2025A`	100.0%
total (χ^2 / d.o.f. = 0.07 / 3)	**99.5%**

Posterior-predicted form factors

The bands show the posterior-predicted form factors (`SSE`); the points show the constraints entering the fit.

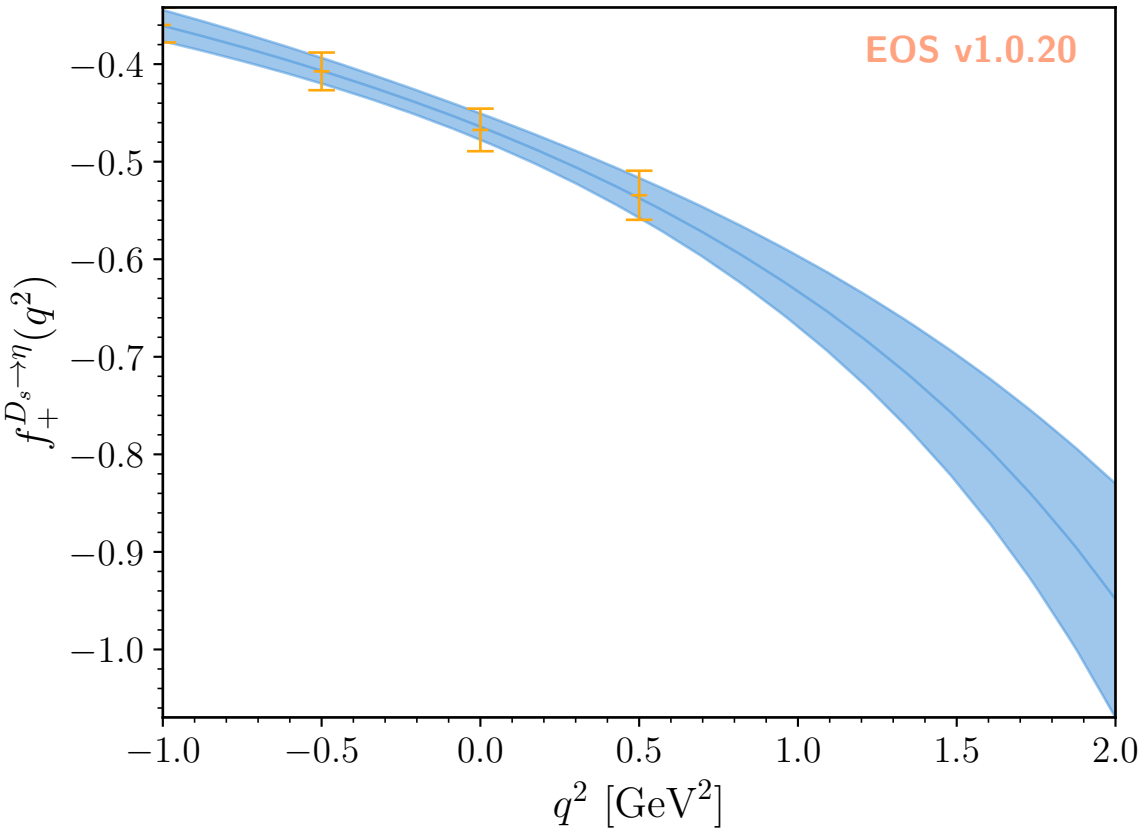

Figure 36: Posterior-predicted form factor for `Ds-to-eta` (`Ds-to-eta-fp`).

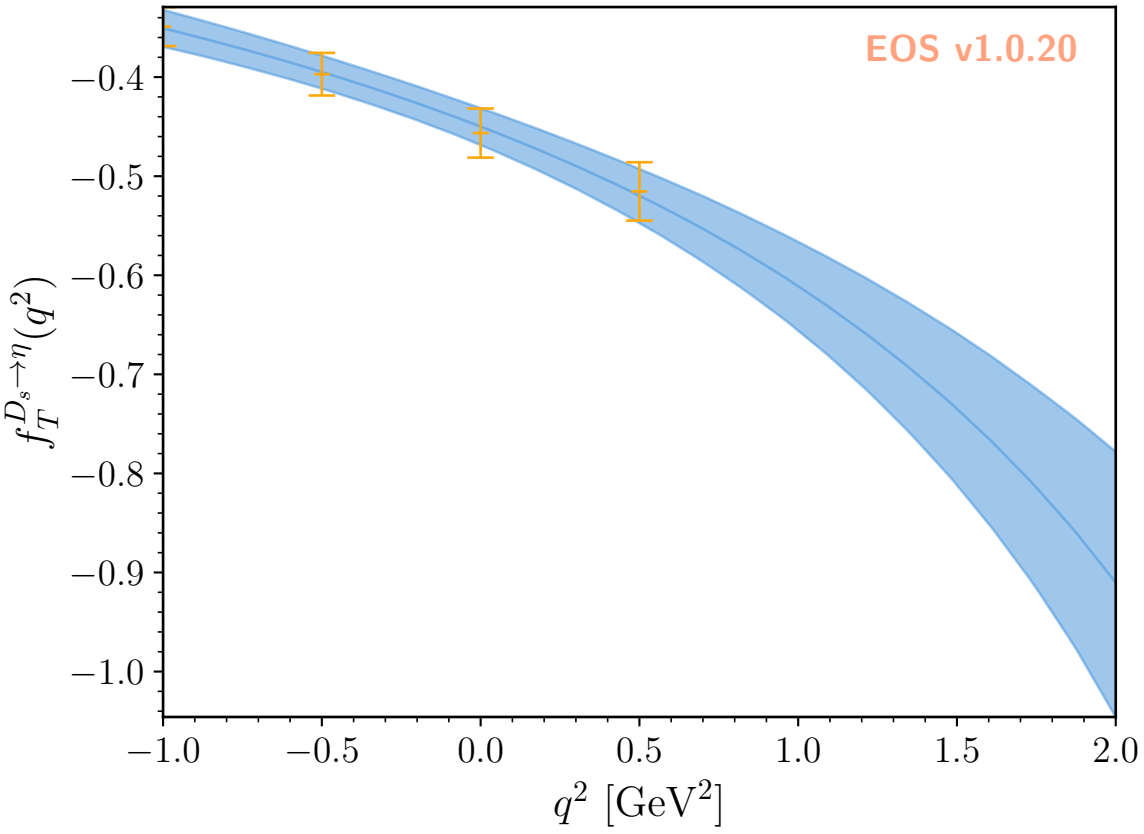

Figure 37: Posterior-predicted form factor for `Ds-to-eta` (`Ds-to-eta-fT`).

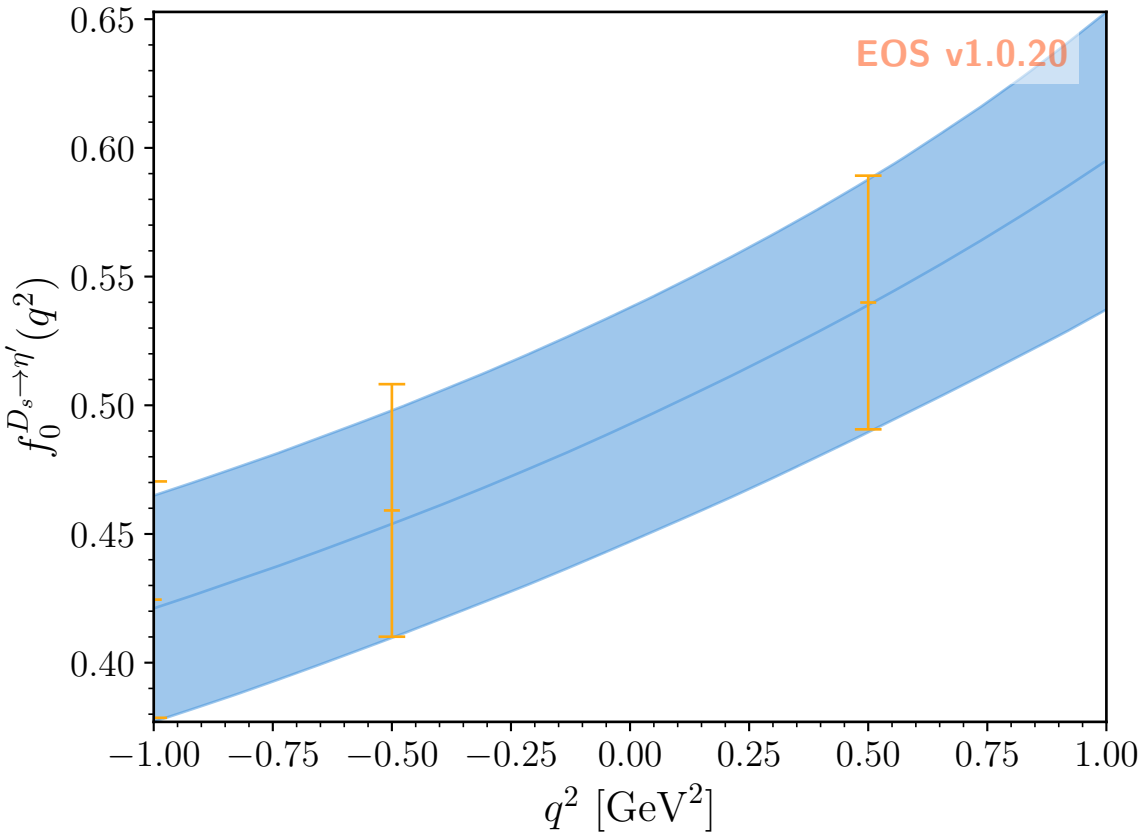

Figure 38: Posterior-predicted form factor for `Ds-to-etaprime` (`Ds-to-etaprime-f0`).

Posterior `Ds-to-etaprime`

This posterior varies 8 `SSE` coefficients. It is constrained by:

- `D_s->eta_prime::f_0+f_++f_T@MR:2025A`

Goodness of fit

The table lists the local p value of each constraint together with the global p value and total χ^2 of the best-fit point.

Constraint	local p value
`D_s->eta_prime::f_0+f_++f_T@MR:2025A`	100.0%
total (χ^2 / d.o.f. $= 0.14$ / 3)	**98.7%**

Posterior-predicted form factors

The bands show the posterior-predicted form factors (`SSE`); the points show the constraints entering the fit.

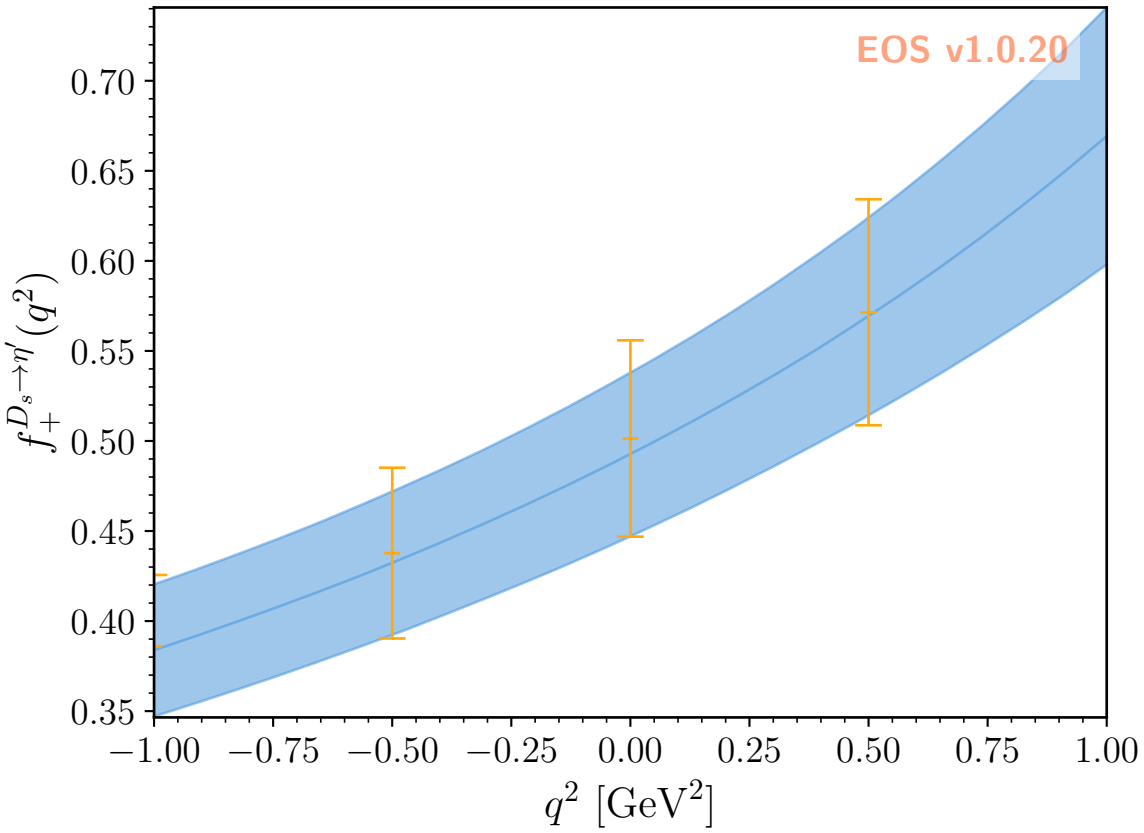

Figure 39: Posterior-predicted form factor for `Ds-to-etaprime` (`Ds-to-etaprime-fp`).

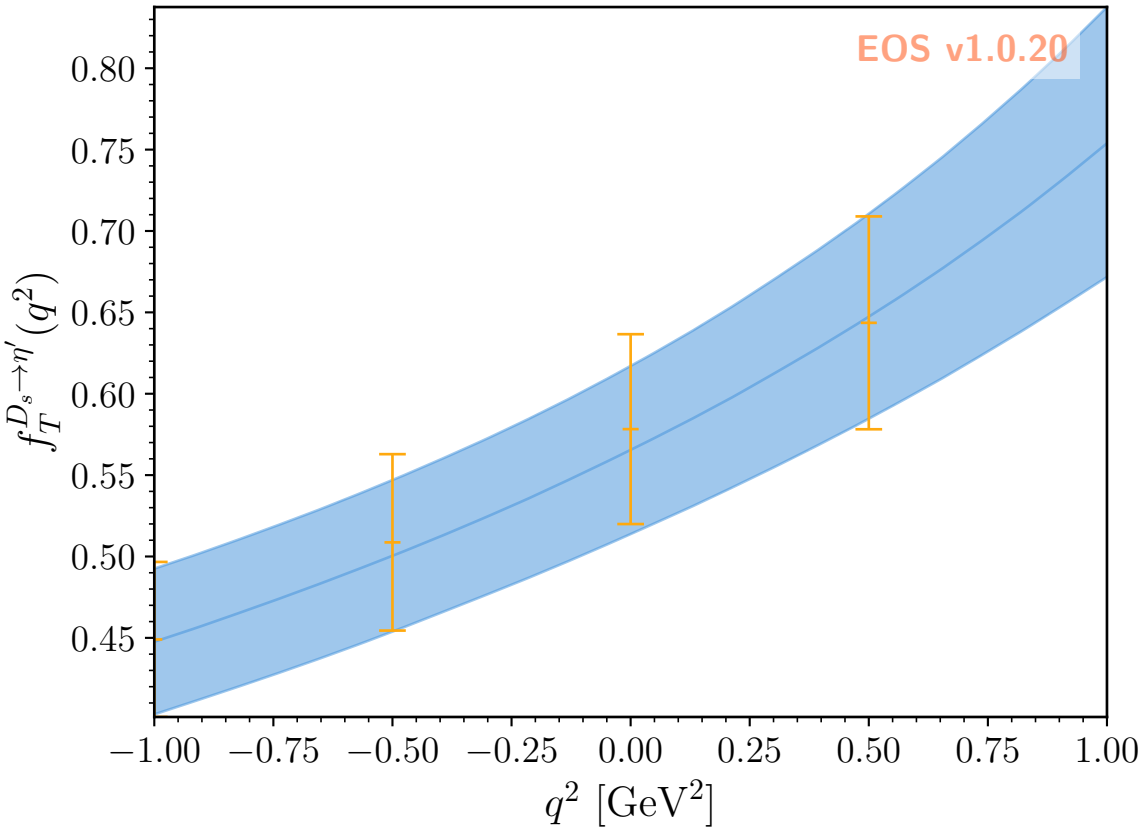

Figure 40: Posterior-predicted form factor for `Ds-to-etaprime` (`Ds-to-etaprime-fT`).

Pseudoscalar to vector ($P \to V$)

Posterior `B-to-Dstar`

This posterior varies 17 `SSE` coefficients. It is constrained by:

- `B->D^*::FormFactors@HPQCD:2023A`
- `B->D^*::FormFactors@FNAL+MILC:2021A`
- `B->D^*::FormFactors@JLQCD:2023A`

Goodness of fit

The table lists the local p value of each constraint together with the global p value and total χ^2 of the best-fit point.

Constraint	local p value
`B->D^*::FormFactors@FNAL+MILC:2021A`	72.5%
`B->D^*::FormFactors@HPQCD:2023A`	100.0%
`B->D^*::FormFactors@JLQCD:2023A`	89.2%
total (χ^2 / d.o.f. = 20.76 / 31)	**91.8%**

Posterior-predicted form factors

The bands show the posterior-predicted form factors (`SSE`); the points show the constraints entering the fit.

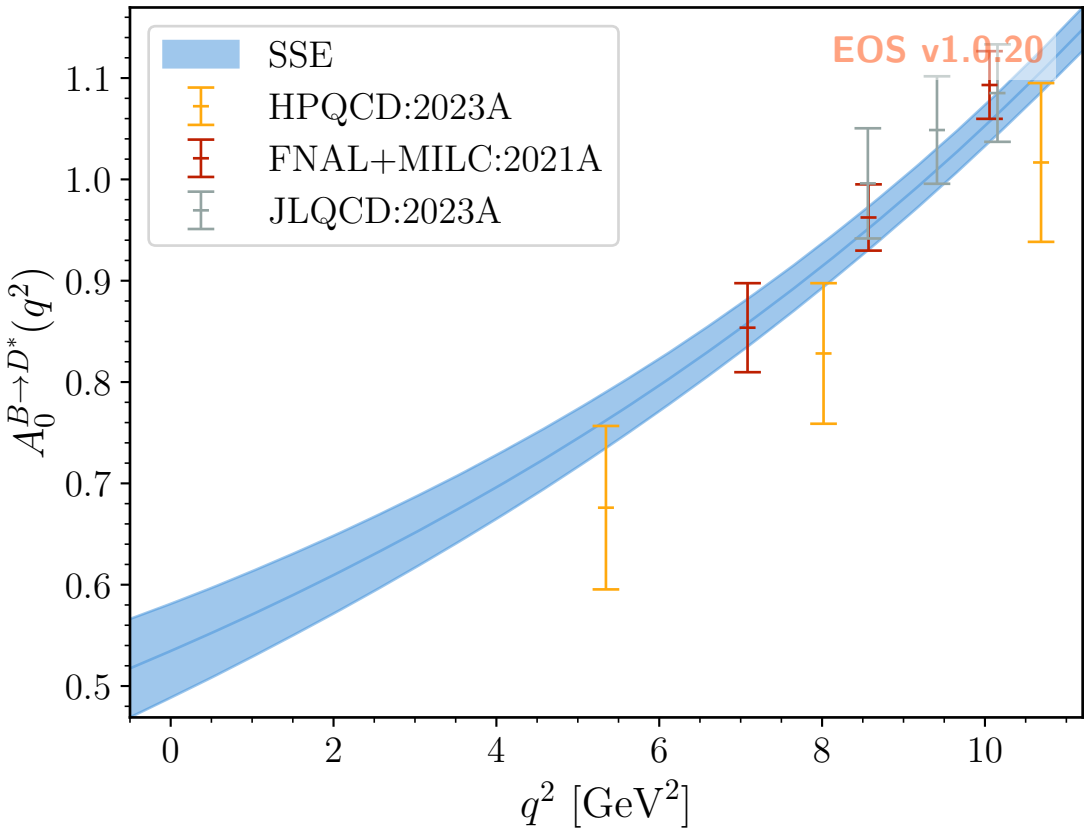

Figure 41: Posterior-predicted form factor for `B-to-Dstar` (`B-to-Dstar-A0`).

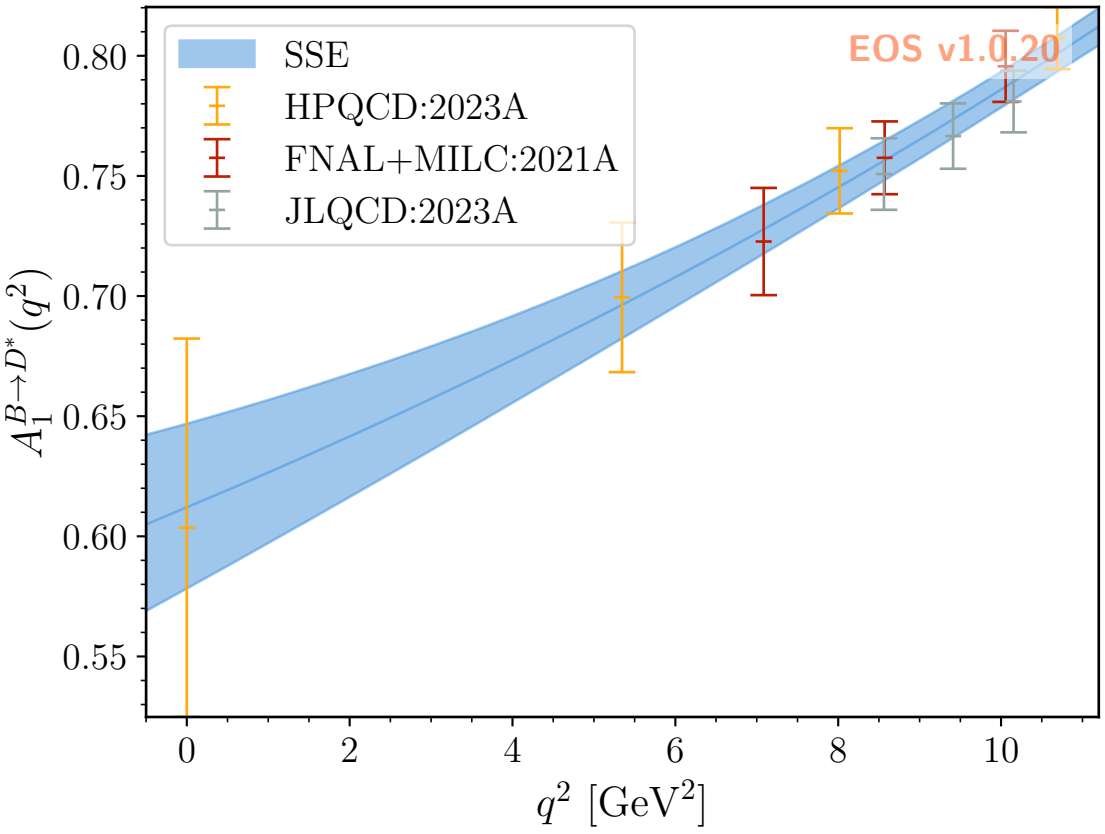

Figure 42: Posterior-predicted form factor for `B-to-Dstar` (`B-to-Dstar-A1`).

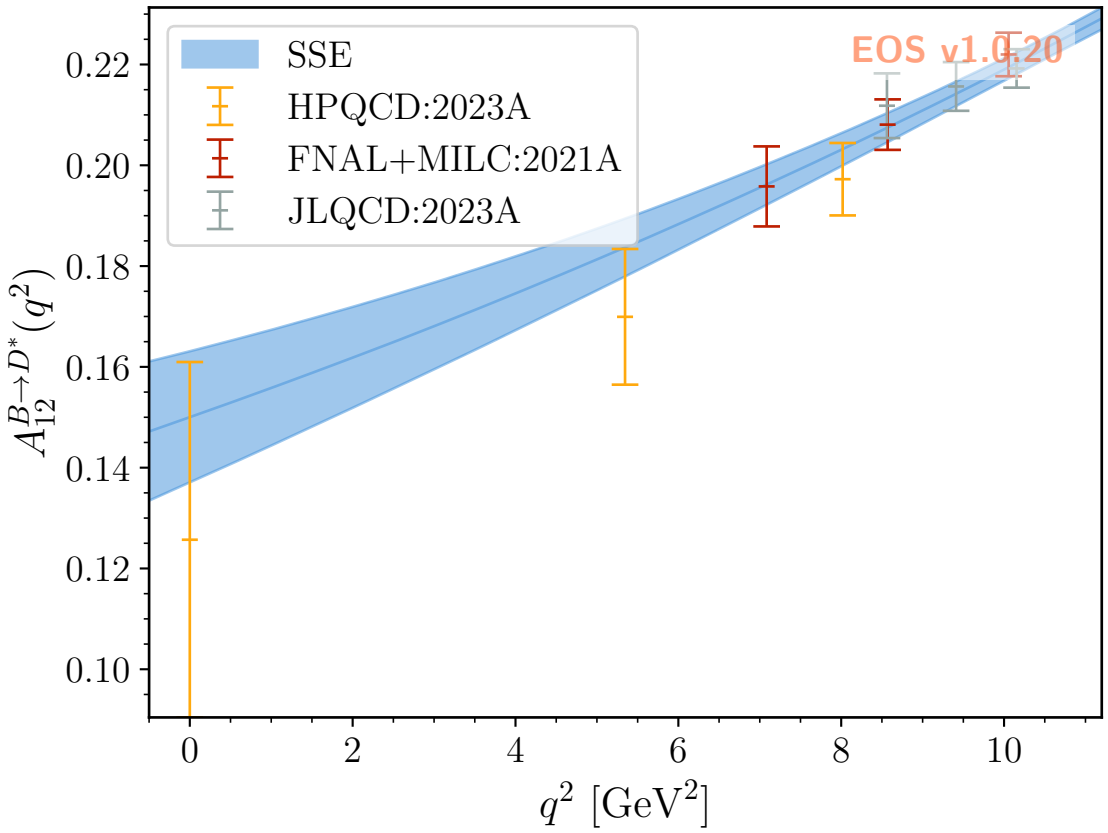

Figure 43: Posterior-predicted form factor for `B-to-Dstar` (`B-to-Dstar-A12`).

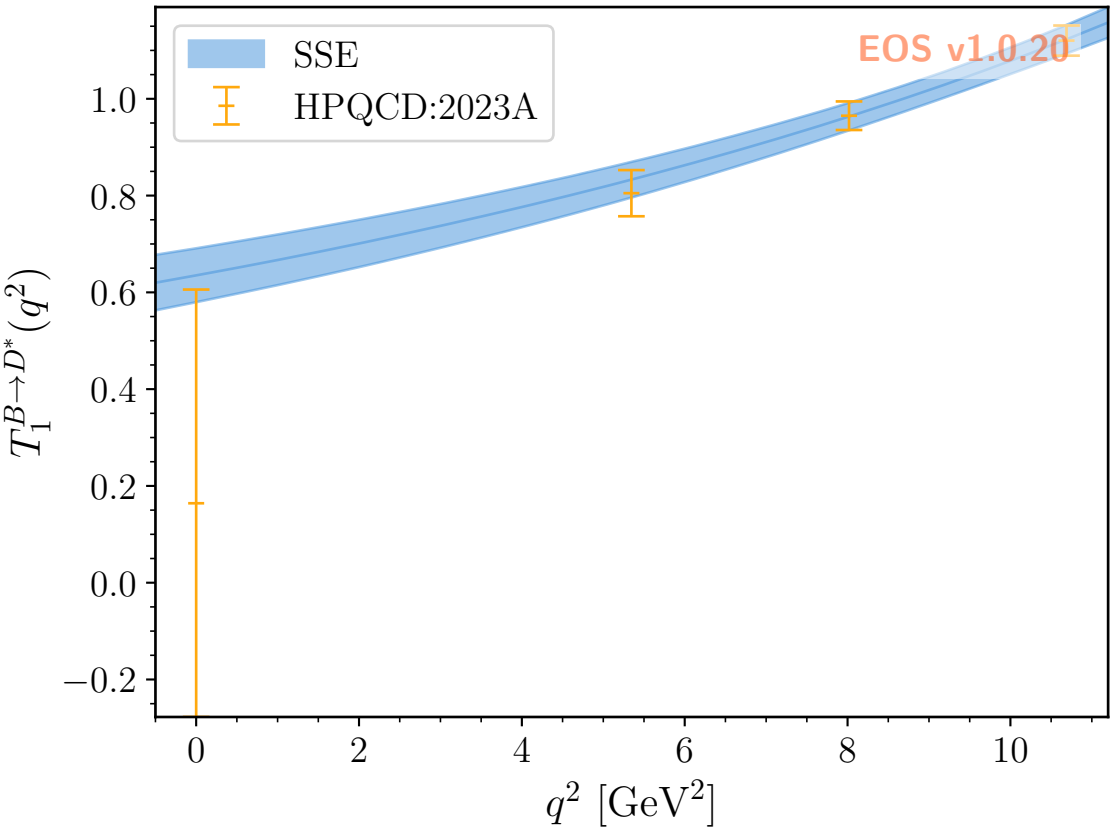

Figure 44: Posterior-predicted form factor for `B-to-Dstar` (`B-to-Dstar-T1`).

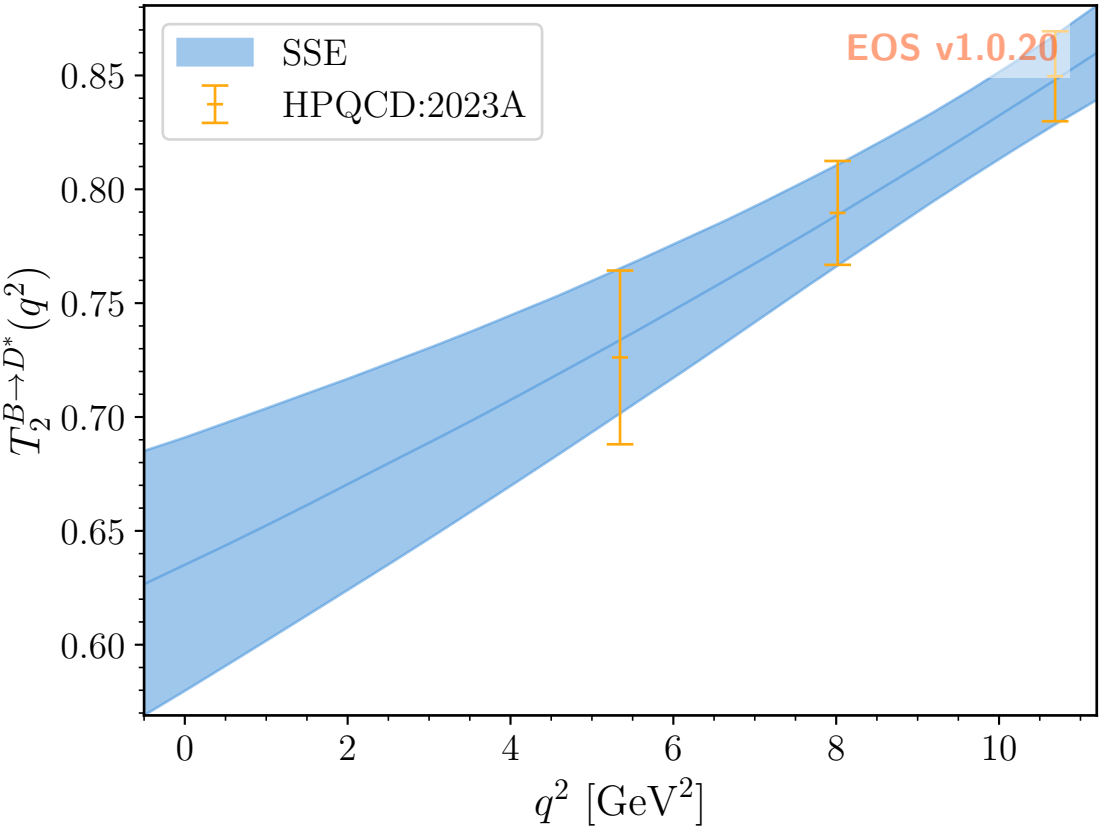

Figure 45: Posterior-predicted form factor for `B-to-Dstar` (`B-to-Dstar-T2`).

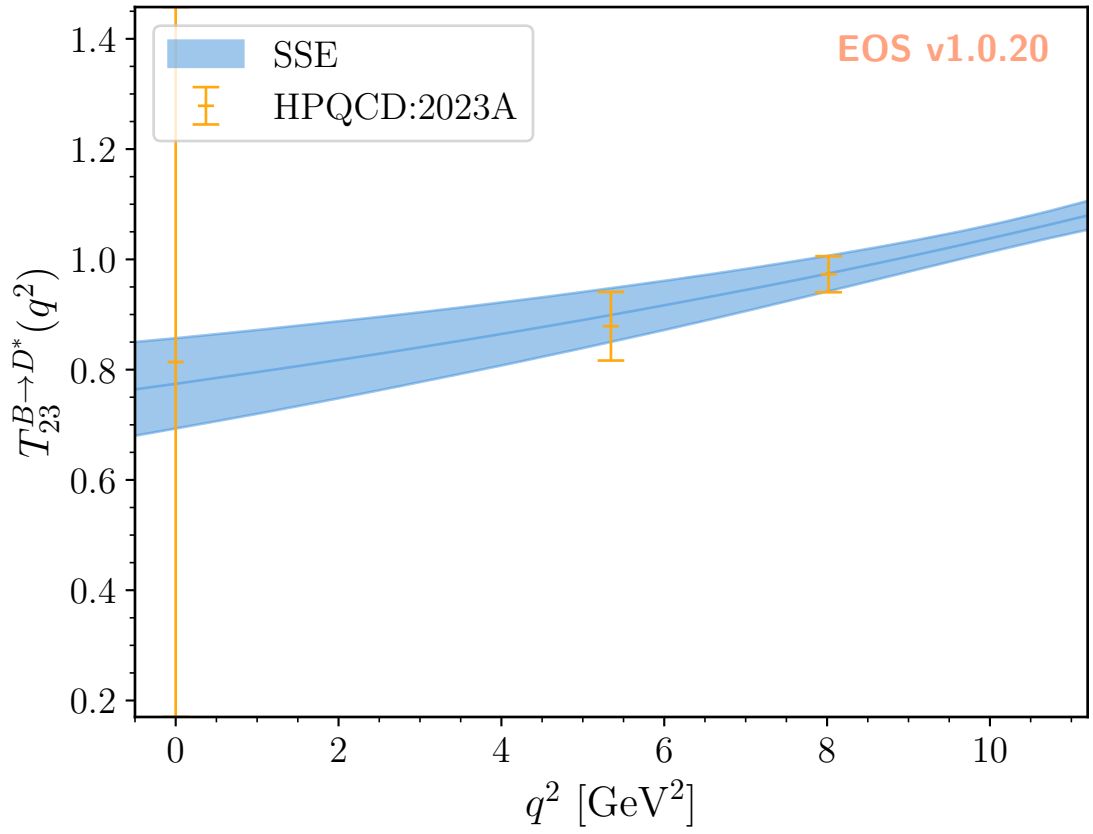

Figure 46: Posterior-predicted form factor for `B-to-Dstar` (`B-to-Dstar-T23`).

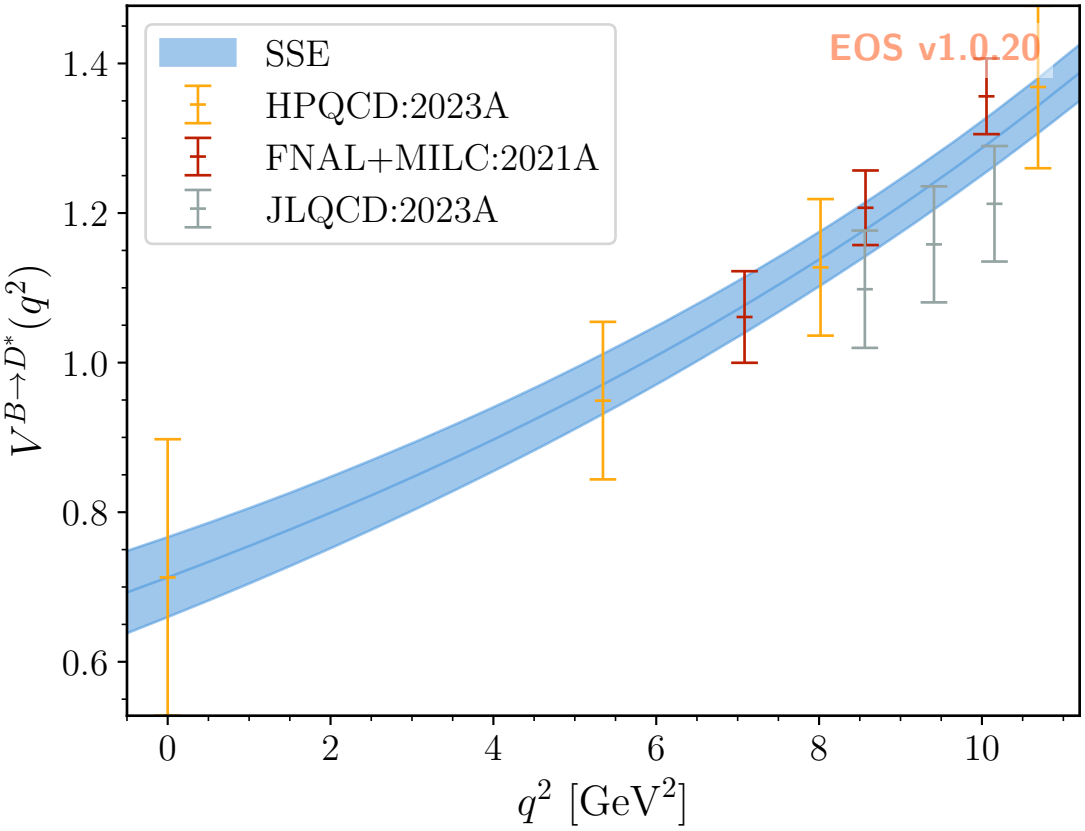

Figure 47: Posterior-predicted form factor for `B-to-Dstar` (`B-to-Dstar-V`).

Posterior B-to-Kstar

This posterior varies 17 SSE coefficients. It is constrained by:

- B->K^*::A_1@HPQCD:2013B
- B->K^*::A_12@HPQCD:2013B
- B->K^*::V@HPQCD:2013B
- B->K^*::FormFactors[A_0,A_1,A_2,V,T_1,T_2,T_23]@GKvD:2018A

Goodness of fit

The table lists the local p value of each constraint together with the global p value and total χ^2 of the best-fit point.

Constraint	local p value
B->K^*::A_12@HPQCD:2013B	83.7%
B->K^*::A_1@HPQCD:2013B	85.8%
B->K^*::FormFactors[A_0,A_1,A_2,V,T_1,T_2,T_23]@GKvD:2018A	100.0%
B->K^*::V@HPQCD:2013B	99.5%
total (χ^2 / d.o.f. = 1.69 / 22)	**100.0%**

Posterior-predicted form factors

The bands show the posterior-predicted form factors (SSE); the points show the constraints entering the fit.

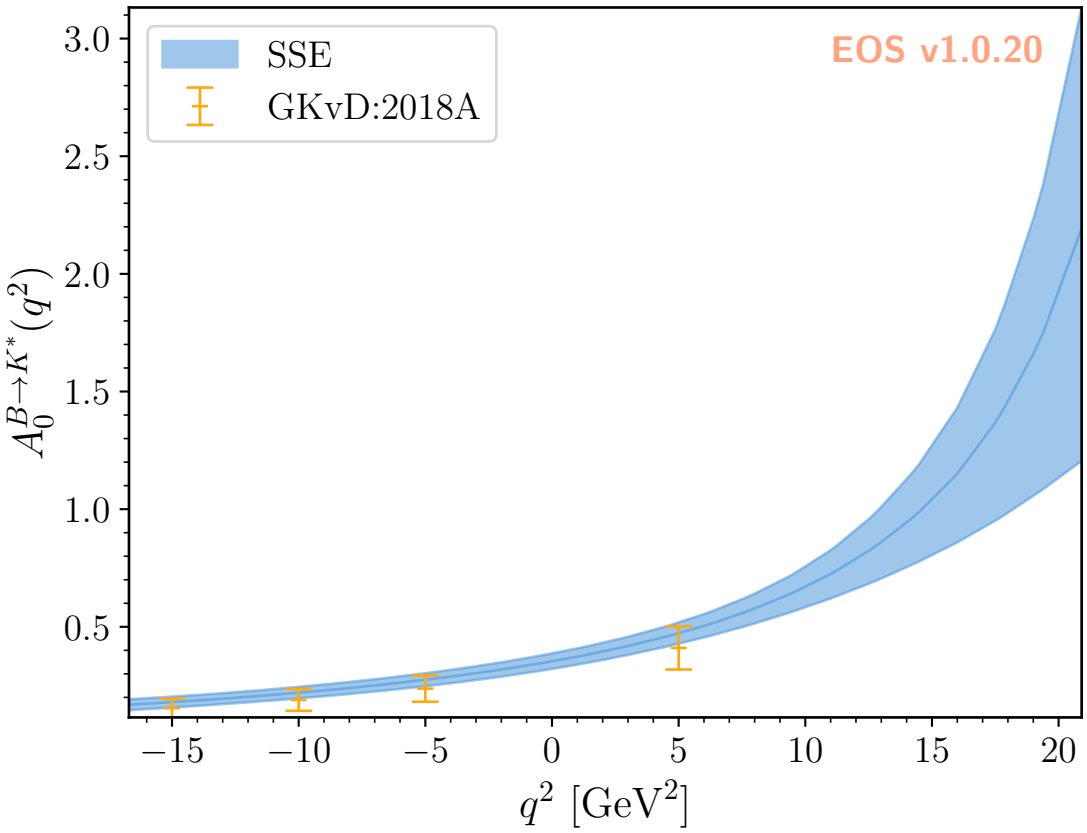

Figure 48: Posterior-predicted form factor for `B-to-Kstar` (`B-to-Kstar-A0`).

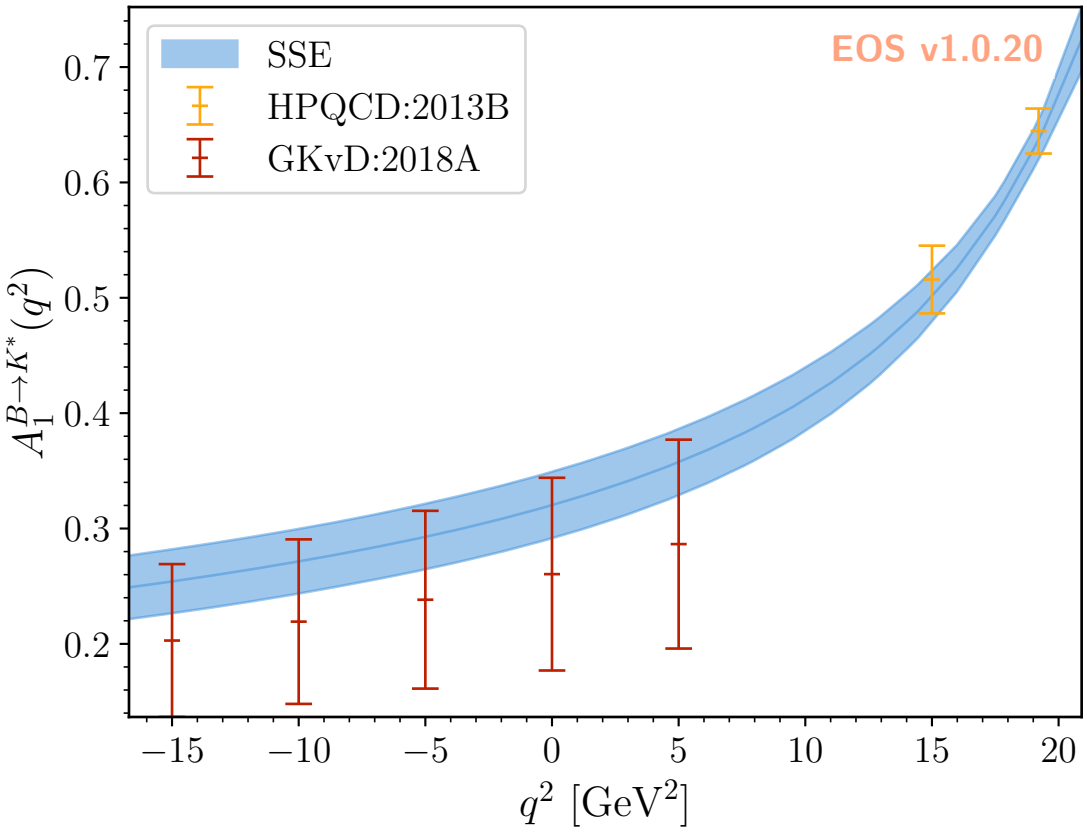

Figure 49: Posterior-predicted form factor for `B-to-Kstar` (`B-to-Kstar-A1`).

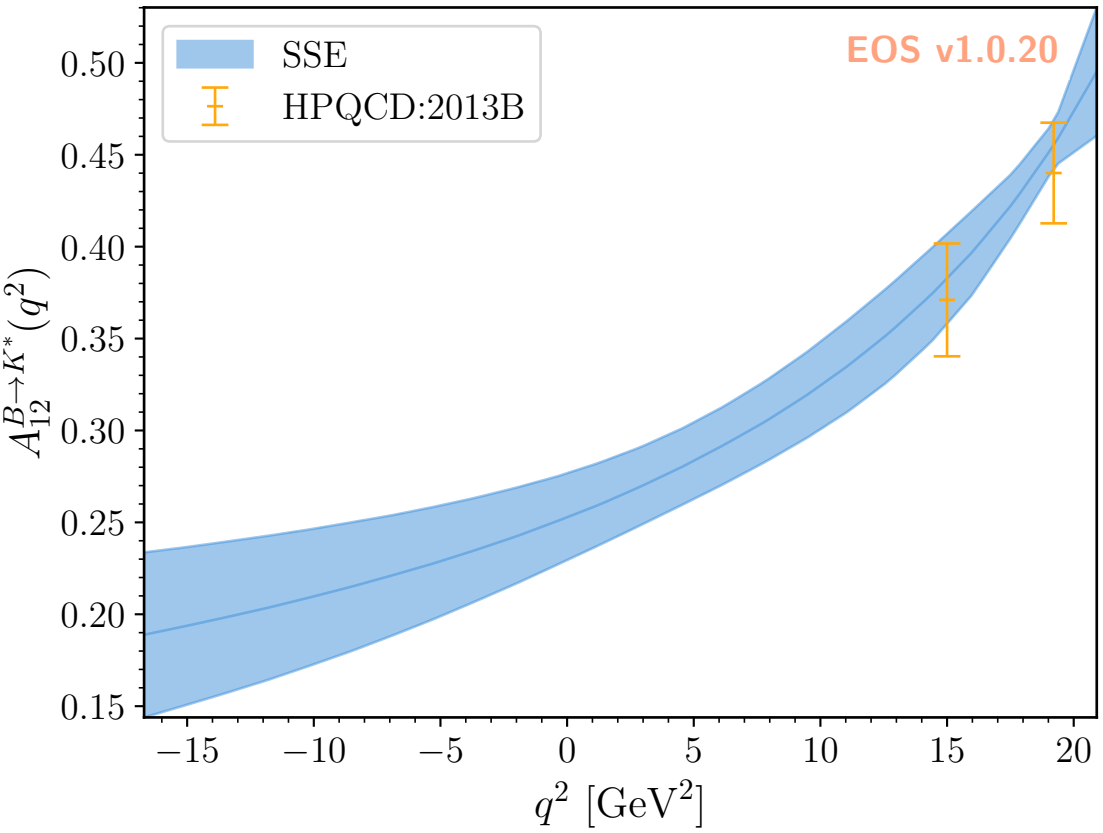

Figure 50: Posterior-predicted form factor for `B-to-Kstar` (`B-to-Kstar-A12`).

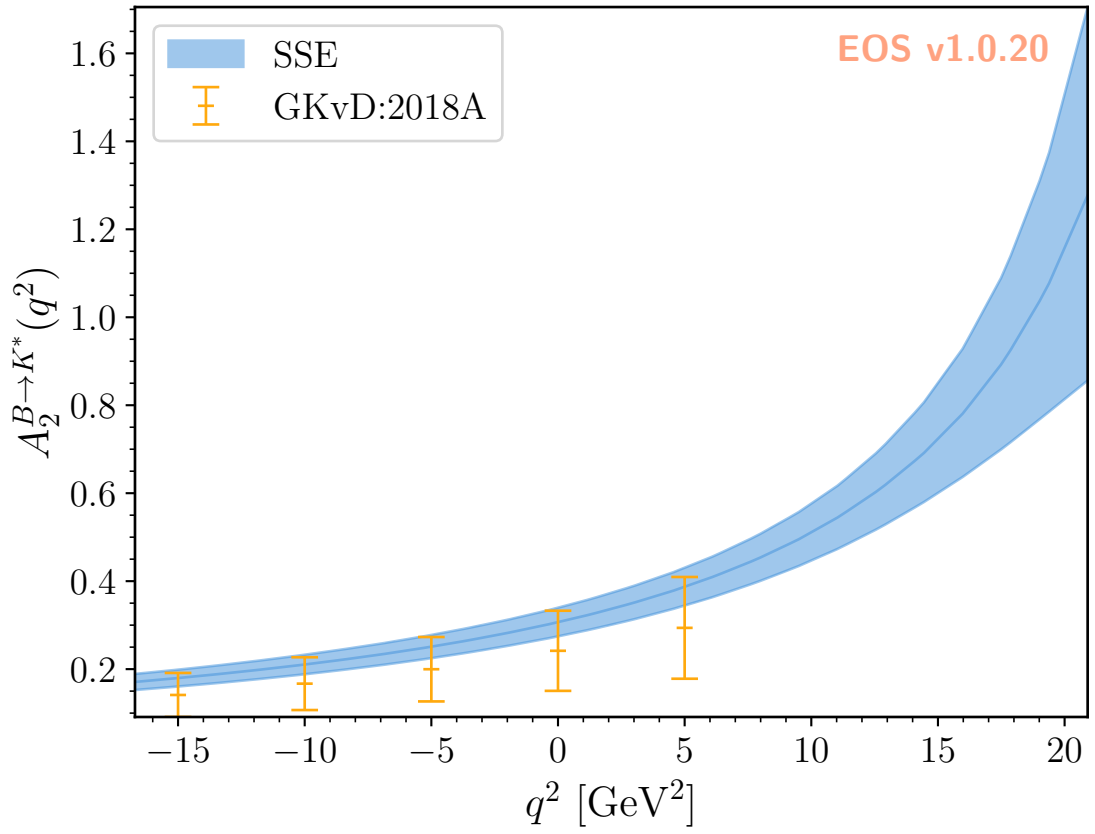

Figure 51: Posterior-predicted form factor for `B-to-Kstar` (`B-to-Kstar-A2`).

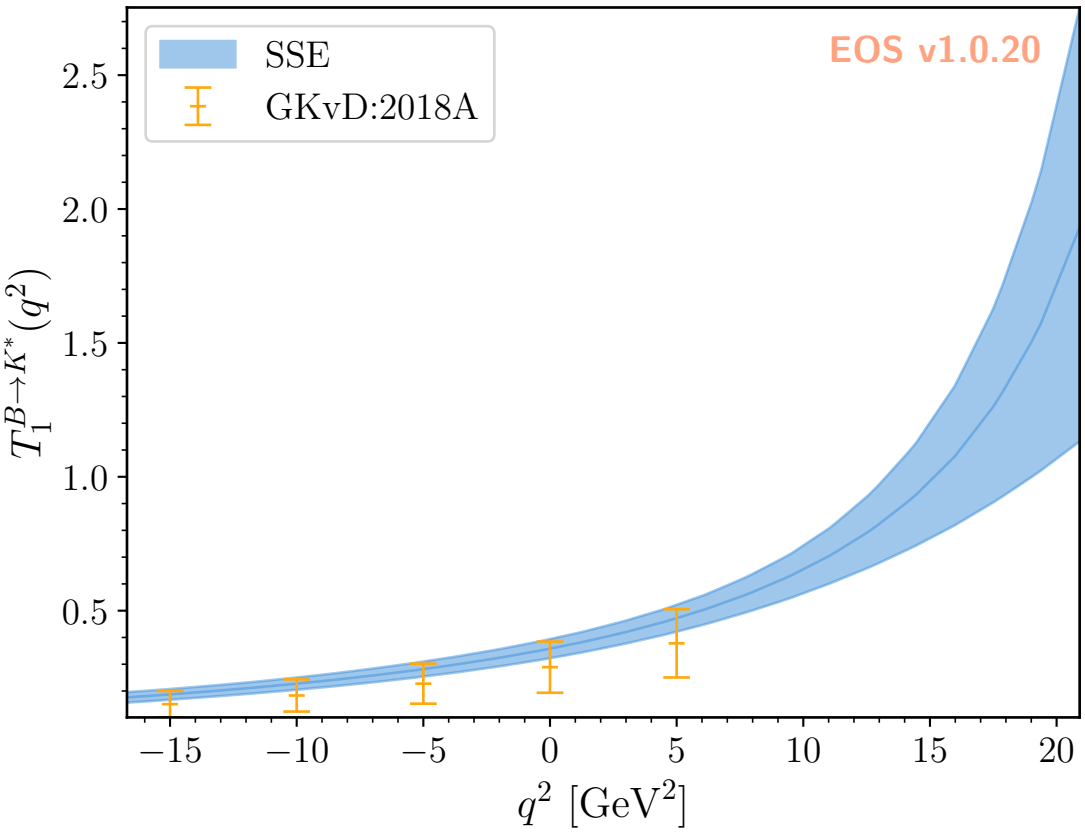

Figure 52: Posterior-predicted form factor for `B-to-Kstar` (`B-to-Kstar-T1`).

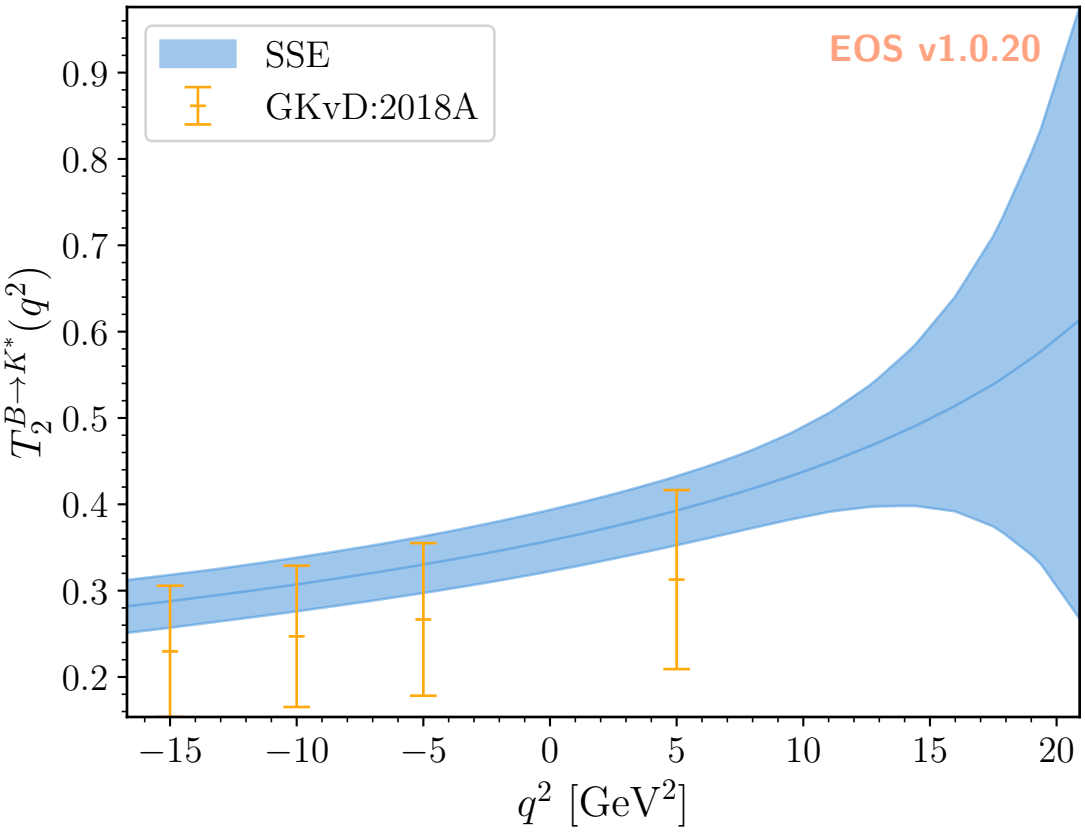

Figure 53: Posterior-predicted form factor for `B-to-Kstar` (`B-to-Kstar-T2`).

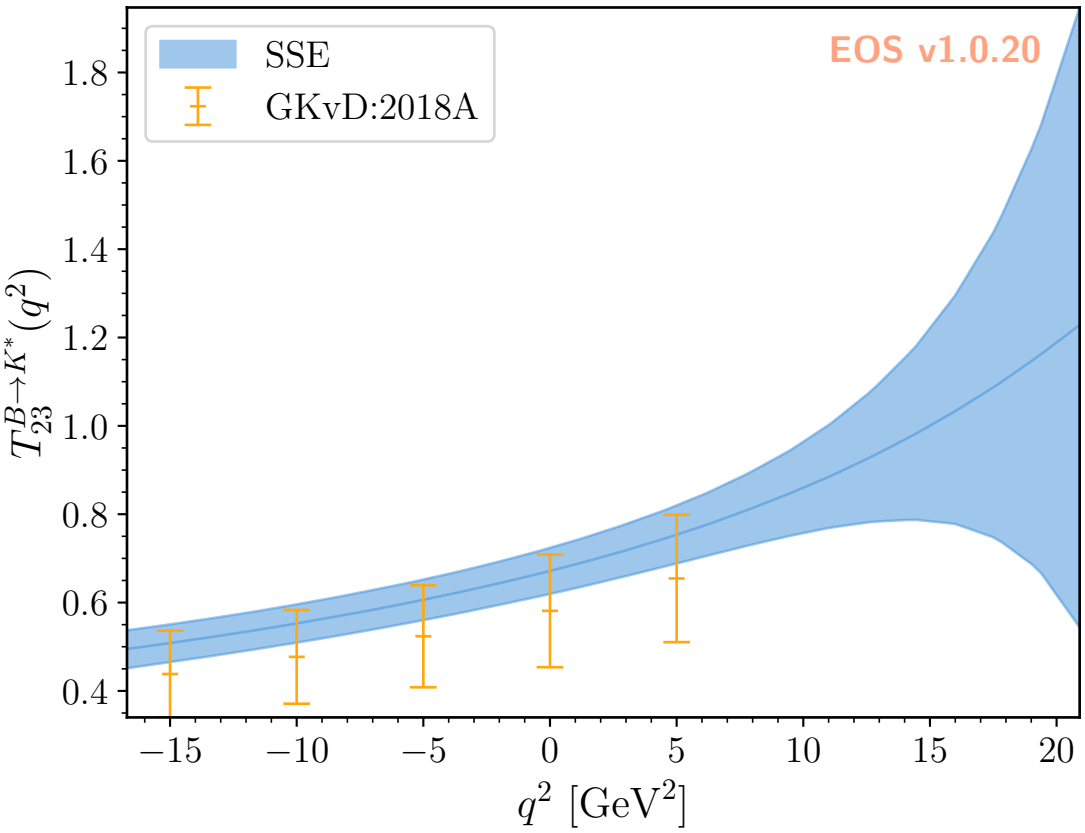

Figure 54: Posterior-predicted form factor for `B-to-Kstar` (`B-to-Kstar-T23`).

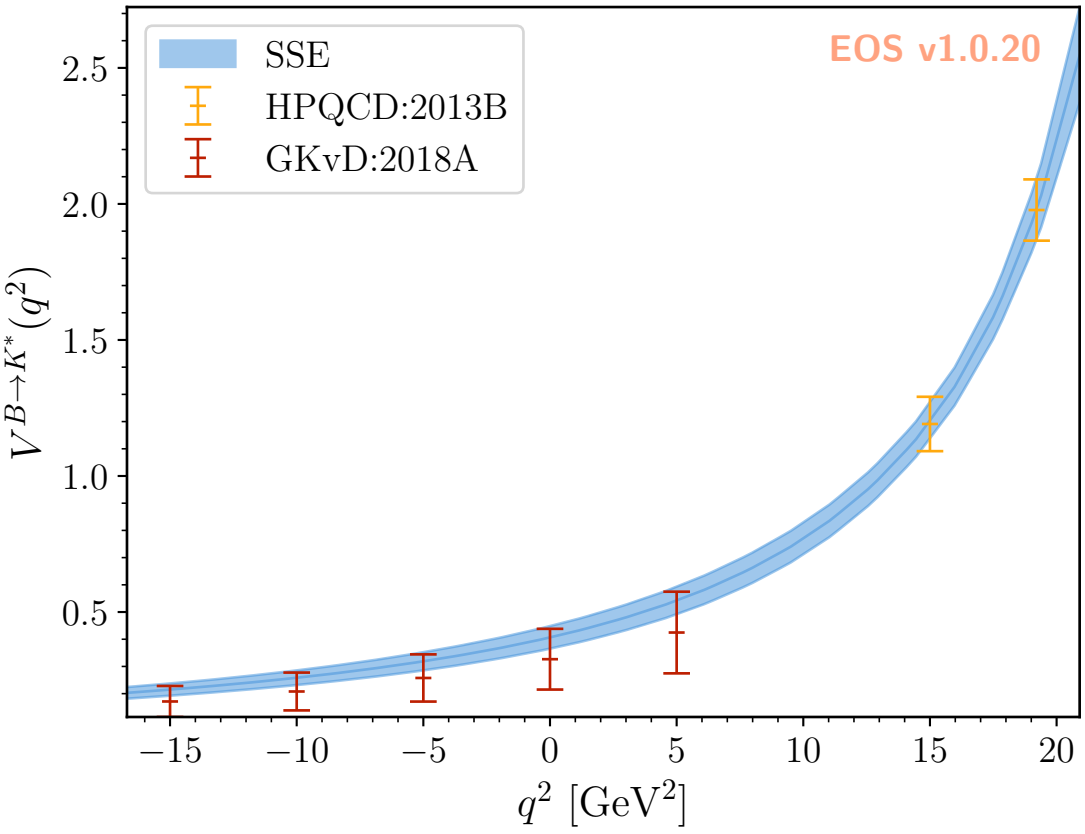

Figure 55: Posterior-predicted form factor for `B-to-Kstar` (`B-to-Kstar-V`).

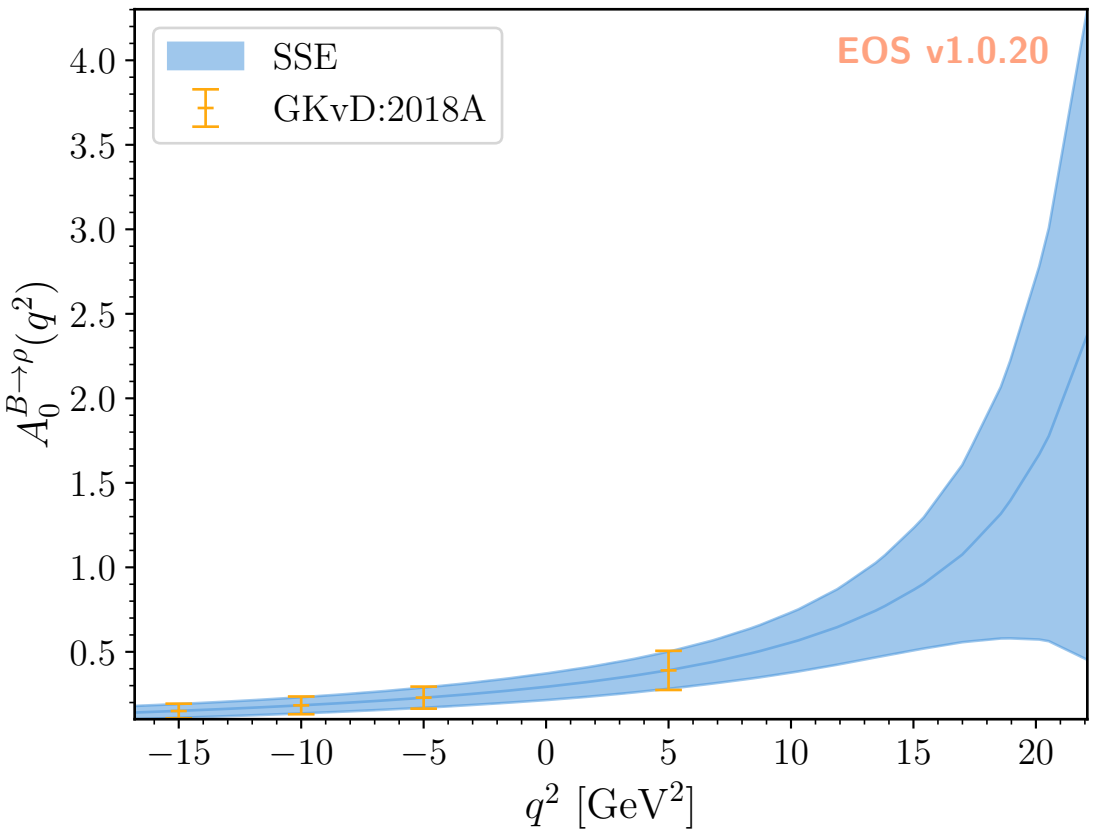

Figure 56: Posterior-predicted form factor for `B-to-rho` (`B-to-rho-A0`).

Posterior `B-to-rho`

This posterior varies 17 `SSE` coefficients. It is constrained by:

- `B->rho::FormFactors[A_0,A_1,A_2,V,T_1,T_2,T_23]@GKvD:2018A`

Goodness of fit

The table lists the local p value of each constraint together with the global p value and total χ^2 of the best-fit point.

Constraint	local p value
`B->rho::FormFactors[A_0,A_1,A_2,V,T_1,T_2,T_23]@GKvD:2018A`	100.0%
total (χ^2 / d.o.f. = 0.01 / 16)	**100.0%**

Posterior-predicted form factors

The bands show the posterior-predicted form factors (`SSE`); the points show the constraints entering the fit.

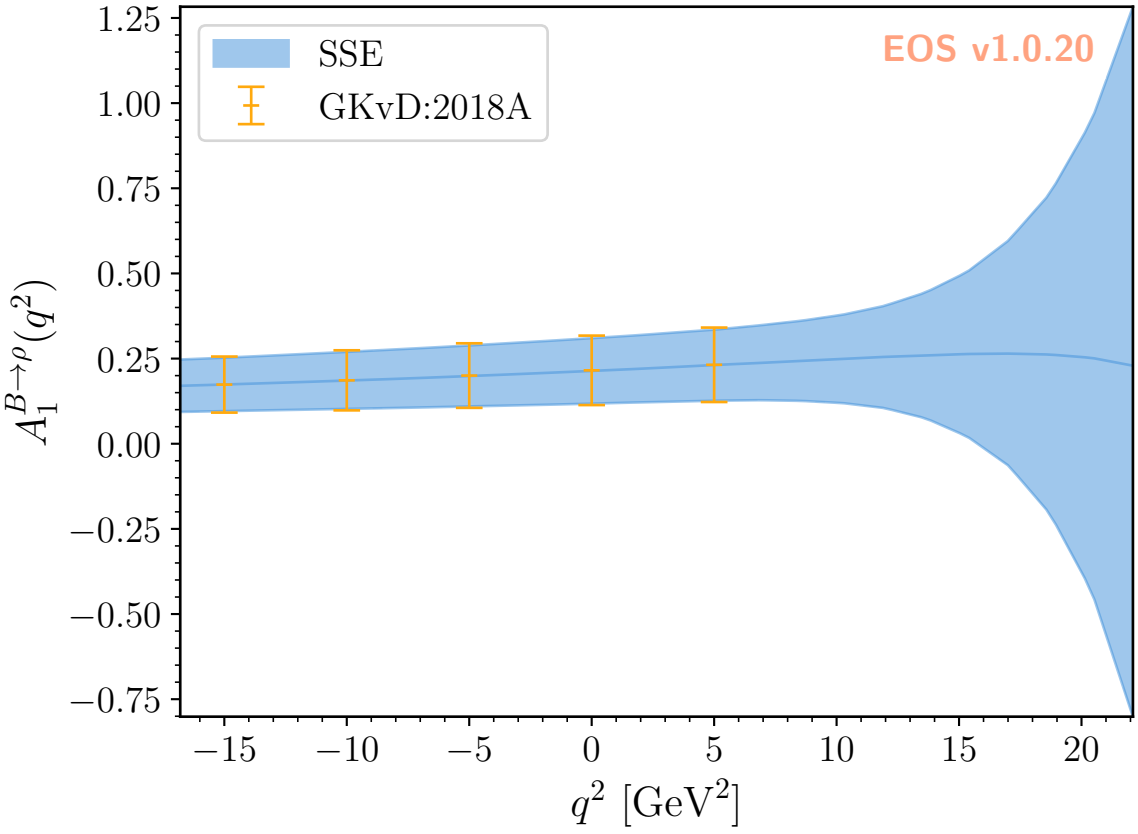

Figure 57: Posterior-predicted form factor for `B-to-rho` (`B-to-rho-A1`).

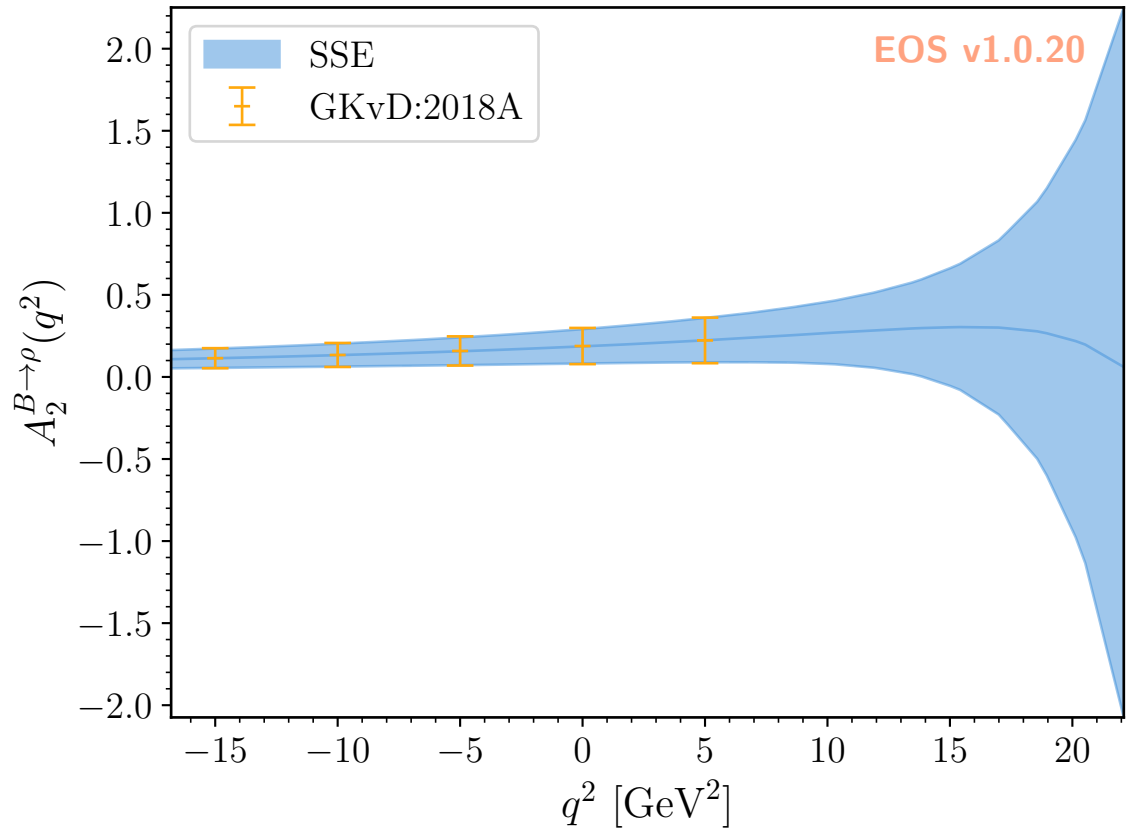

Figure 58: Posterior-predicted form factor for `B-to-rho` (`B-to-rho-A2`).

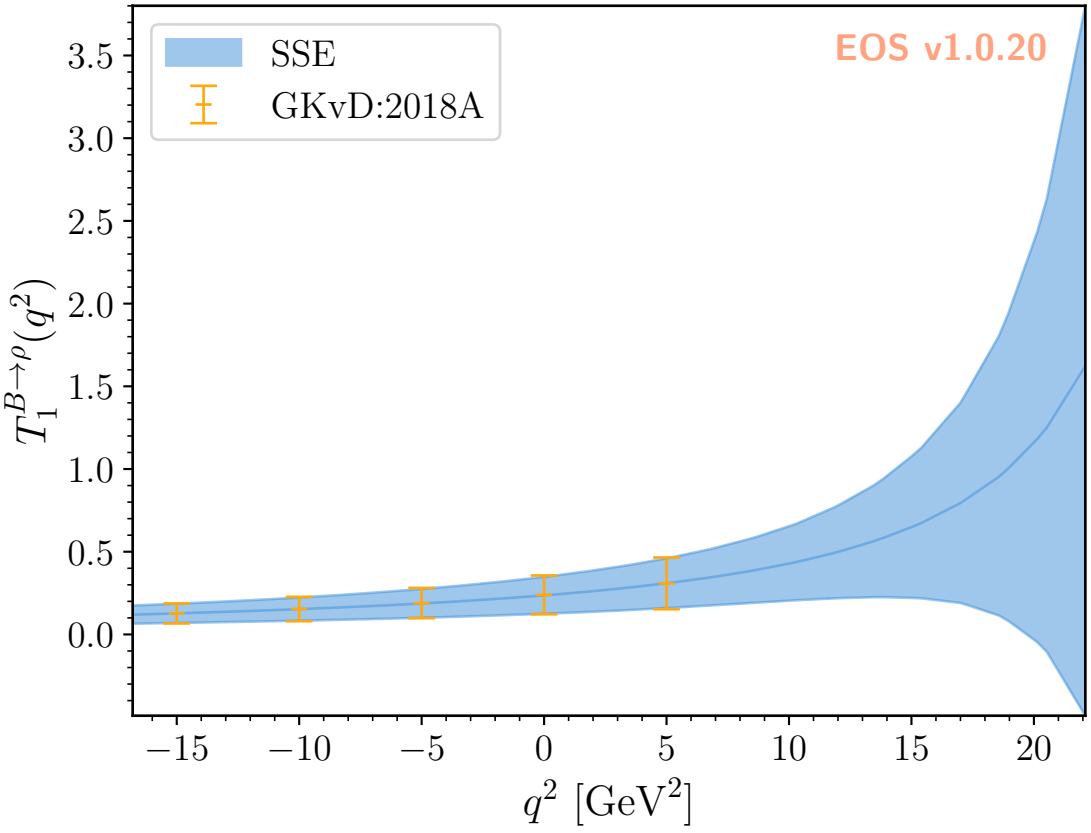

Figure 59: Posterior-predicted form factor for `B-to-rho` (`B-to-rho-T1`).

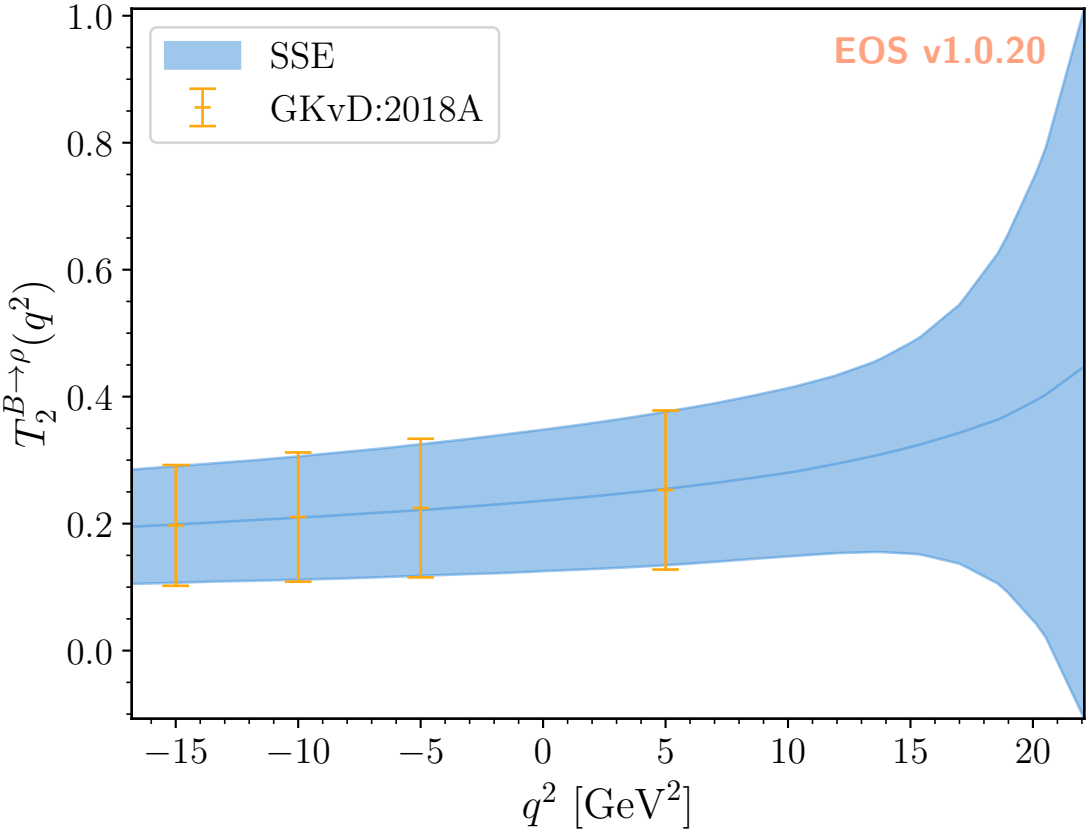

Figure 60: Posterior-predicted form factor for `B-to-rho` (`B-to-rho-T2`).

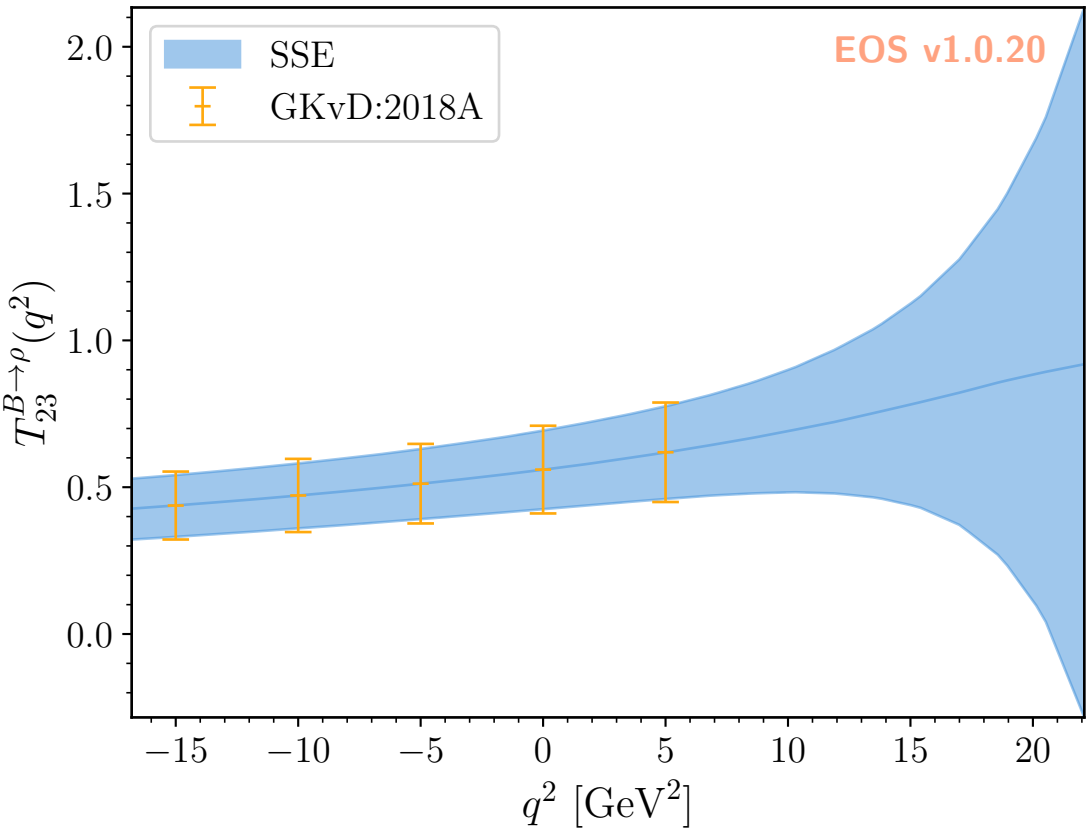

Figure 61: Posterior-predicted form factor for `B-to-rho` (`B-to-rho-T23`).

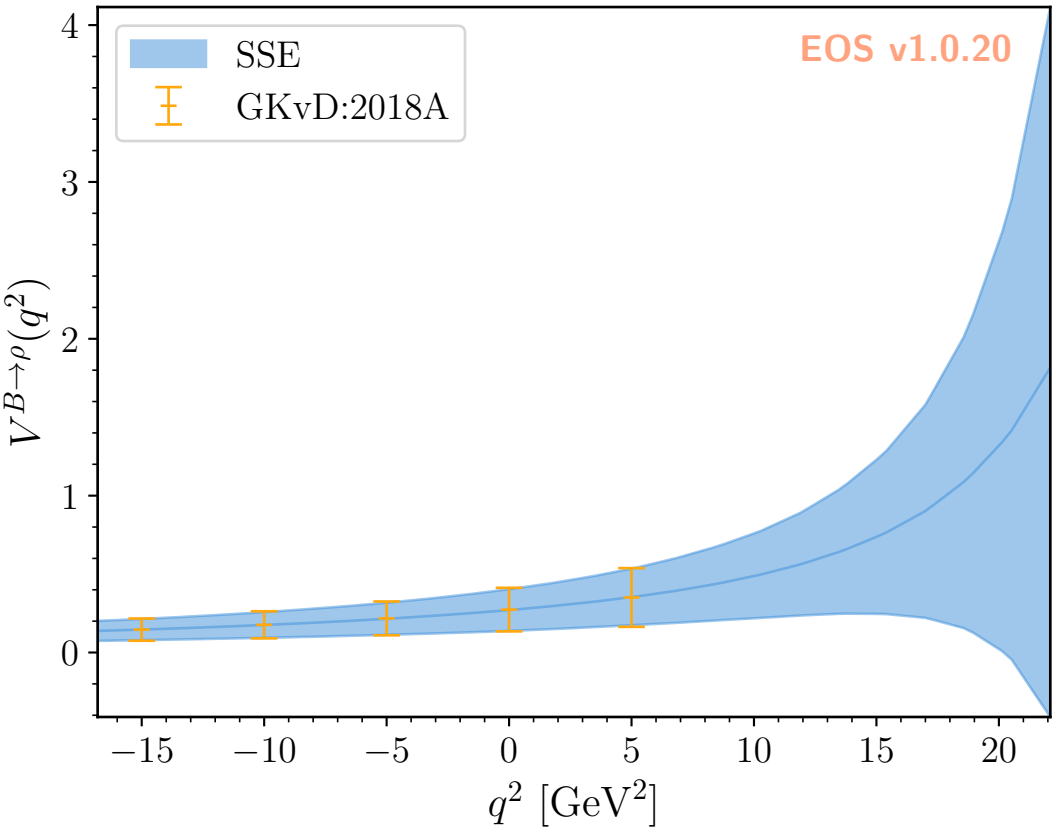

Figure 62: Posterior-predicted form factor for `B-to-rho` (`B-to-rho-V`).

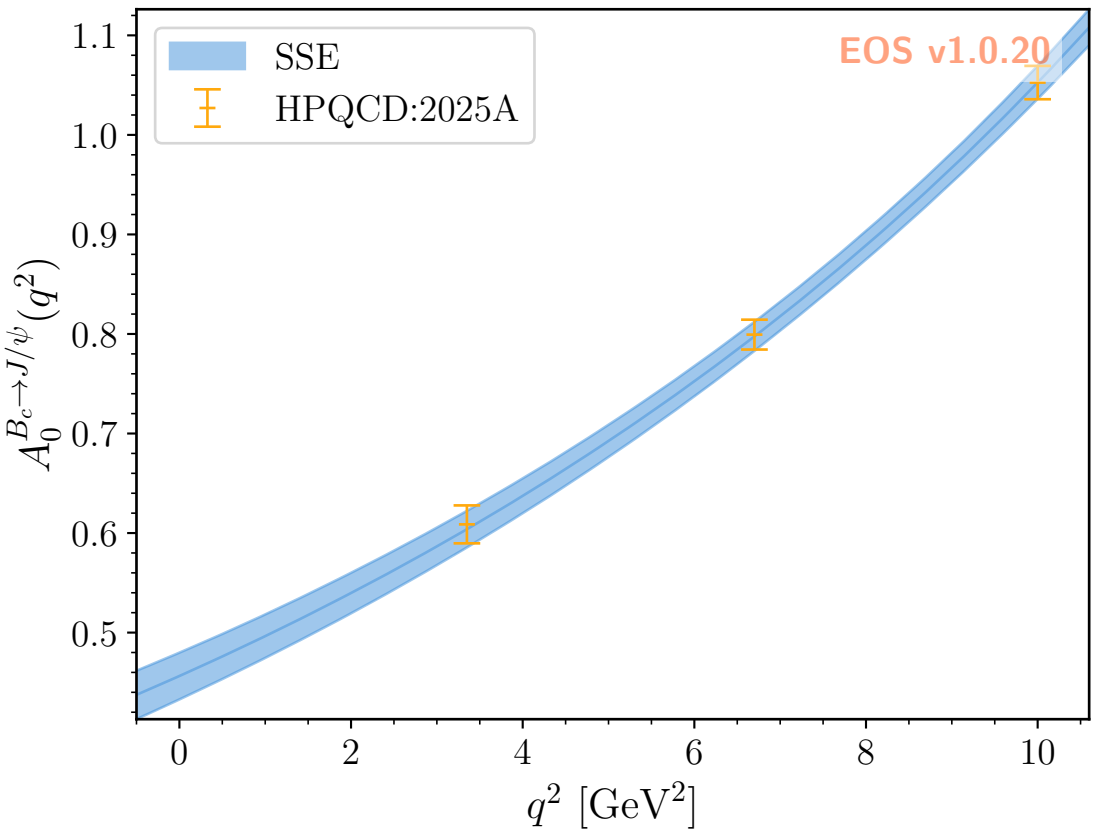

Figure 63: Posterior-predicted form factor for `Bc-to-Jpsi` (`Bc-to-Jpsi-A0`).

Posterior `Bc-to-Jpsi`

This posterior varies 17 `SSE` coefficients. It is constrained by:

- `B_c->J/psi::FormFactors@HPQCD:2025A`

Goodness of fit

The table lists the local p value of each constraint together with the global p value and total χ^2 of the best-fit point.

Constraint	local p value
`B_c->J/psi::FormFactors@HPQCD:2025A`	100.0%
total (χ^2 / d.o.f. = 4.78 / 7)	**68.7%**

Posterior-predicted form factors

The bands show the posterior-predicted form factors (`SSE`); the points show the constraints entering the fit.

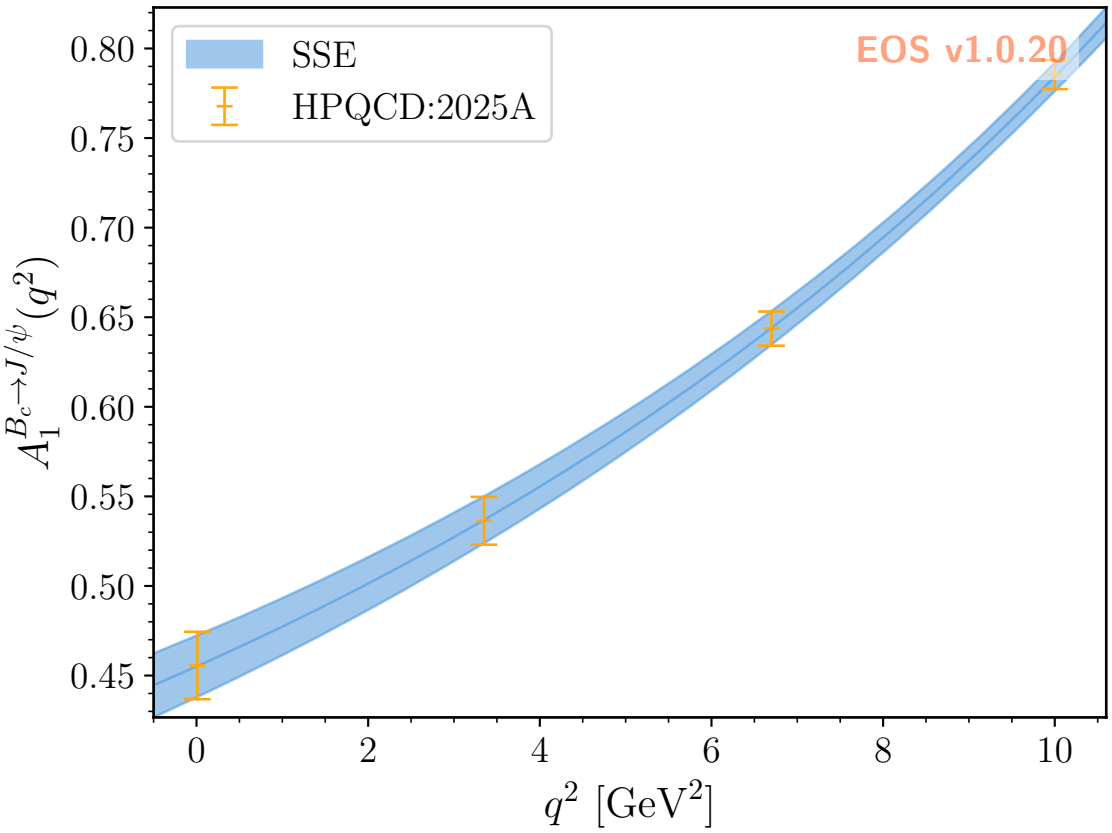

Figure 64: Posterior-predicted form factor for `Bc-to-Jpsi` (`Bc-to-Jpsi-A1`).

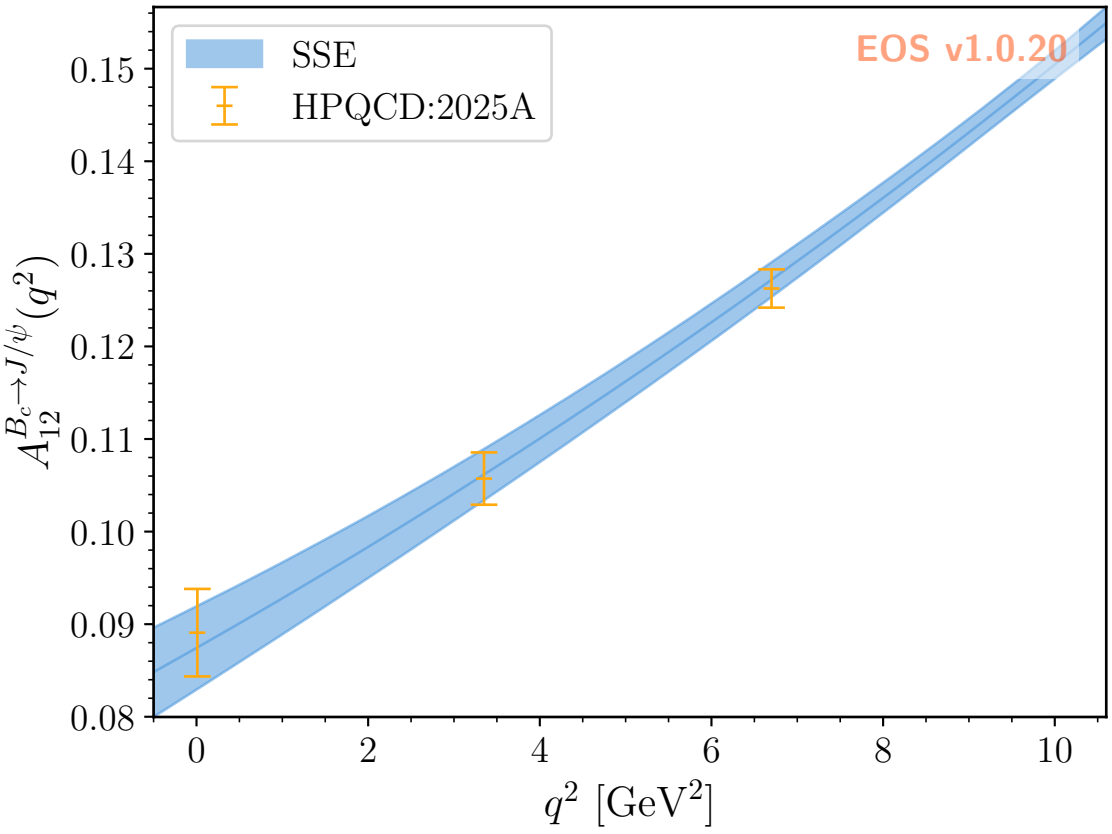

Figure 65: Posterior-predicted form factor for `Bc-to-Jpsi` (`Bc-to-Jpsi-A12`).

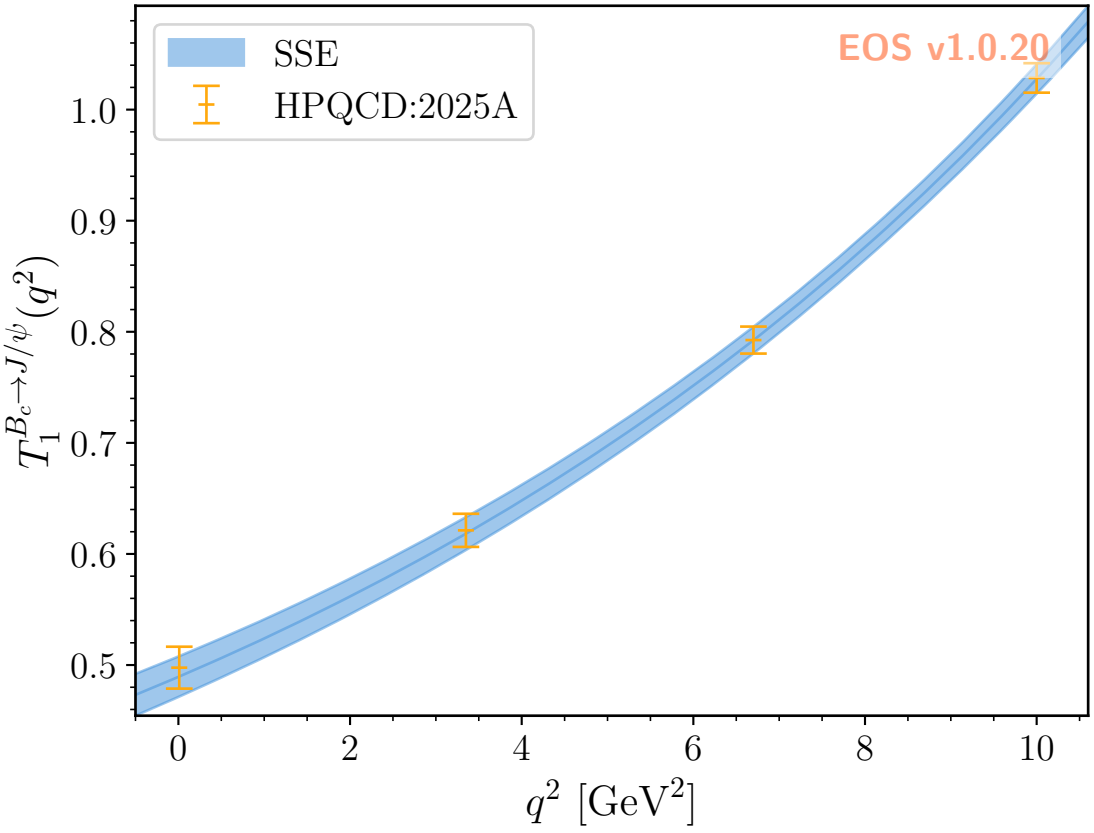

Figure 66: Posterior-predicted form factor for `Bc-to-Jpsi` (`Bc-to-Jpsi-T1`).

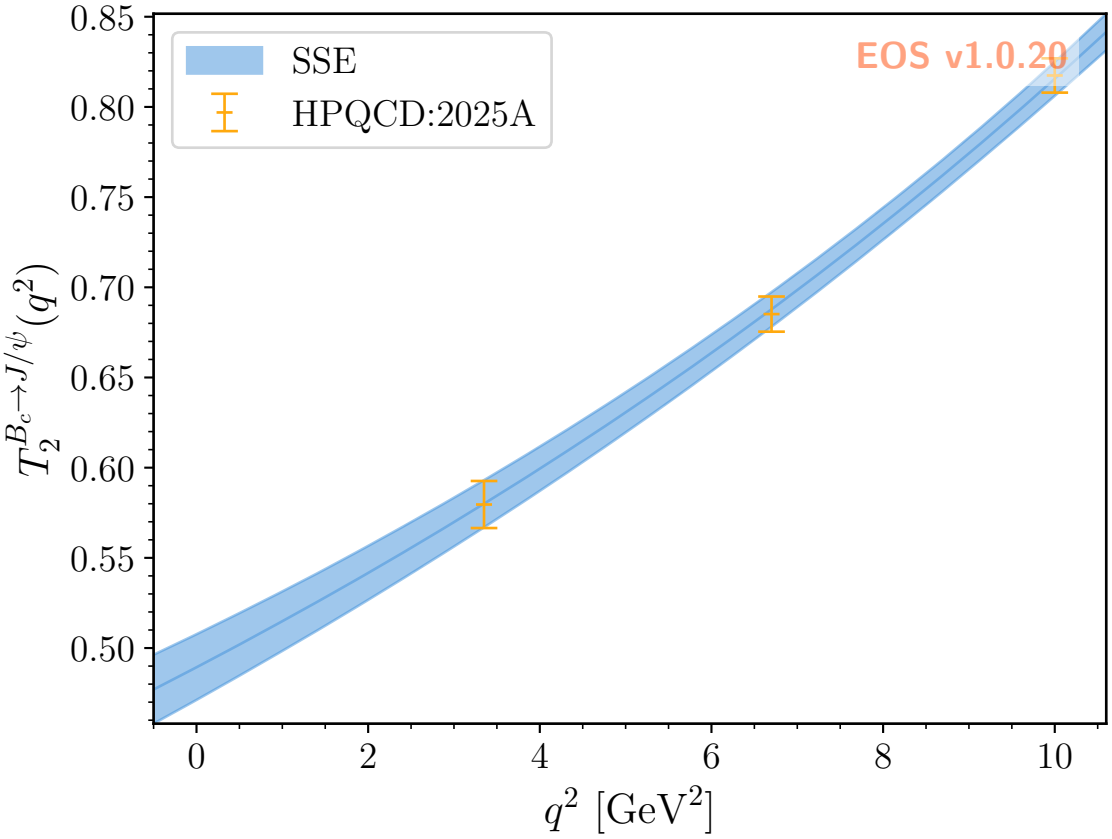

Figure 67: Posterior-predicted form factor for `Bc-to-Jpsi` (`Bc-to-Jpsi-T2`).

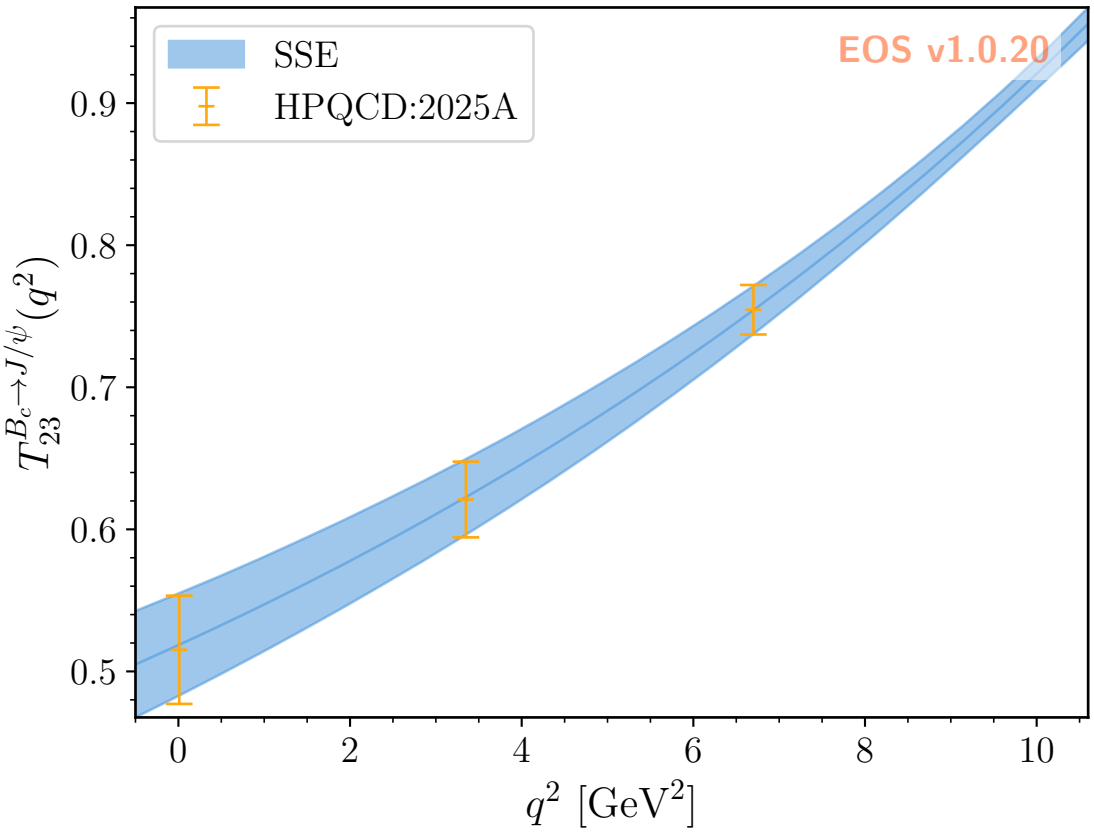

Figure 68: Posterior-predicted form factor for `Bc-to-Jpsi` (`Bc-to-Jpsi-T23`).

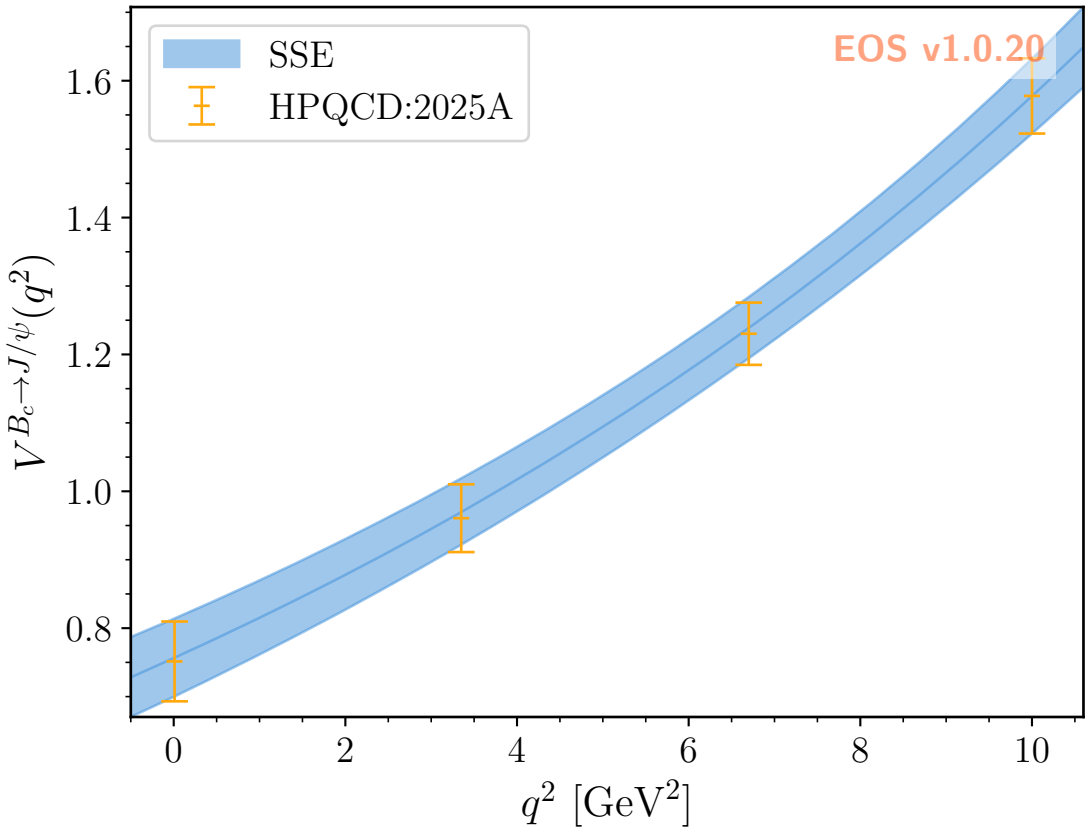

Figure 69: Posterior-predicted form factor for `Bc-to-Jpsi` (`Bc-to-Jpsi-V`).

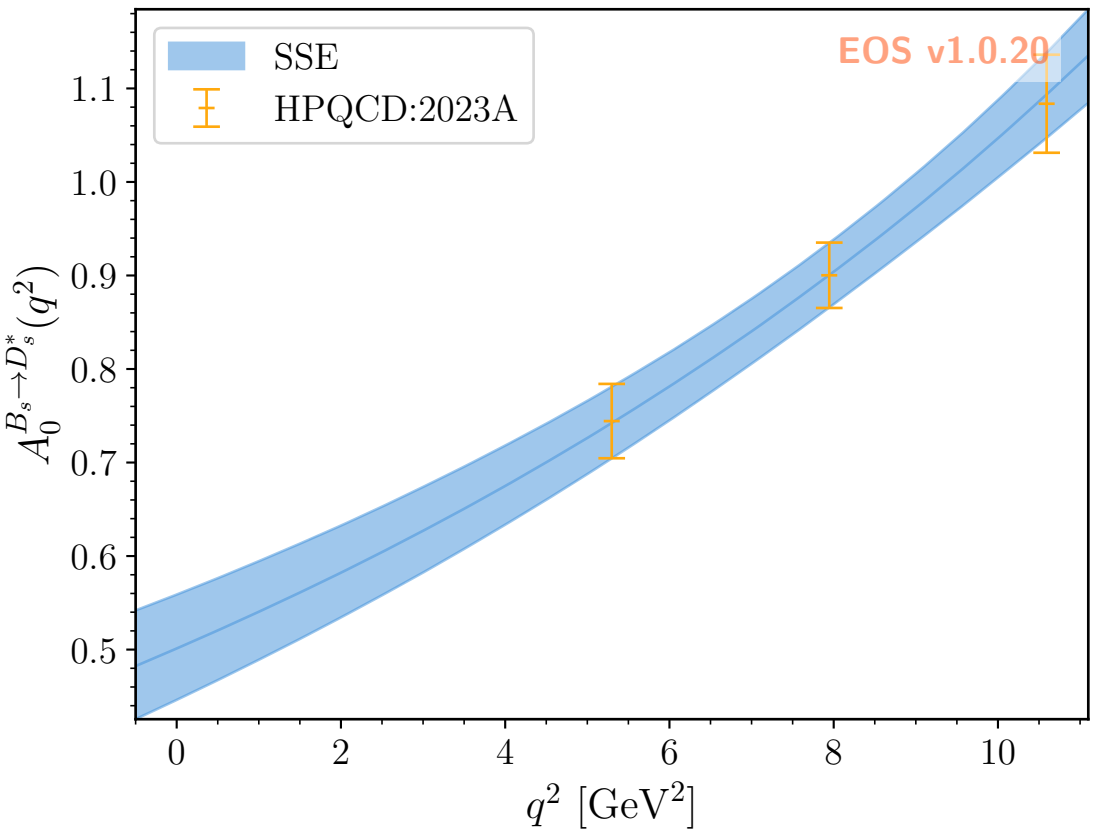

Figure 70: Posterior-predicted form factor for `Bs-to-Dsstar` (`Bs-to-Dsstar-A0`).

Posterior `Bs-to-Dsstar`

This posterior varies 17 `SSE` coefficients. It is constrained by:

- `B_s->D_s^*::FormFactors@HPQCD:2023A`

Goodness of fit

The table lists the local p value of each constraint together with the global p value and total χ^2 of the best-fit point.

Constraint	local p value
`B_s->D_s^*::FormFactors@HPQCD:2023A`	100.0%
total (χ^2 / d.o.f. = 3.19 / 7)	**86.7%**

Posterior-predicted form factors

The bands show the posterior-predicted form factors (`SSE`); the points show the constraints entering the fit.

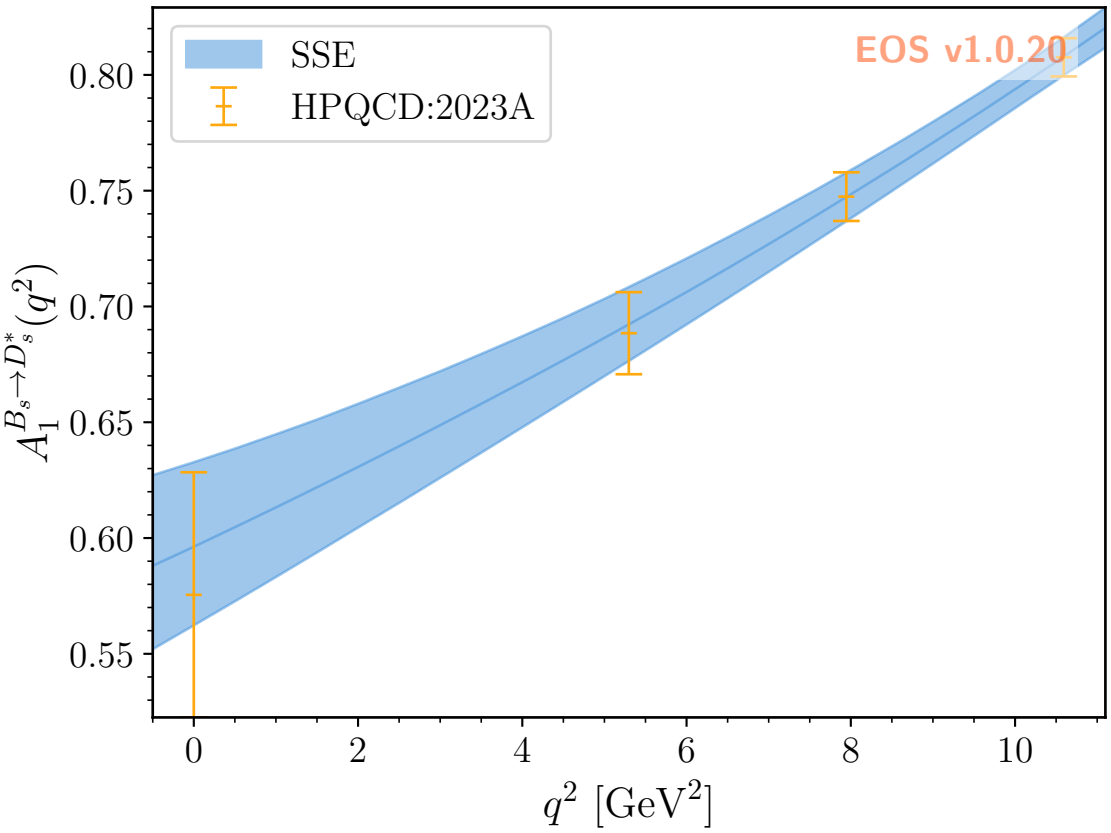

Figure 71: Posterior-predicted form factor for `Bs-to-Dsstar` (`Bs-to-Dsstar-A1`).

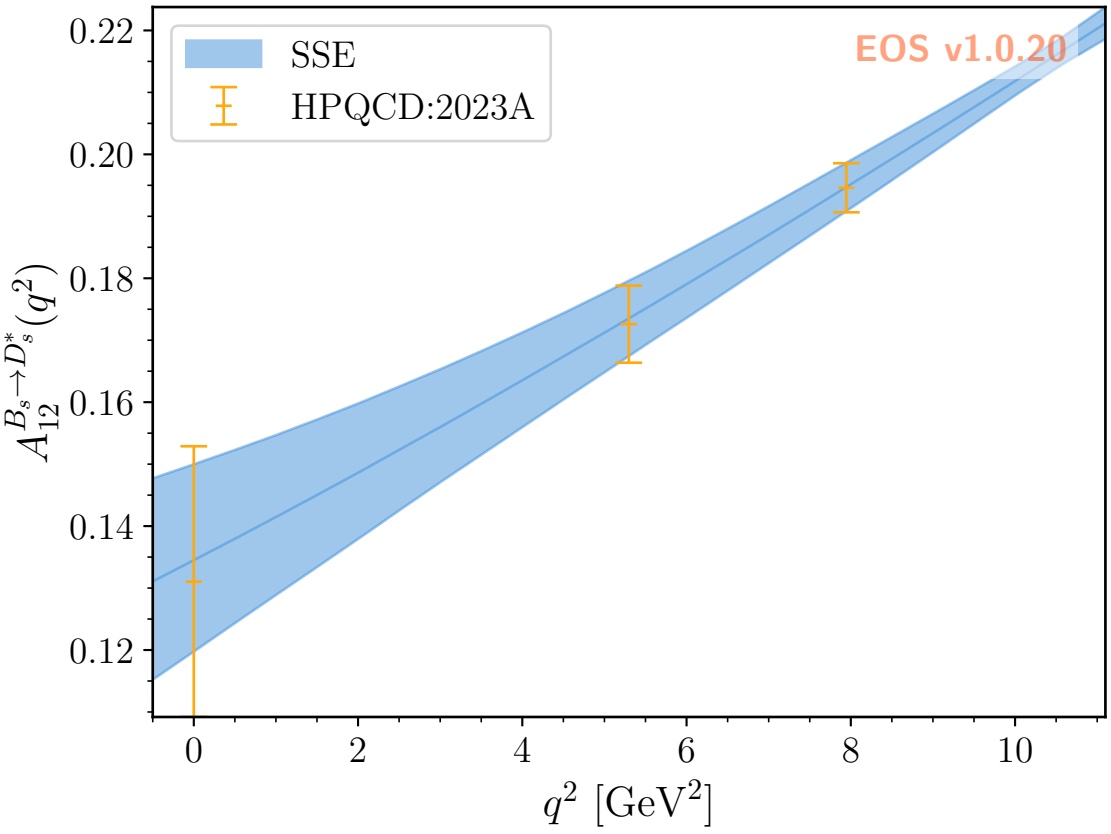

Figure 72: Posterior-predicted form factor for `Bs-to-Dsstar` (`Bs-to-Dsstar-A12`).

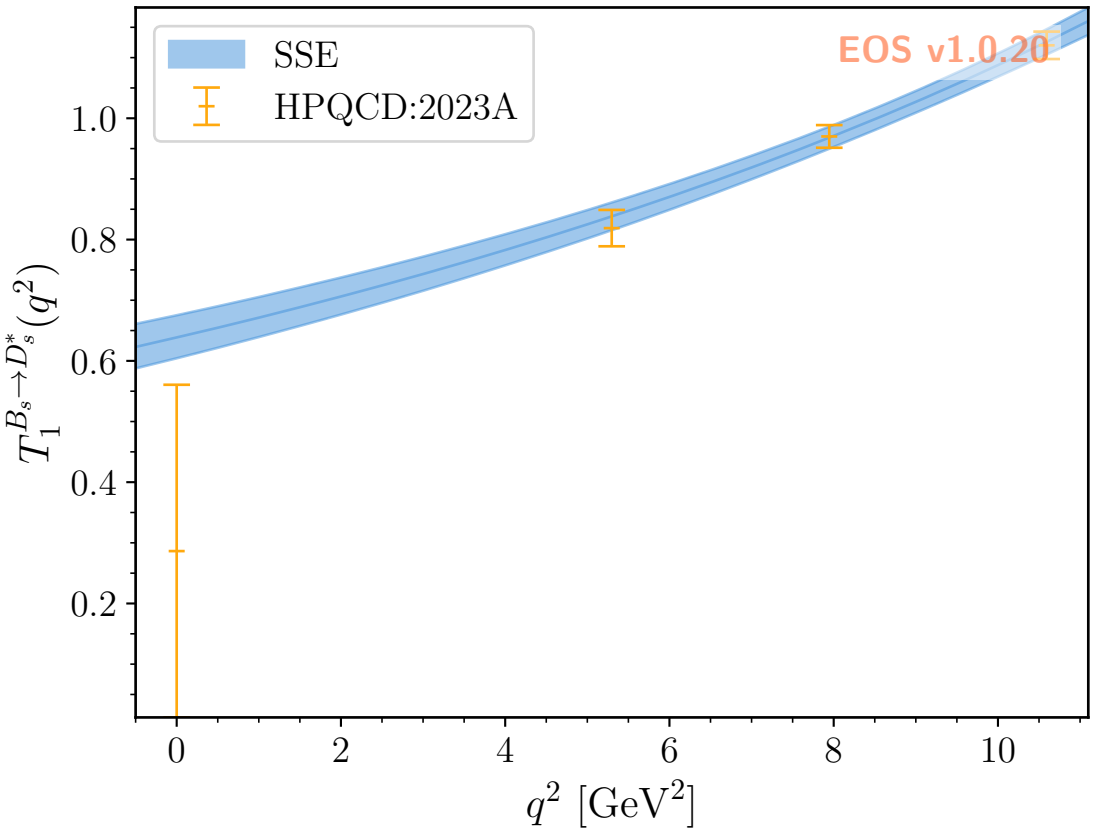

Figure 73: Posterior-predicted form factor for `Bs-to-Dsstar` (`Bs-to-Dsstar-T1`).

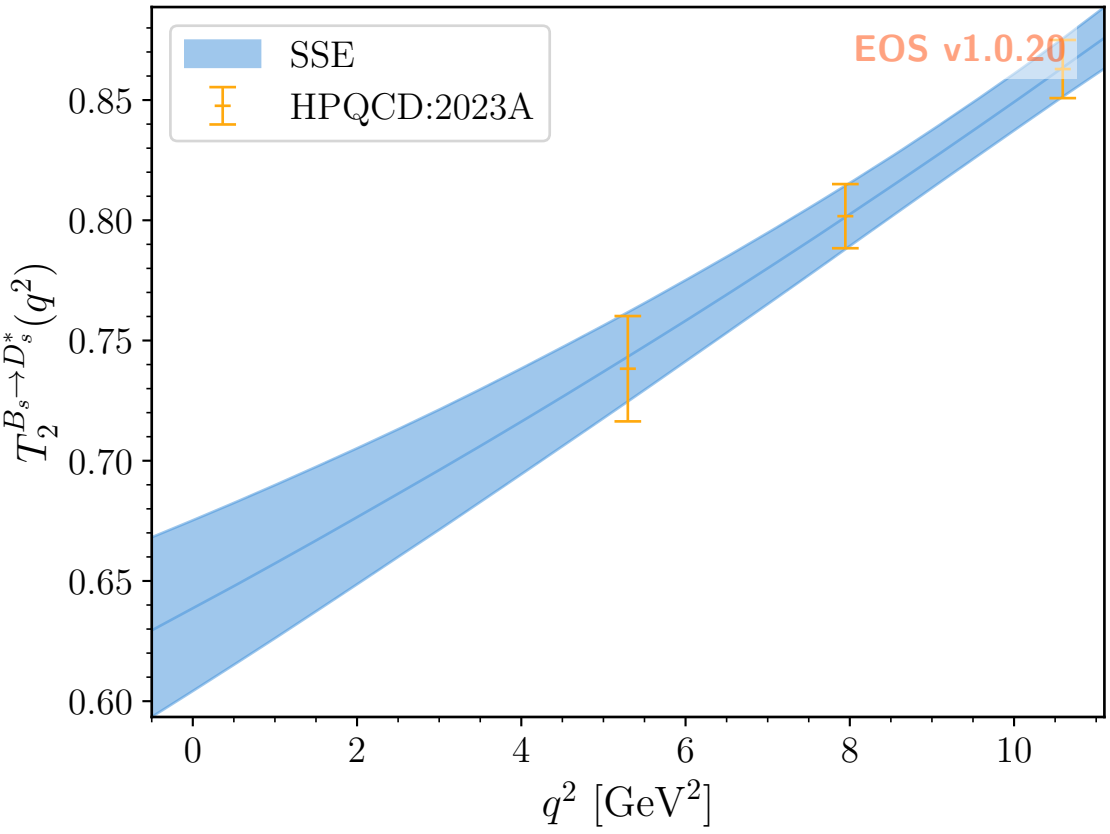

Figure 74: Posterior-predicted form factor for `Bs-to-Dsstar` (`Bs-to-Dsstar-T2`).

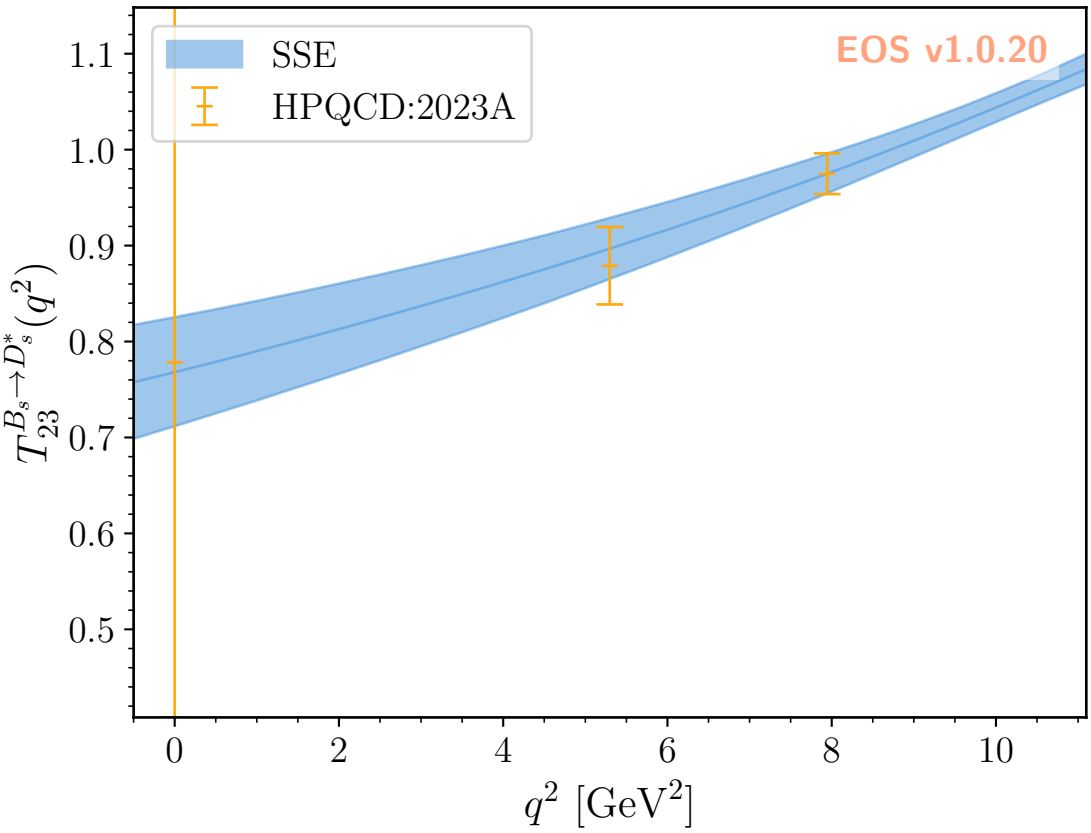

Figure 75: Posterior-predicted form factor for `Bs-to-Dsstar` (`Bs-to-Dsstar-T23`).

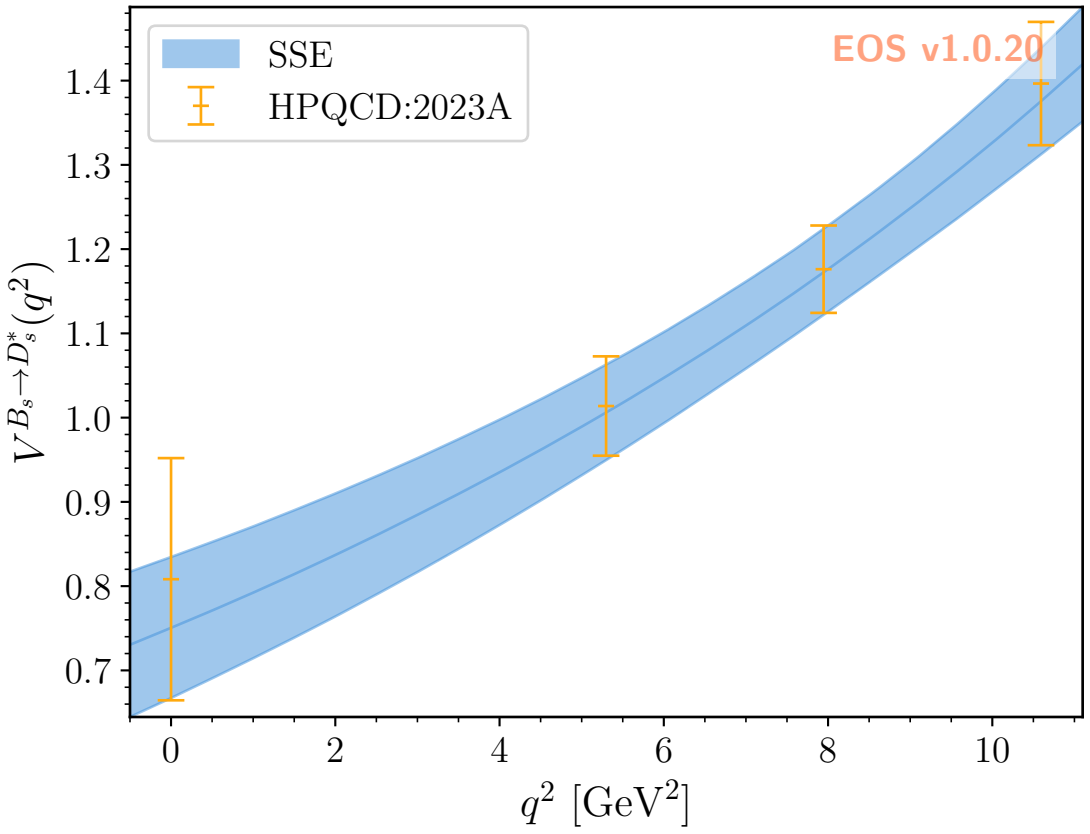

Figure 76: Posterior-predicted form factor for `Bs-to-Dsstar` (`Bs-to-Dsstar-V`).

Posterior Bs-to-Kstar

This posterior varies 17 SSE coefficients. It is constrained by:

- B_s->K^*::A_1@HPQCD:2013B
- B_s->K^*::A_12@HPQCD:2013B
- B_s->K^*::V@HPQCD:2013B
- B_s->K^*::FormFactors[A_0,A_1,A_2,V,T_1,T_2,T_23]@GvDV:2020A

Goodness of fit

The table lists the local p value of each constraint together with the global p value and total χ^2 of the best-fit point.

Constraint	local p value
B_s->K^*::A_12@HPQCD:2013B	96.0%
B_s->K^*::A_1@HPQCD:2013B	63.3%
B_s->K^*::FormFactors[A_0,A_1,A_2,V,T_1,T_2,T_23]@GvDV:2020A	100.0%
B_s->K^*::V@HPQCD:2013B	29.5%
total (χ^2 / d.o.f. = 5.40 / 22)	**100.0%**

Posterior-predicted form factors

The bands show the posterior-predicted form factors (SSE); the points show the constraints entering the fit.

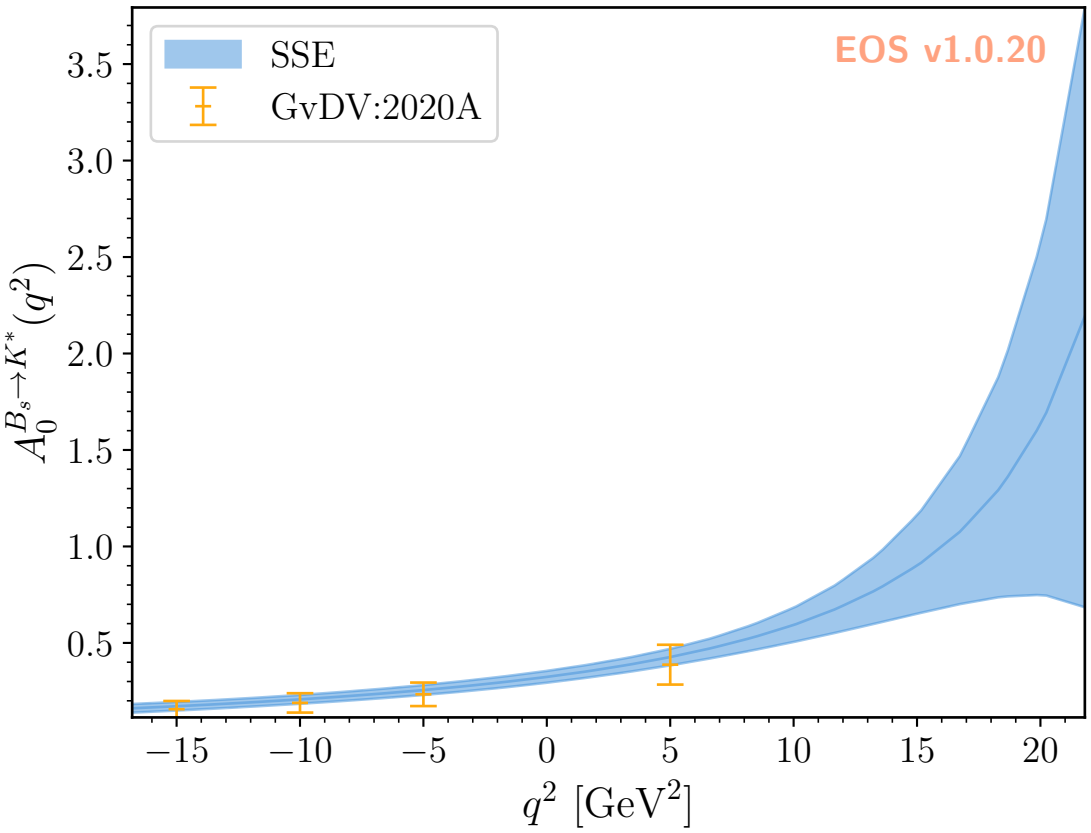

Figure 77: Posterior-predicted form factor for `Bs-to-Kstar` (`Bs-to-Kstar-A0`).

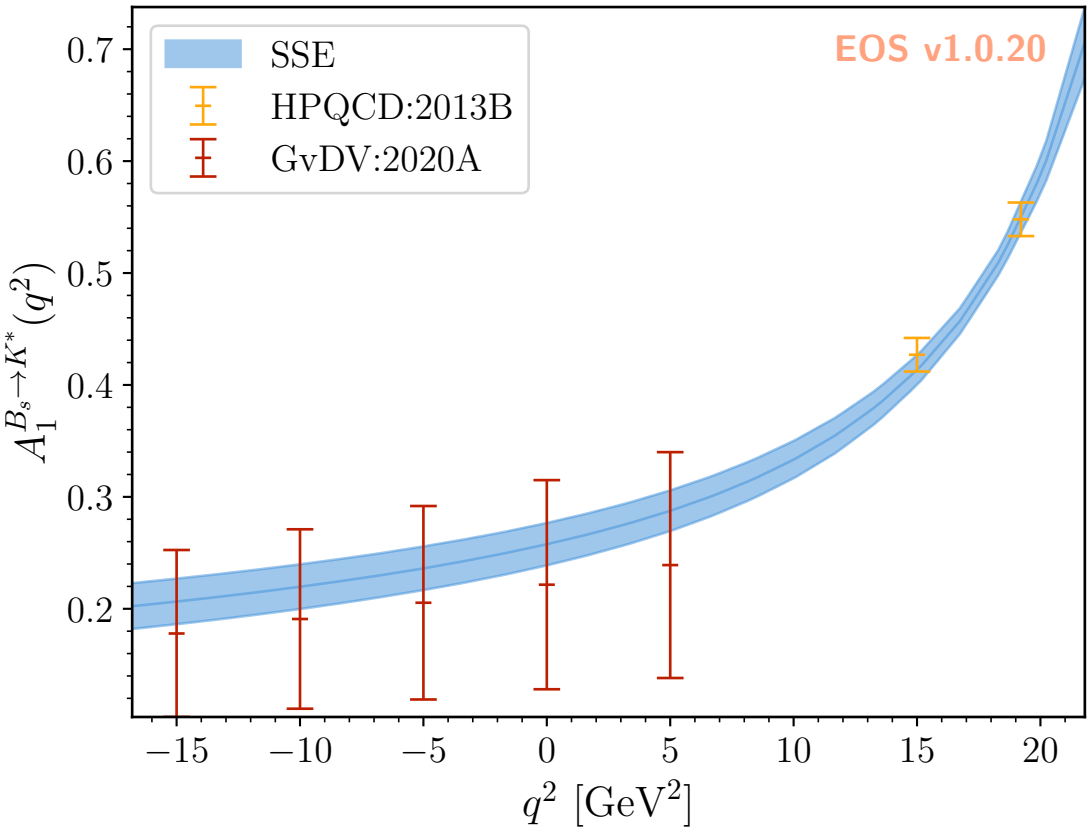

Figure 78: Posterior-predicted form factor for `Bs-to-Kstar` (`Bs-to-Kstar-A1`).

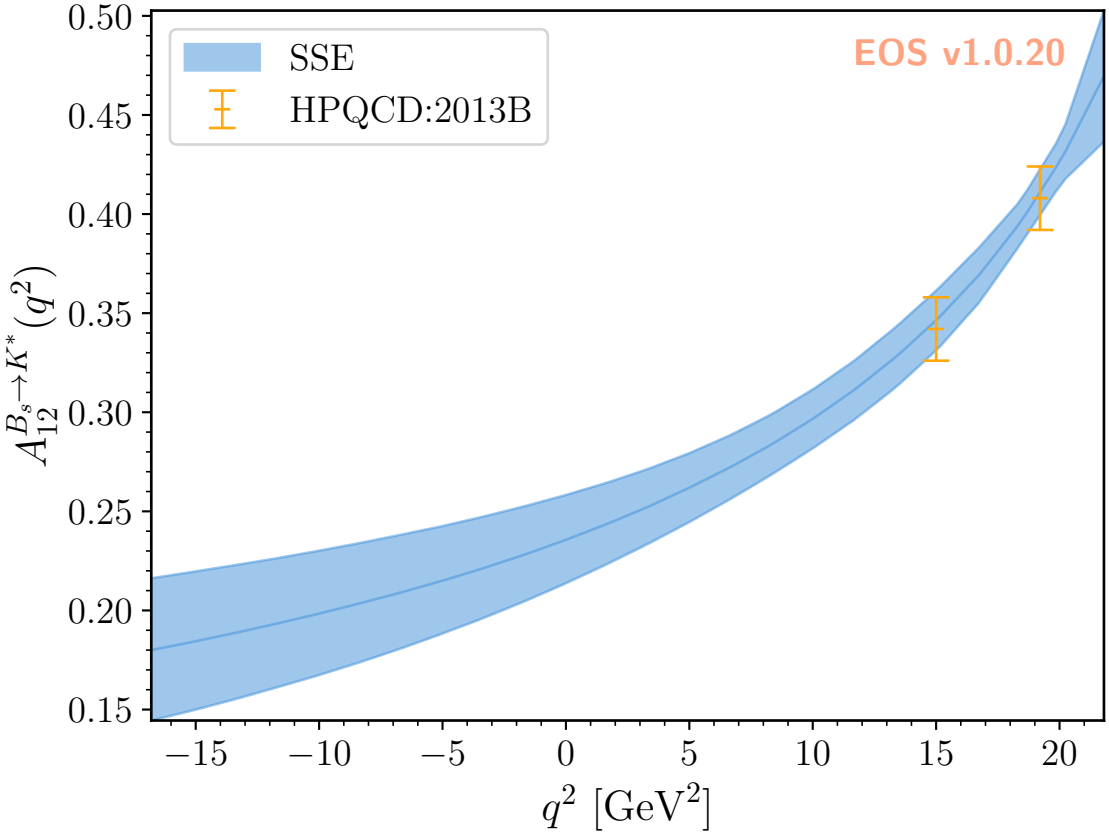

Figure 79: Posterior-predicted form factor for `Bs-to-Kstar` (`Bs-to-Kstar-A12`).

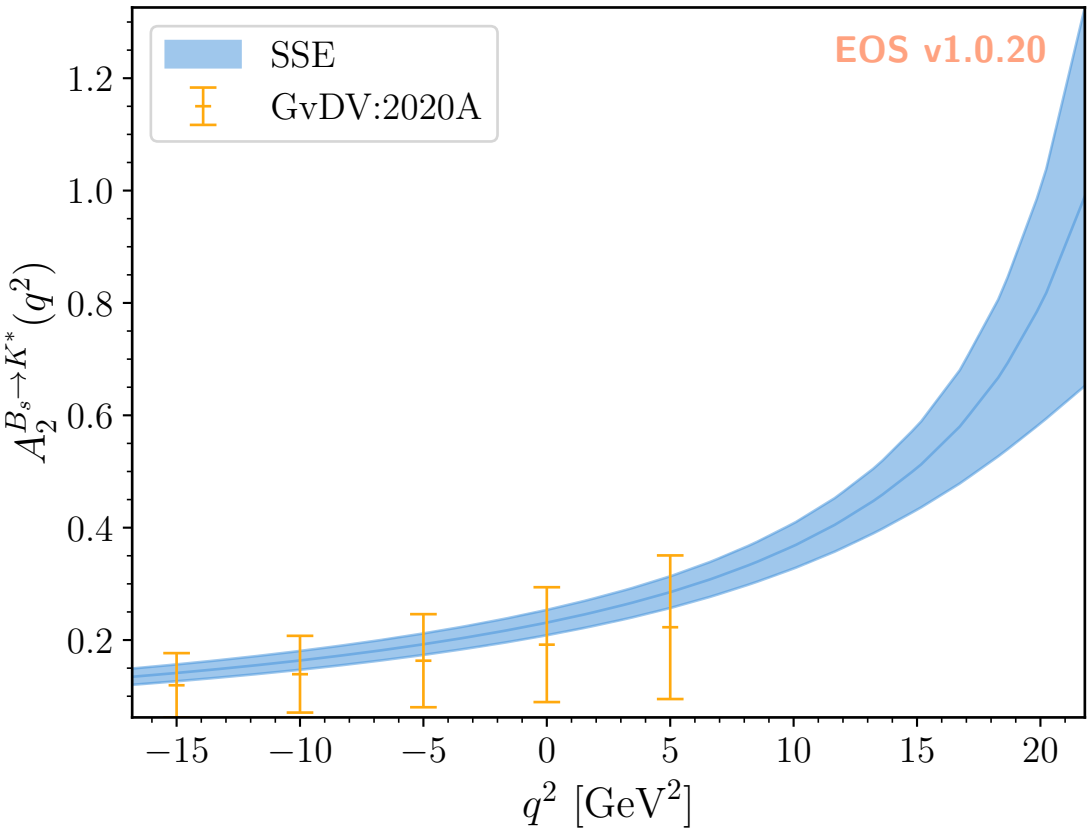

Figure 80: Posterior-predicted form factor for `Bs-to-Kstar` (`Bs-to-Kstar-A2`).

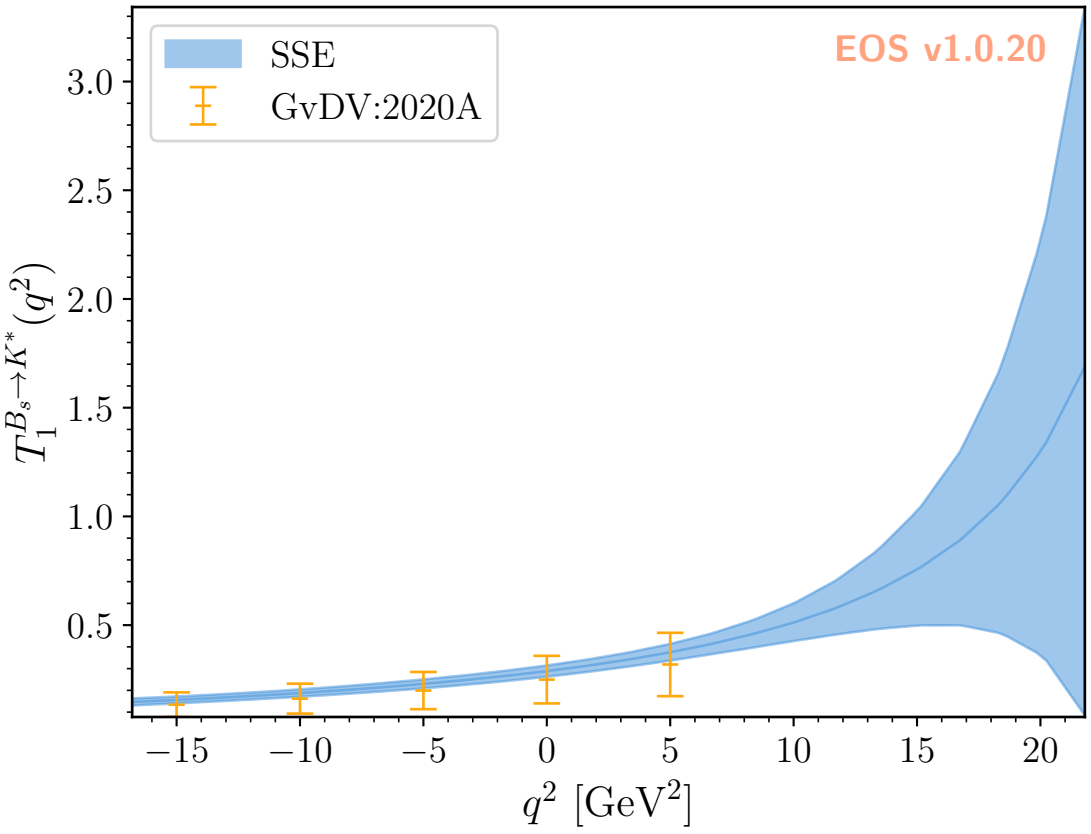

Figure 81: Posterior-predicted form factor for `Bs-to-Kstar` (`Bs-to-Kstar-T1`).

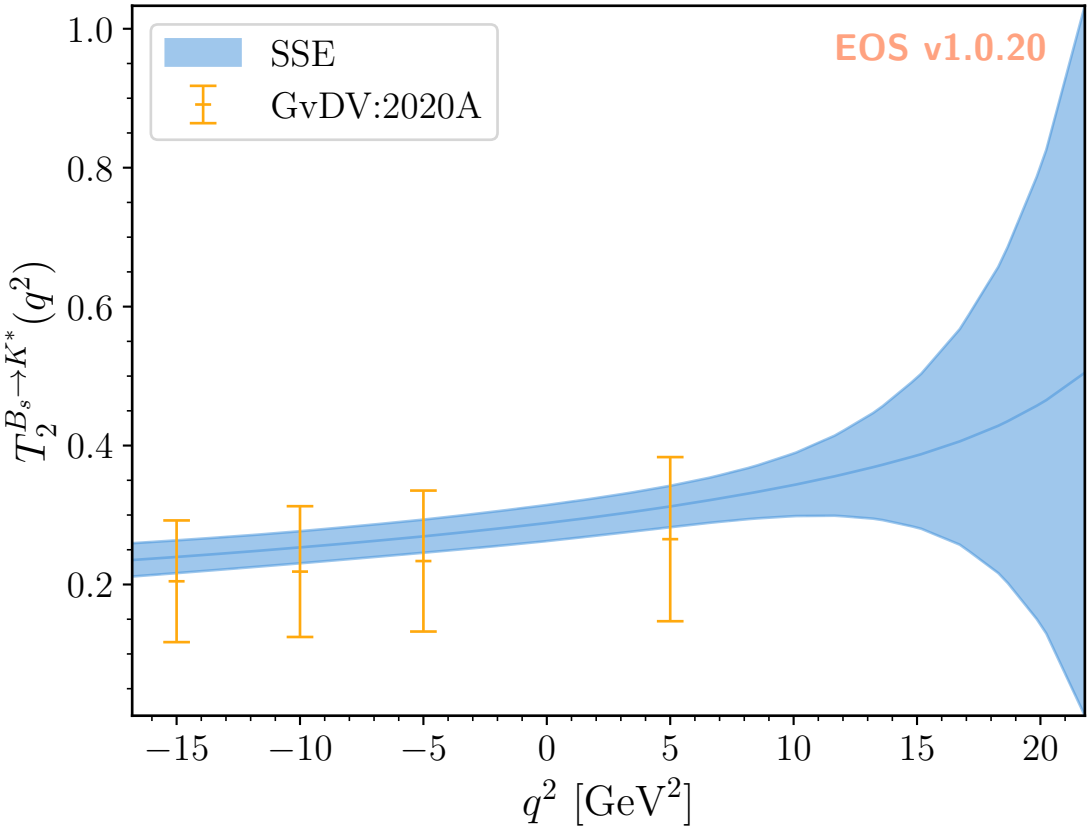

Figure 82: Posterior-predicted form factor for `Bs-to-Kstar` (`Bs-to-Kstar-T2`).

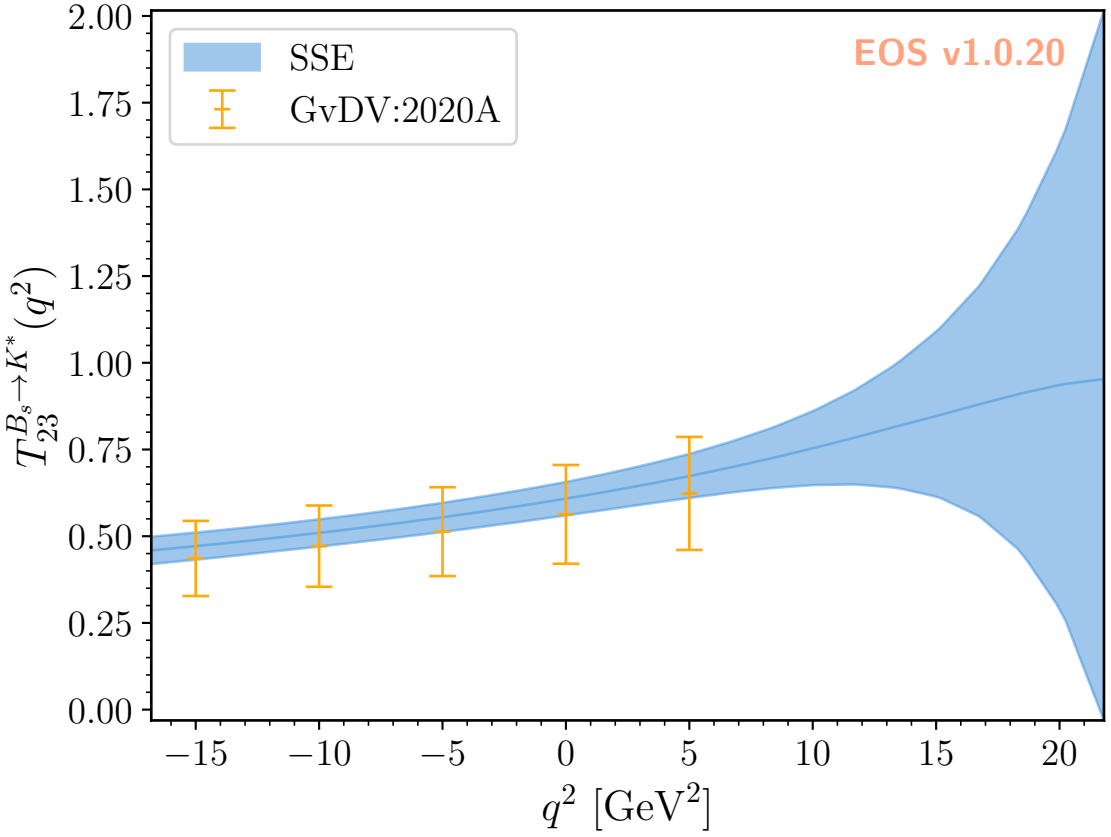

Figure 83: Posterior-predicted form factor for `Bs-to-Kstar` (`Bs-to-Kstar-T23`).

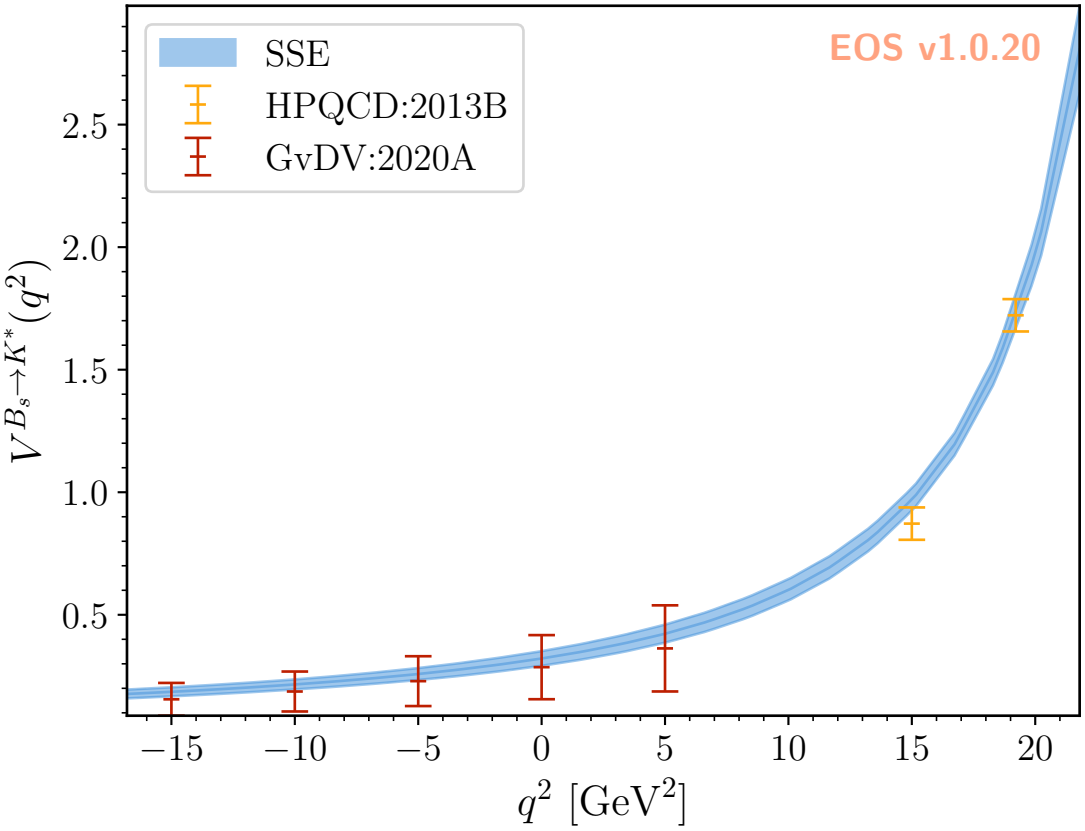

Figure 84: Posterior-predicted form factor for `Bs-to-Kstar` (`Bs-to-Kstar-V`).

Posterior Bs-to-phi

This posterior varies 17 SSE coefficients. It is constrained by:

- B_s->phi::V+A_0+A_1+A_12@HLMW:2015A
- B_s->phi::T_1+T_2+T_23@HLMW:2015A
- B_s->phi::FormFactors[A_0,A_1,A_2,V,T_1,T_2,T_23]@GvDV:2020A

Goodness of fit

The table lists the local p value of each constraint together with the global p value and total χ^2 of the best-fit point.

Constraint	local p value
B_s->phi::FormFactors[A_0,A_1,A_2,V,T_1,T_2,T_23]@GvDV:2020A	100.0%
B_s->phi::T_1+T_2+T_23@HLMW:2015A	100.0%
B_s->phi::V+A_0+A_1+A_12@HLMW:2015A	100.0%
total (χ^2 / d.o.f. = 0.91 / 30)	**100.0%**

Posterior-predicted form factors

The bands show the posterior-predicted form factors (SSE); the points show the constraints entering the fit.

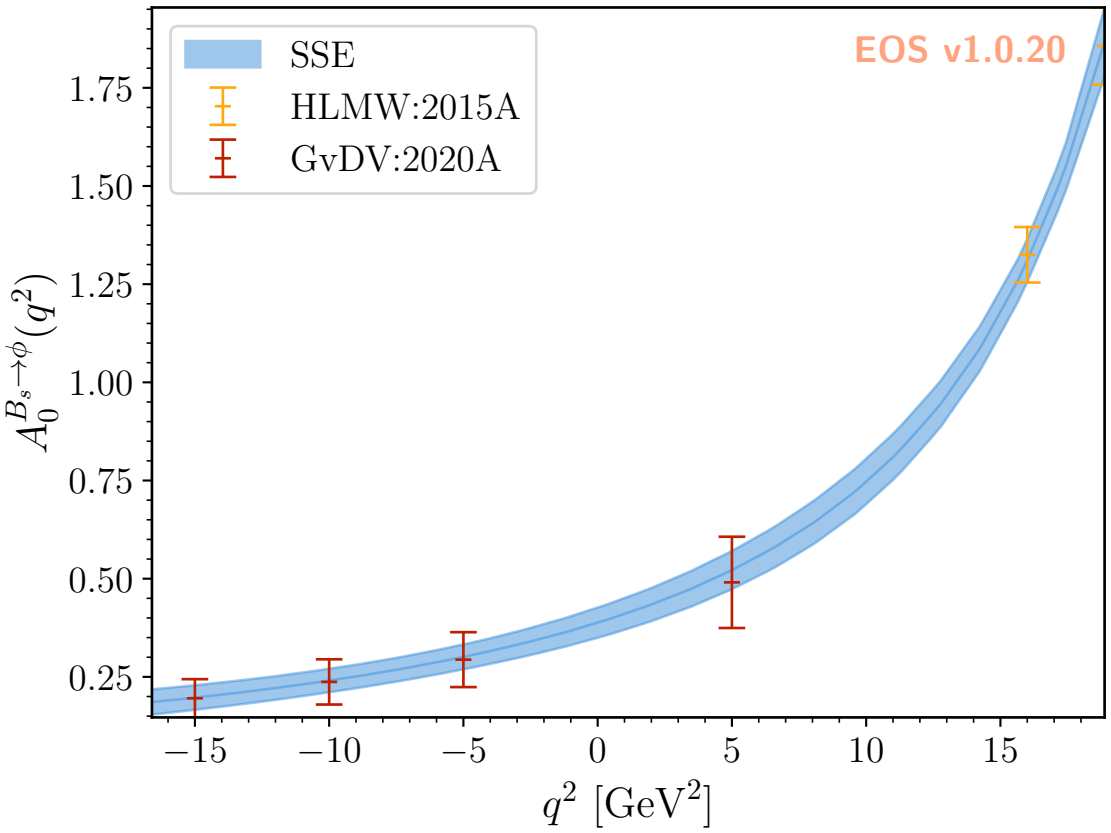

Figure 85: Posterior-predicted form factor for `Bs-to-phi` (`Bs-to-phi-A0`).

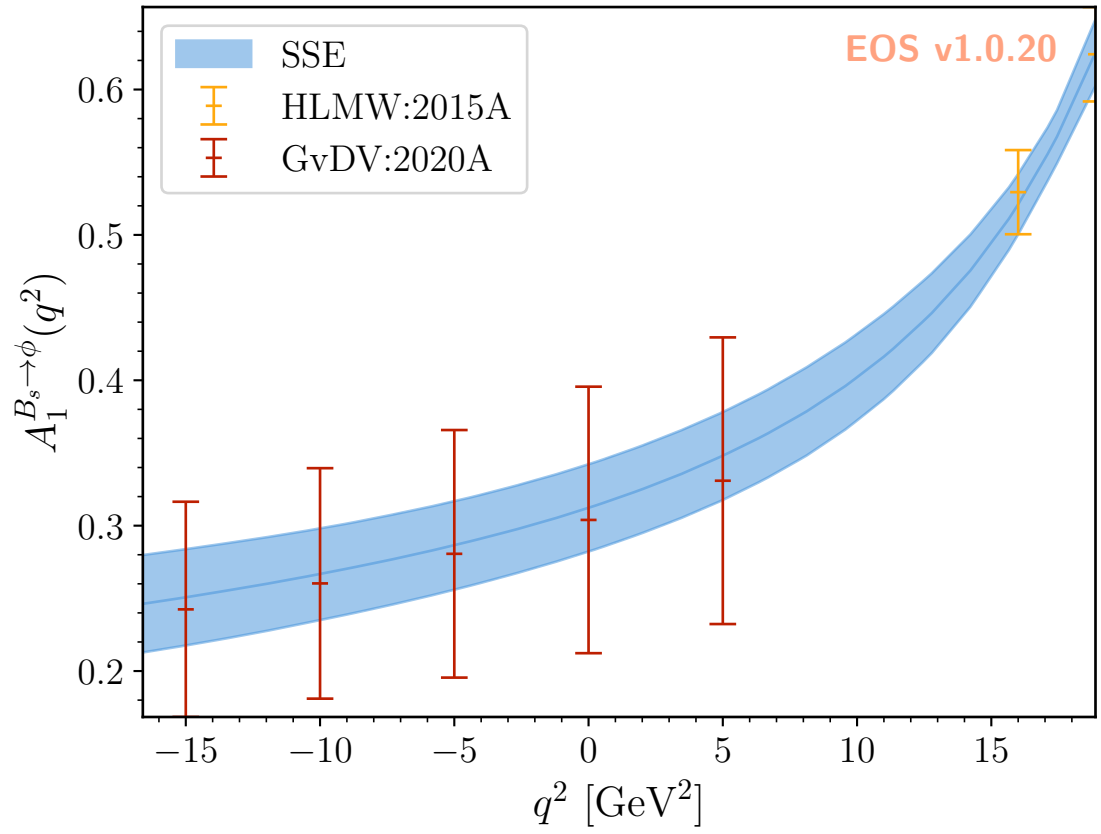

Figure 86: Posterior-predicted form factor for `Bs-to-phi` (`Bs-to-phi-A1`).

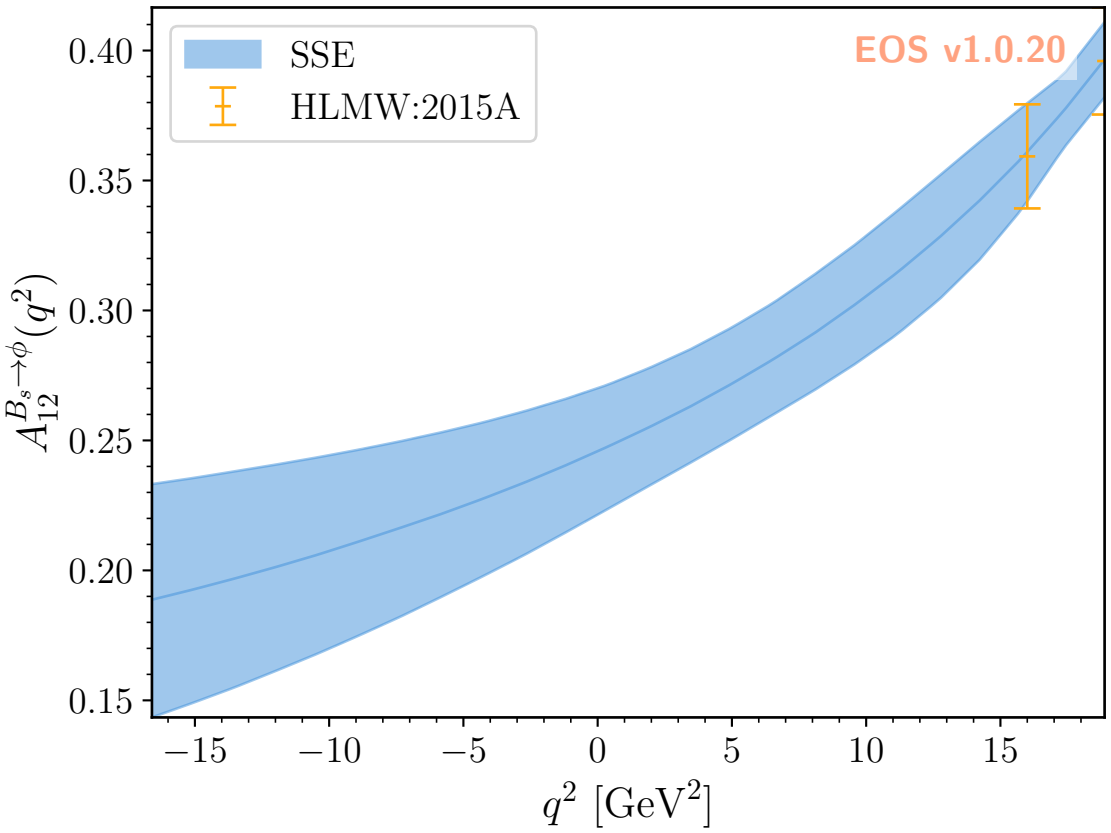

Figure 87: Posterior-predicted form factor for `Bs-to-phi` (`Bs-to-phi-A12`).

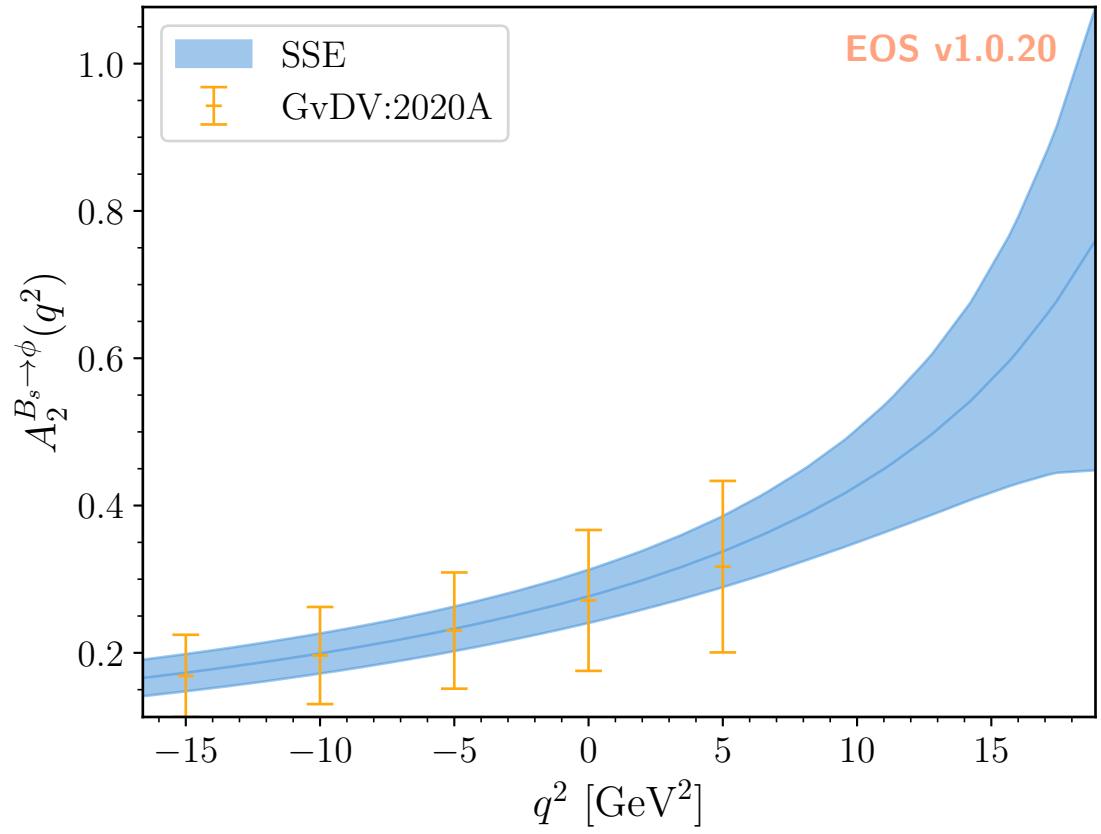

Figure 88: Posterior-predicted form factor for `Bs-to-phi` (`Bs-to-phi-A2`).

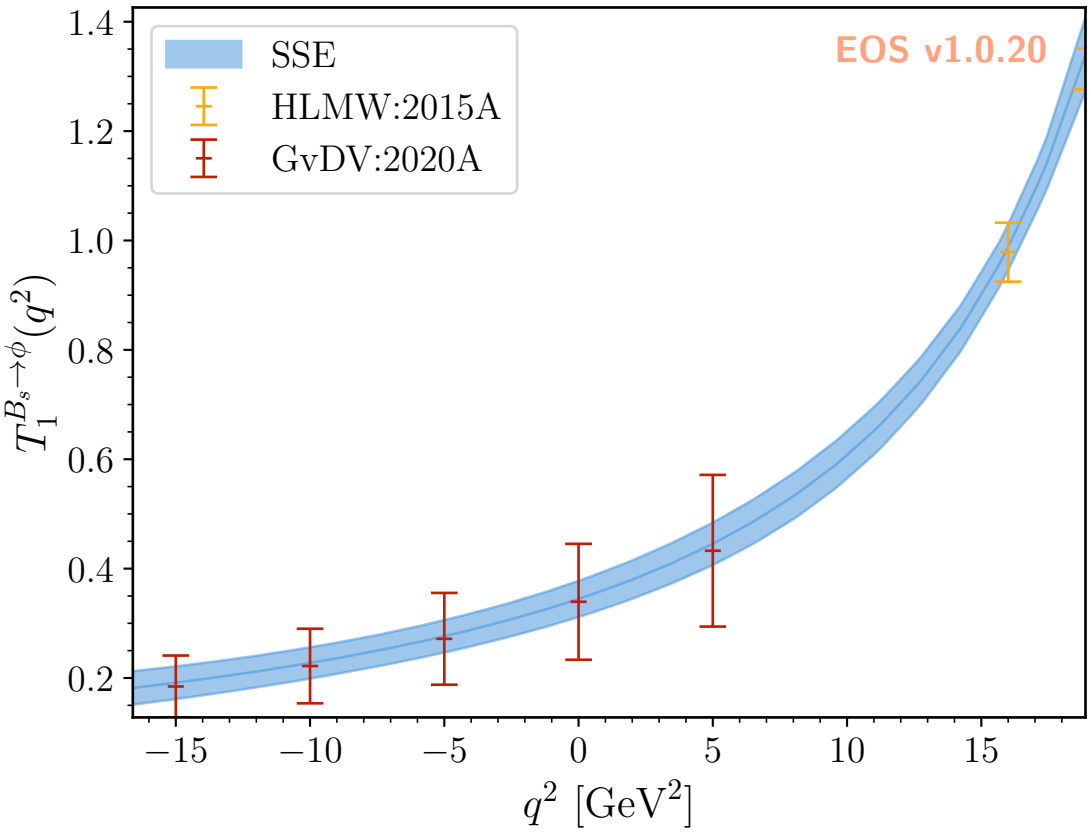

Figure 89: Posterior-predicted form factor for `Bs-to-phi` (`Bs-to-phi-T1`).

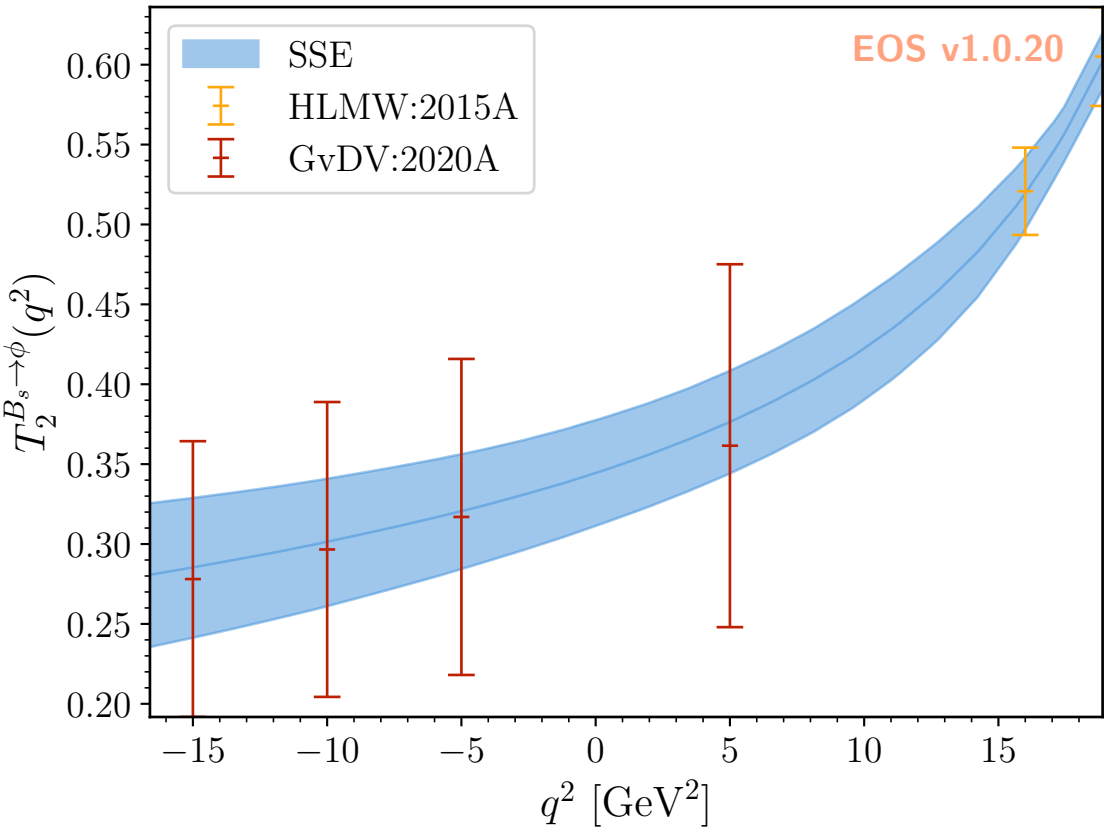

Figure 90: Posterior-predicted form factor for `Bs-to-phi` (`Bs-to-phi-T2`).

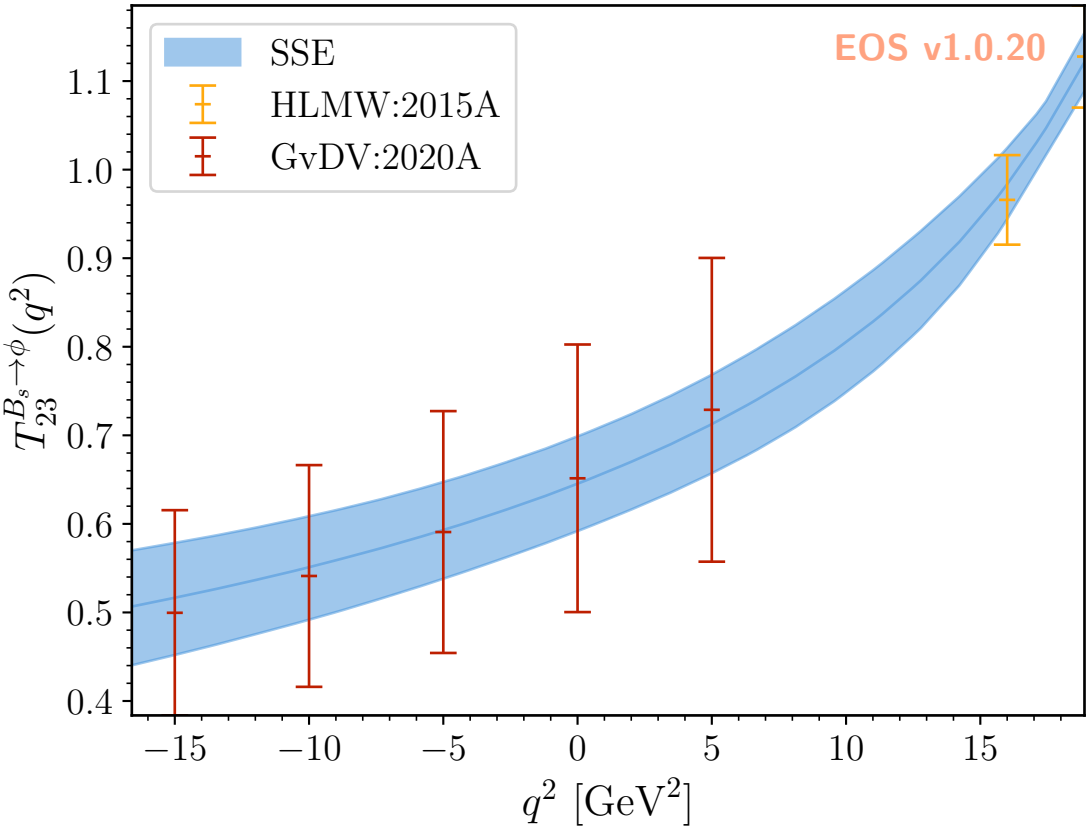

Figure 91: Posterior-predicted form factor for `Bs-to-phi` (`Bs-to-phi-T23`).

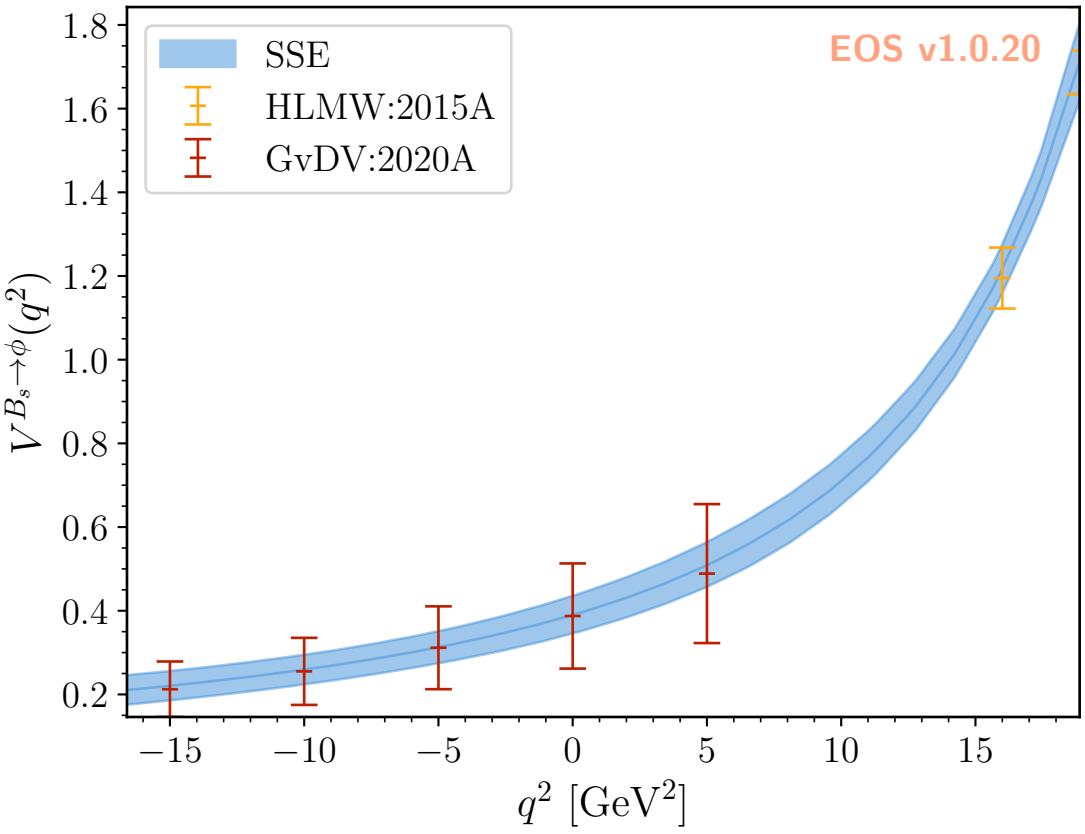

Figure 92: Posterior-predicted form factor for `Bs-to-phi` (`Bs-to-phi-V`).

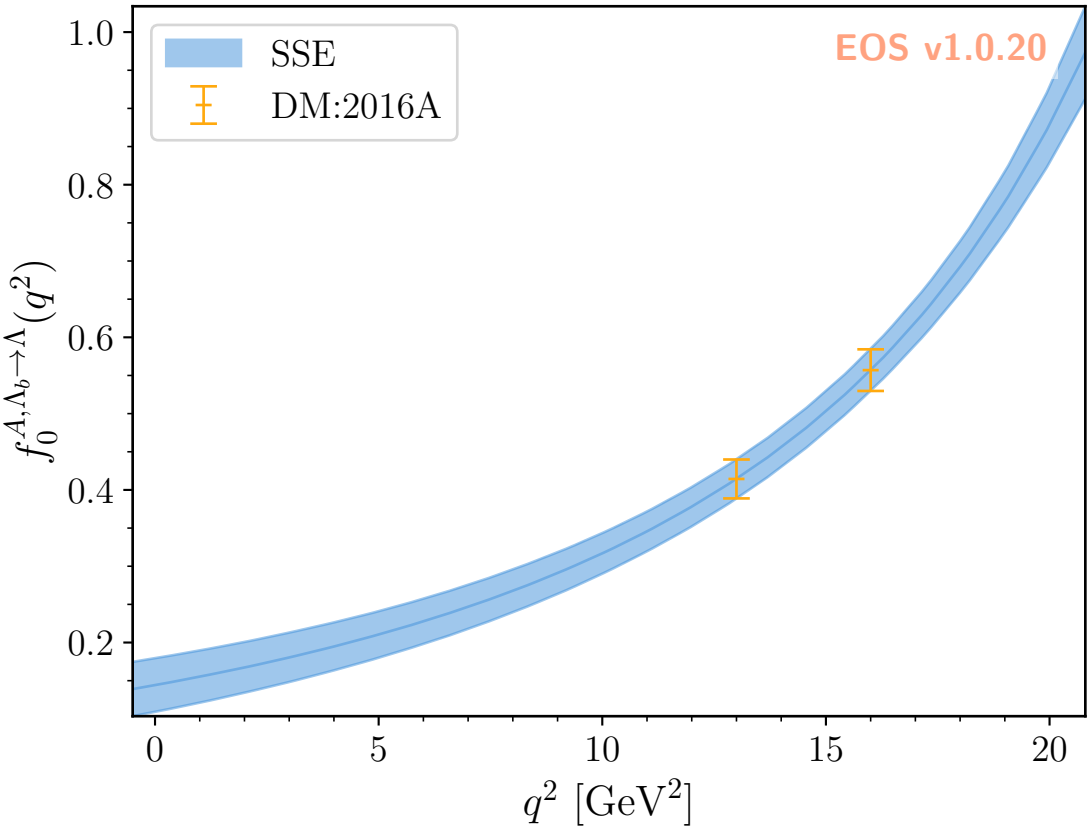

Figure 93: Posterior-predicted form factor for `Lambdab-to-Lambda` (`Lambdab-to-Lambda-f0A`).

Spin-1/2 to spin-1/2 ($1/2^+ \to 1/2^+$)

Posterior `Lambdab-to-Lambda`

This posterior varies 25 `SSE` coefficients. It is constrained by:

- `Lambda_b->Lambda::f_time+long+perp^V+A+T+T5[nominal,no-prior]@DM:2016A`

Goodness of fit

The table lists the local p value of each constraint. The fit is saturated (the number of data points equals the number of coefficients, d.o.f. $= 0$), so no global p value is defined.

Constraint	local p value
`Lambda_b->Lambda::f_time+long+perp^V+A+T+T5[nominal,no-prior]@DM:2016A`	100.0%

Posterior-predicted form factors

The bands show the posterior-predicted form factors (`SSE`); the points show the constraints entering the fit.

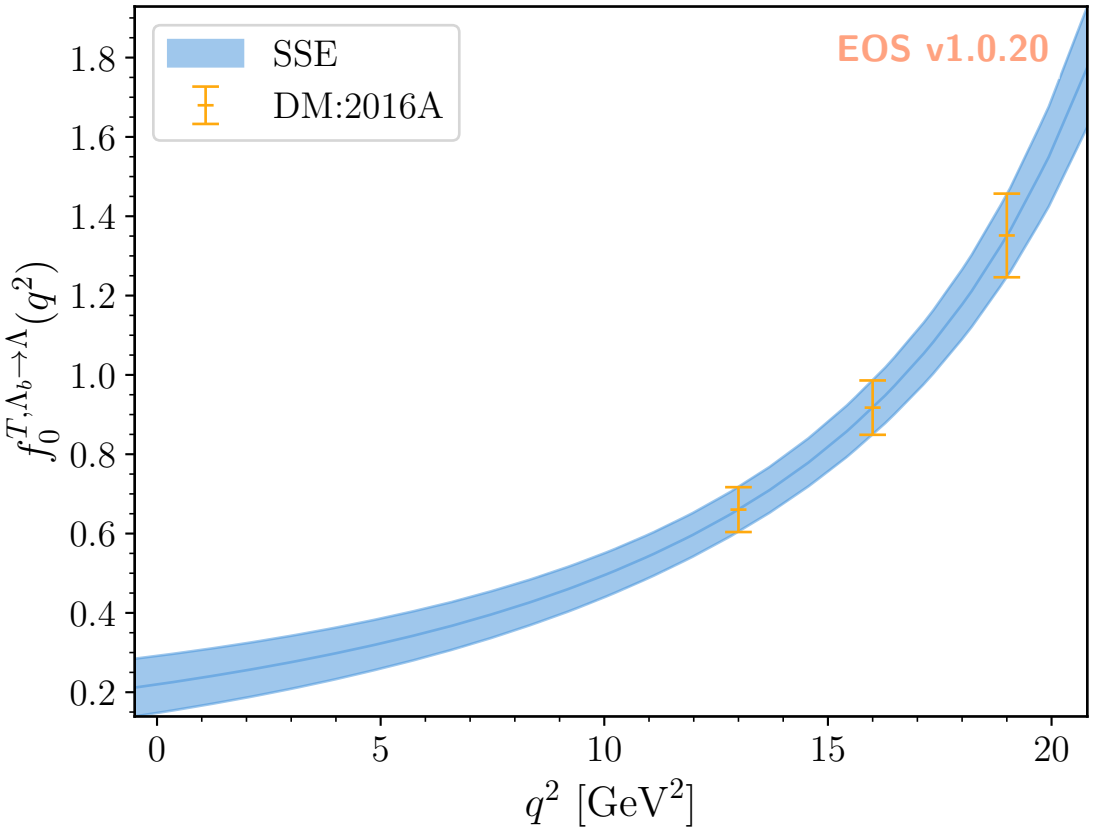

Figure 94: Posterior-predicted form factor for `Lambdab-to-Lambda` (`Lambdab-to-Lambda-f0T`).

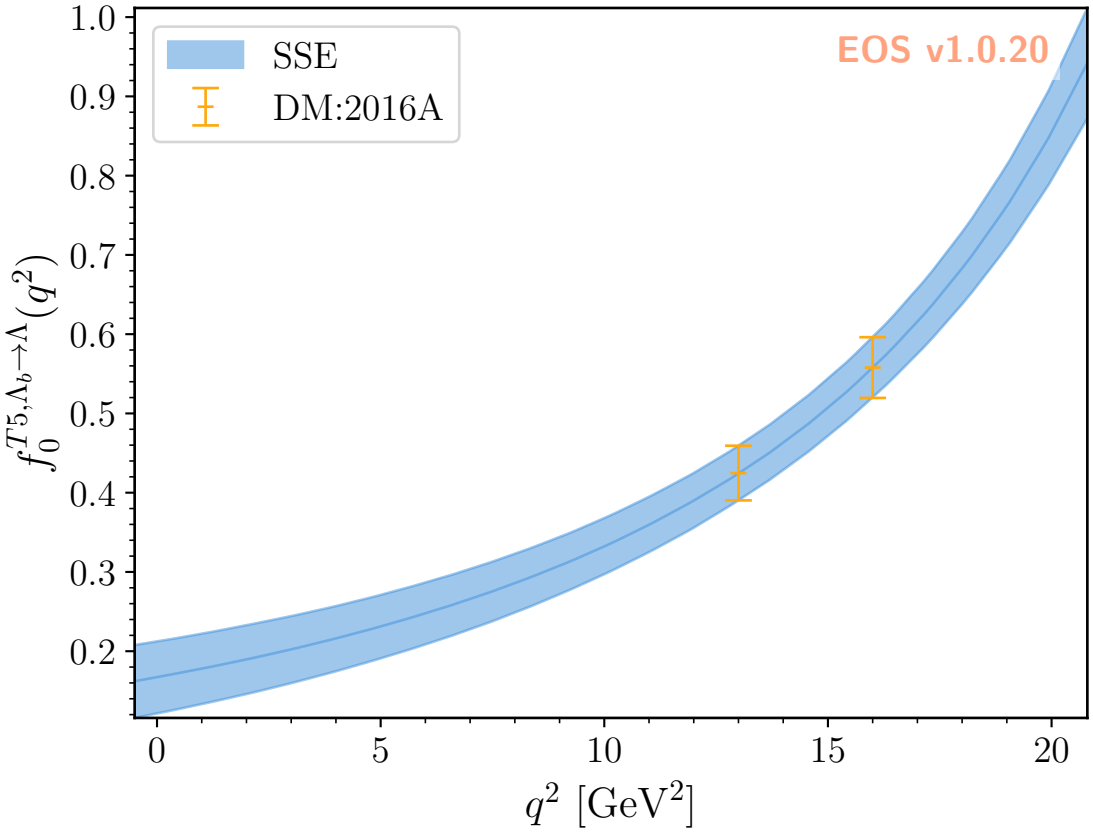

Figure 95: Posterior-predicted form factor for `Lambdab-to-Lambda` (`Lambdab-to-Lambda-f0T5`).

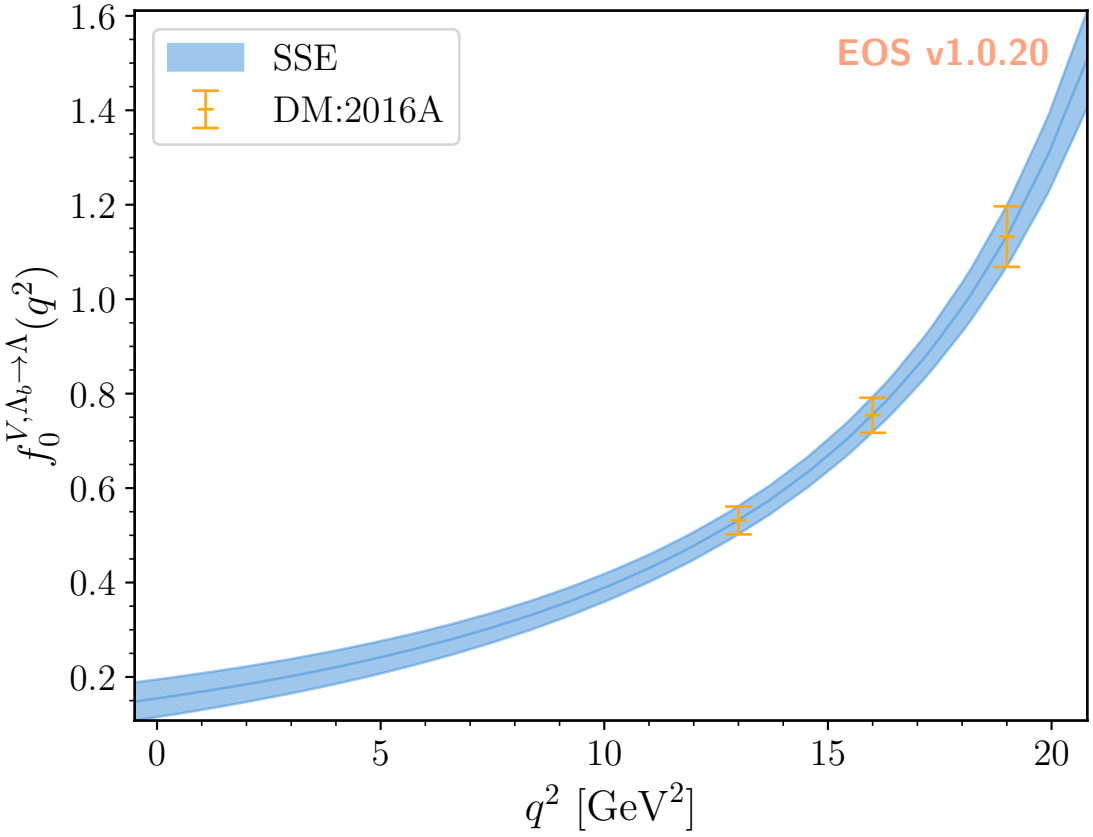

Figure 96: Posterior-predicted form factor for `Lambdab-to-Lambda` (`Lambdab-to-Lambda-f0V`).

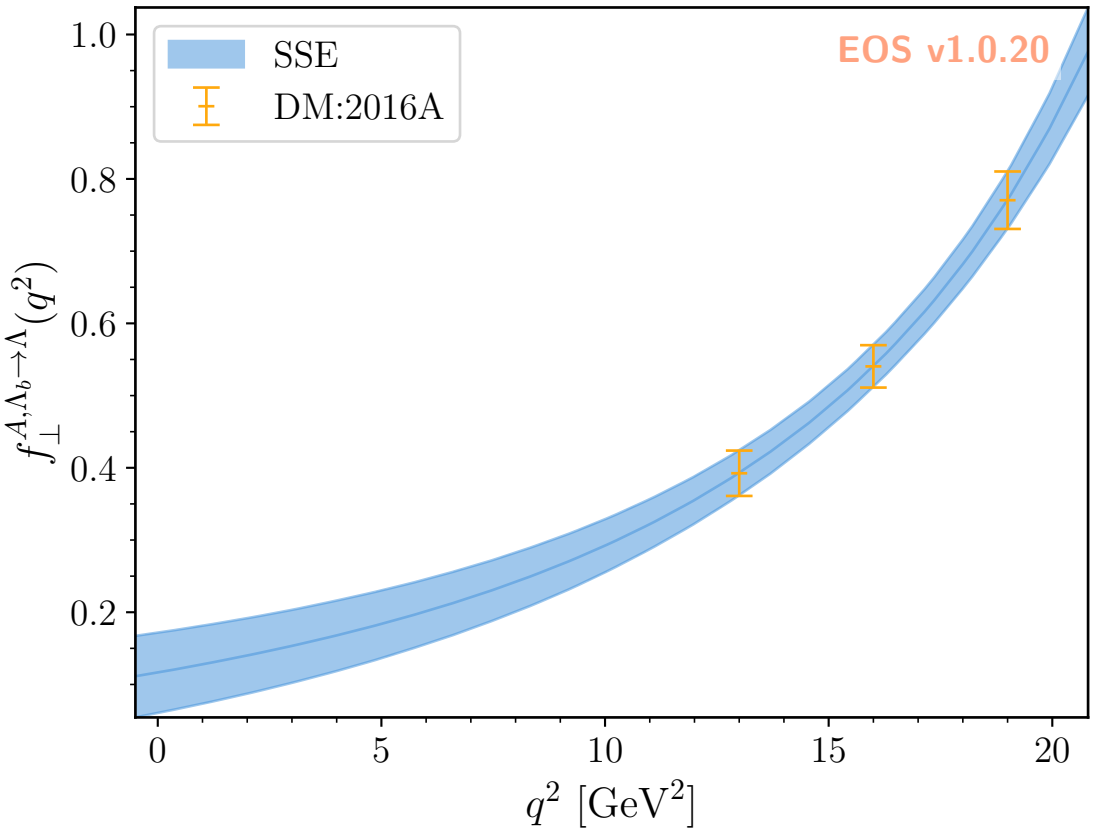

Figure 97: Posterior-predicted form factor for `Lambdab-to-Lambda` (`Lambdab-to-Lambda-fperpA`).

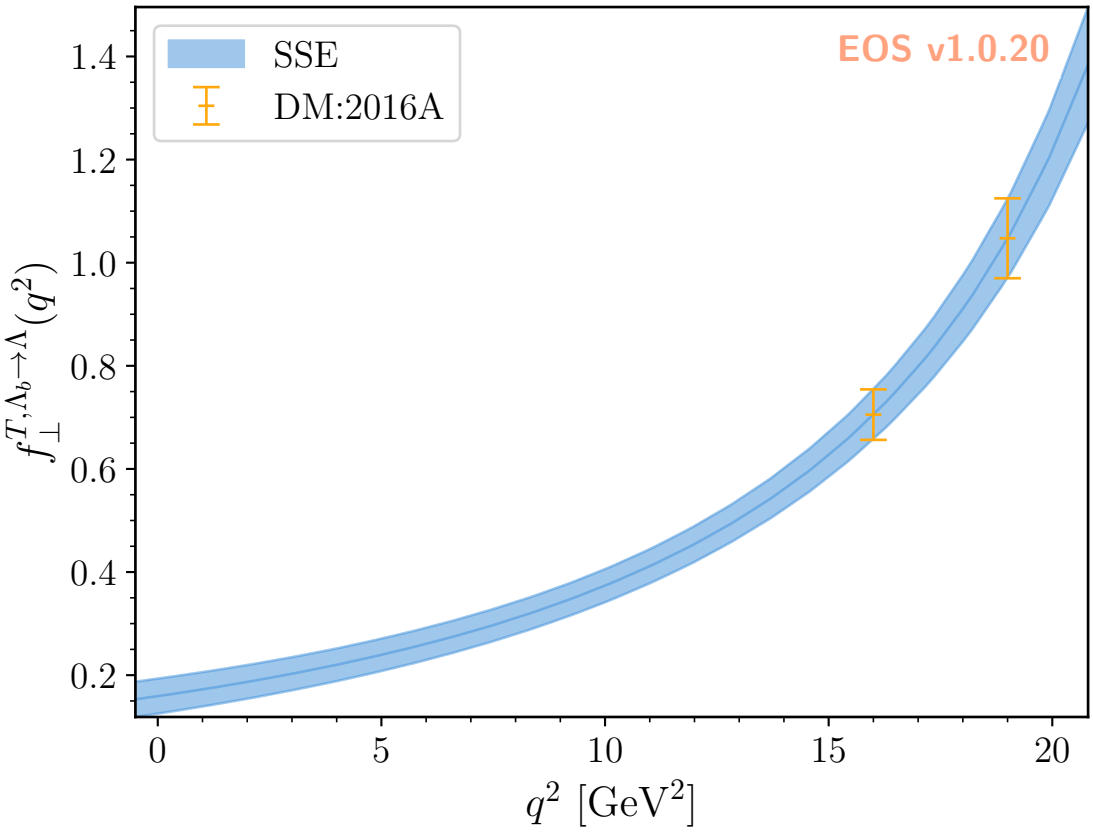

Figure 98: Posterior-predicted form factor for `Lambdab-to-Lambda` (`Lambdab-to-Lambda-fperpT`).

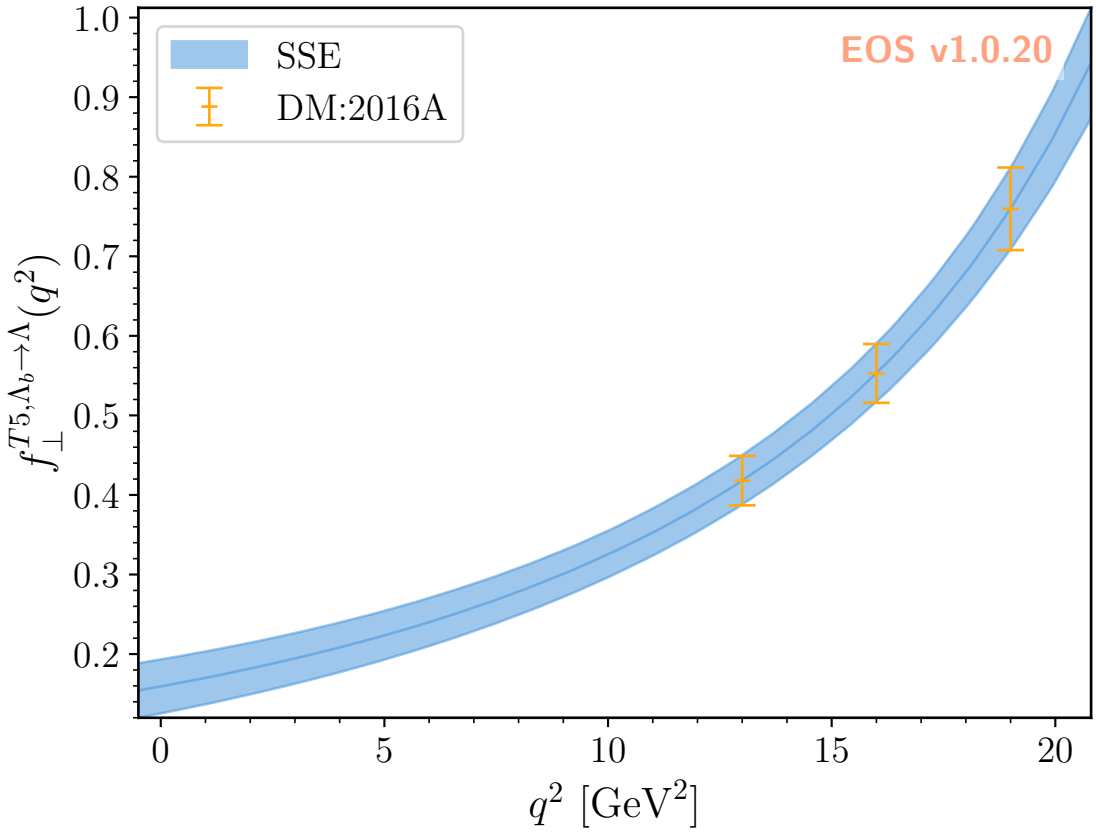

Figure 99: Posterior-predicted form factor for `Lambdab-to-Lambda` (`Lambdab-to-Lambda-fperpT5`).

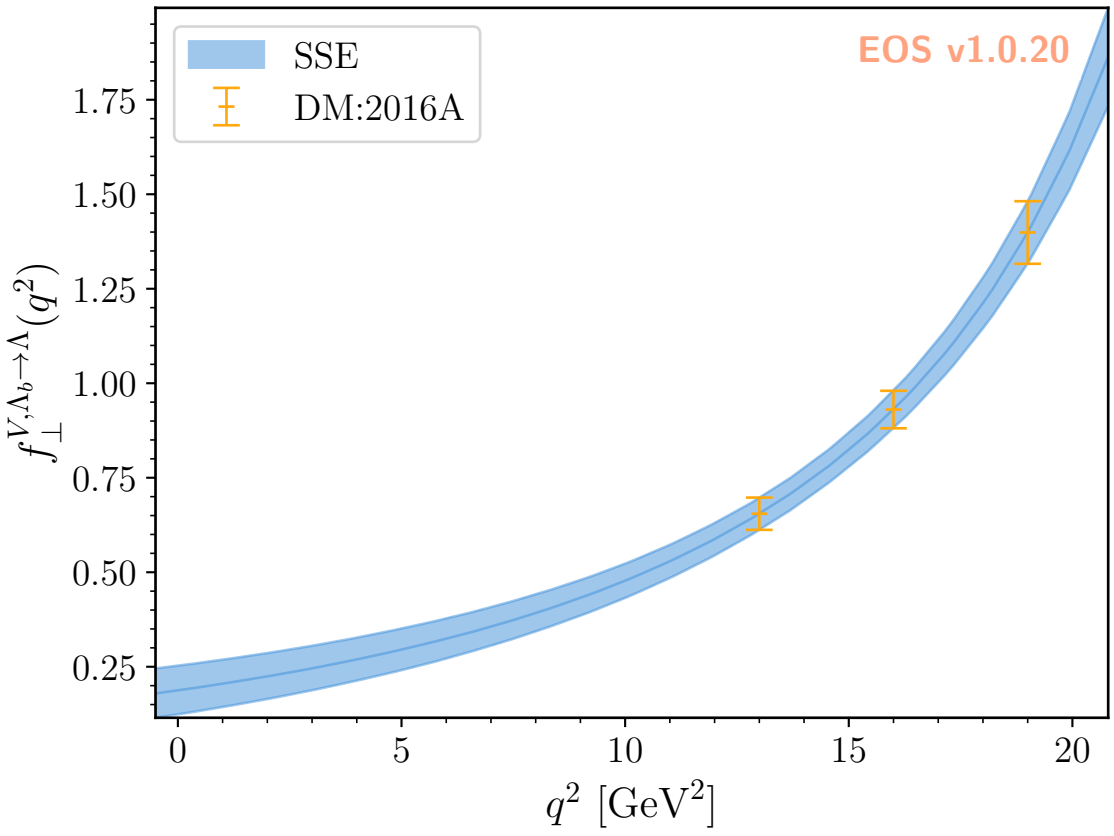

Figure 100: Posterior-predicted form factor for `Lambdab-to-Lambda` (`Lambdab-to-Lambda-fperpV`).

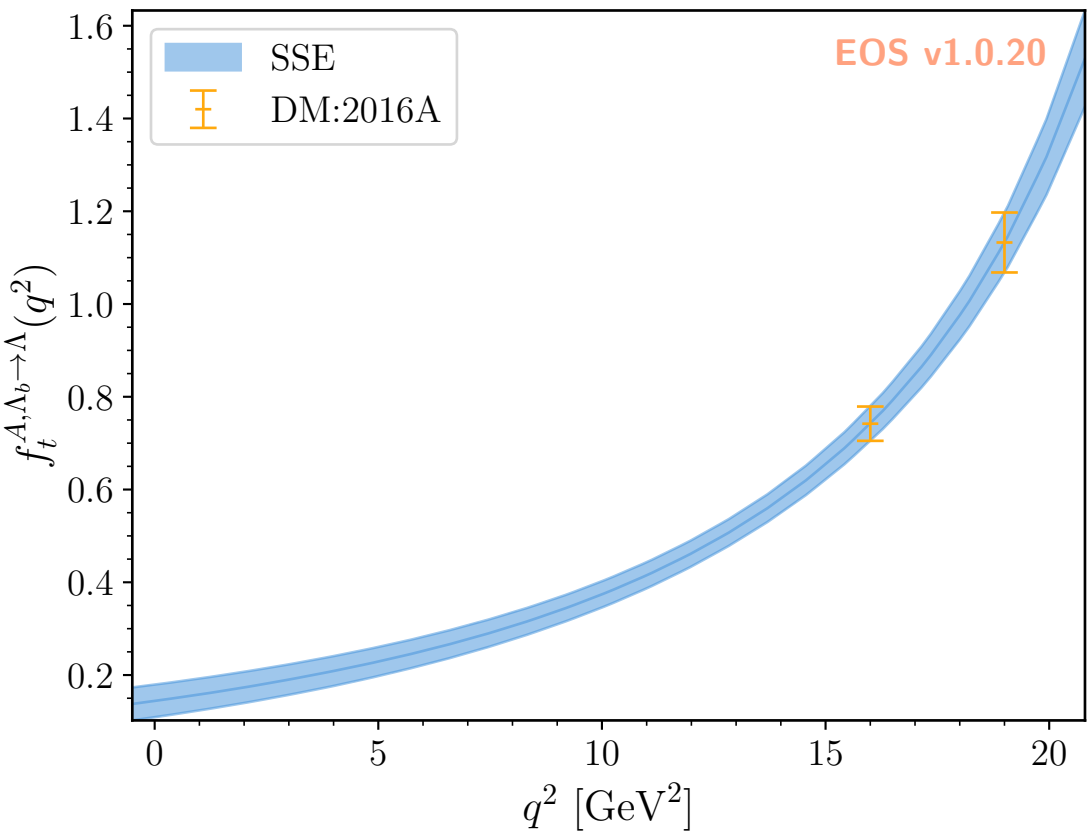

Figure 101: Posterior-predicted form factor for `Lambdab-to-Lambda` (`Lambdab-to-Lambda-ftA`).

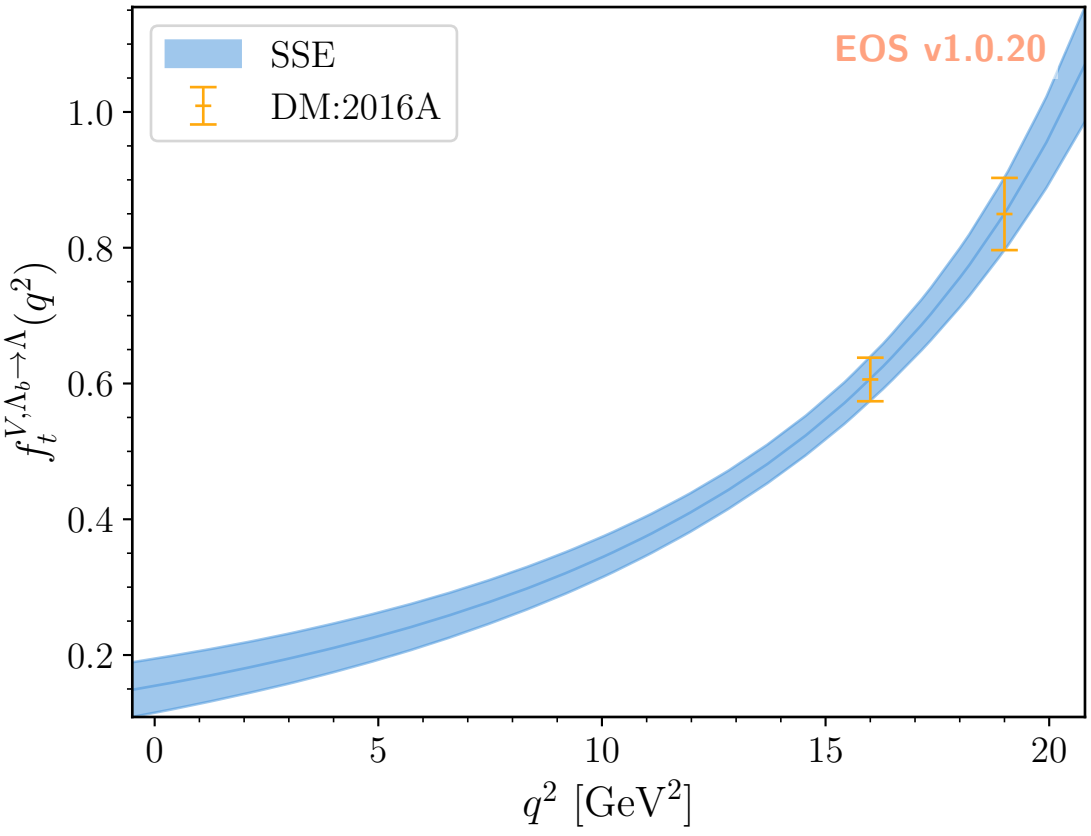

Figure 102: Posterior-predicted form factor for `Lambdab-to-Lambda` (`Lambdab-to-Lambda-ftV`).

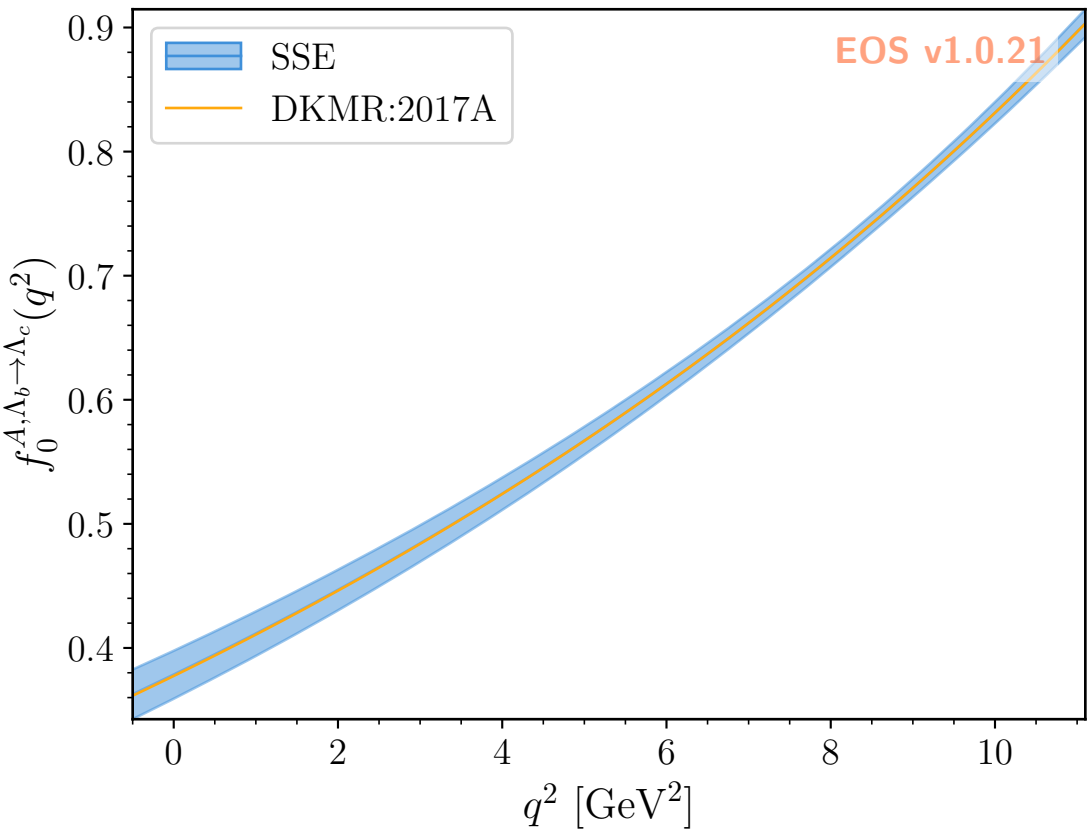

Figure 103: Posterior-predicted form factor for `Lambdab-to-Lambdac` (`Lambdab-to-Lambdac-f0A`).

Posterior `Lambdab-to-Lambdac`

This posterior varies 15 `SSE` coefficients. It is constrained by:

- `Lambda_b->Lambda_c::f_time+long+perp^V+A+T+T5@DKMR:2017A`

Goodness of fit

The table lists the local p value of each constraint together with the global p value and total χ^2 of the best-fit point.

Constraint	local p value
`Lambda_b->Lambda_c::f_time+long+perp^V+A+T+T5@DKMR:2017A`	100.0%
total (χ^2 / d.o.f. $= 0.12$ / 3)	**99.0%**

Posterior-predicted form factors

The bands show the posterior-predicted form factors (`SSE`); the points show the constraints entering the fit.

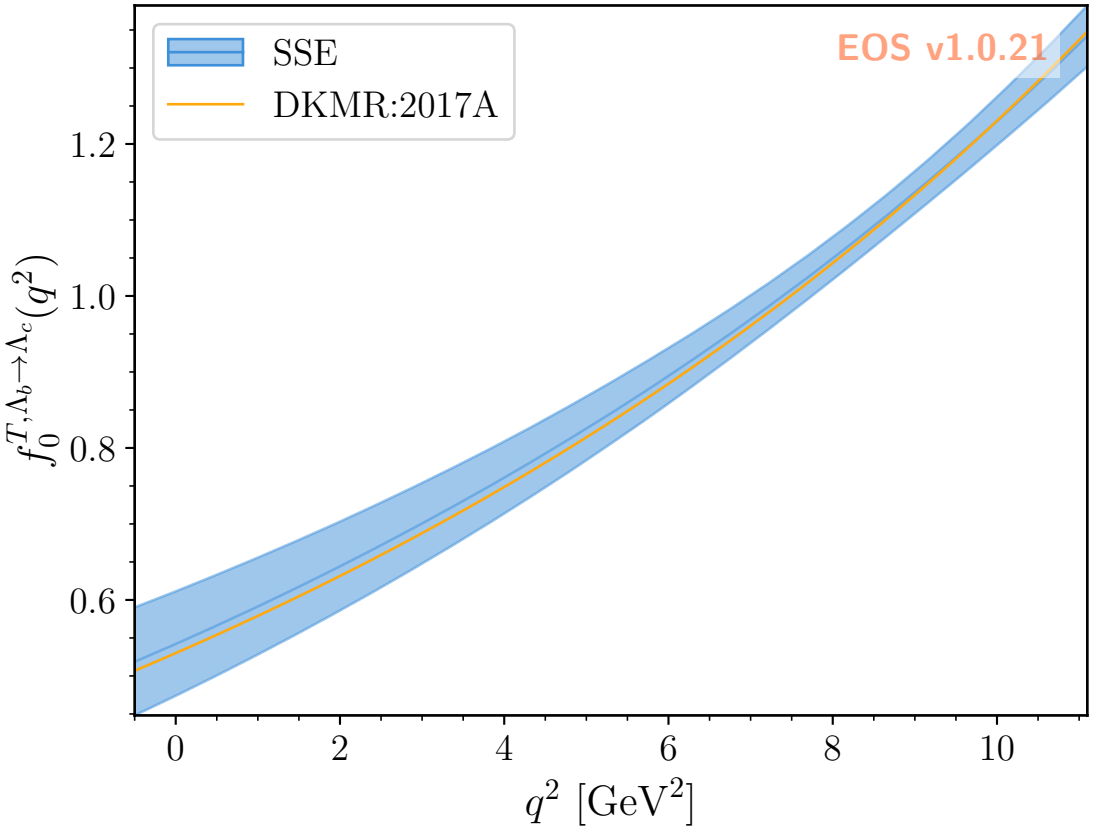

Figure 104: Posterior-predicted form factor for `Lambdab-to-Lambdac` (`Lambdab-to-Lambdac-f0T`).

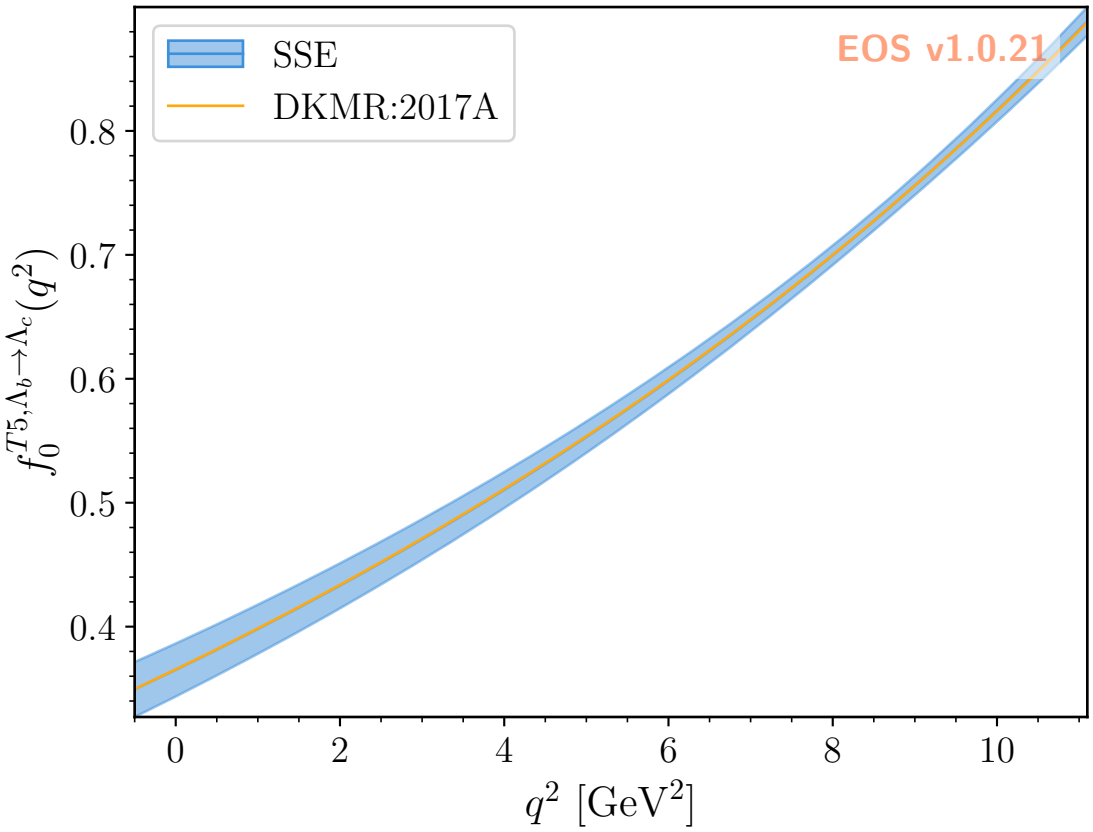

Figure 105: Posterior-predicted form factor for `Lambdab-to-Lambdac` (`Lambdab-to-Lambdac-f0T5`).

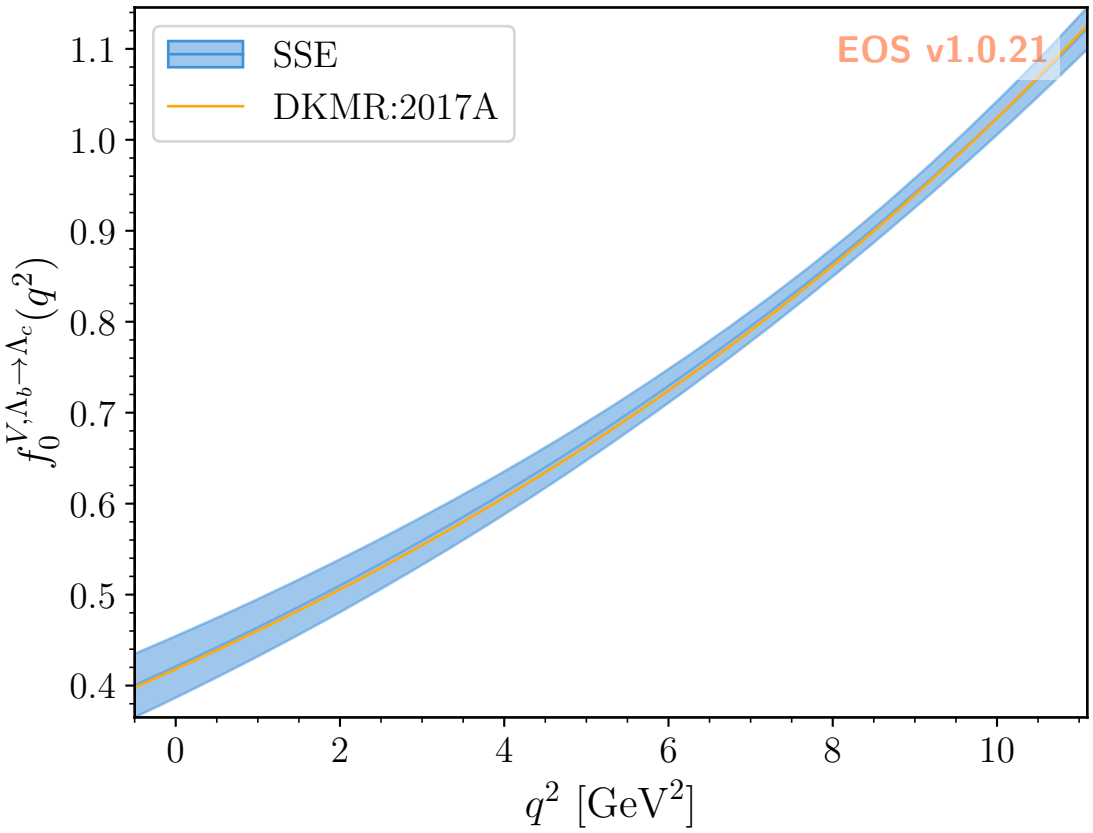

Figure 106: Posterior-predicted form factor for `Lambdab-to-Lambdac` (`Lambdab-to-Lambdac-f0V`).

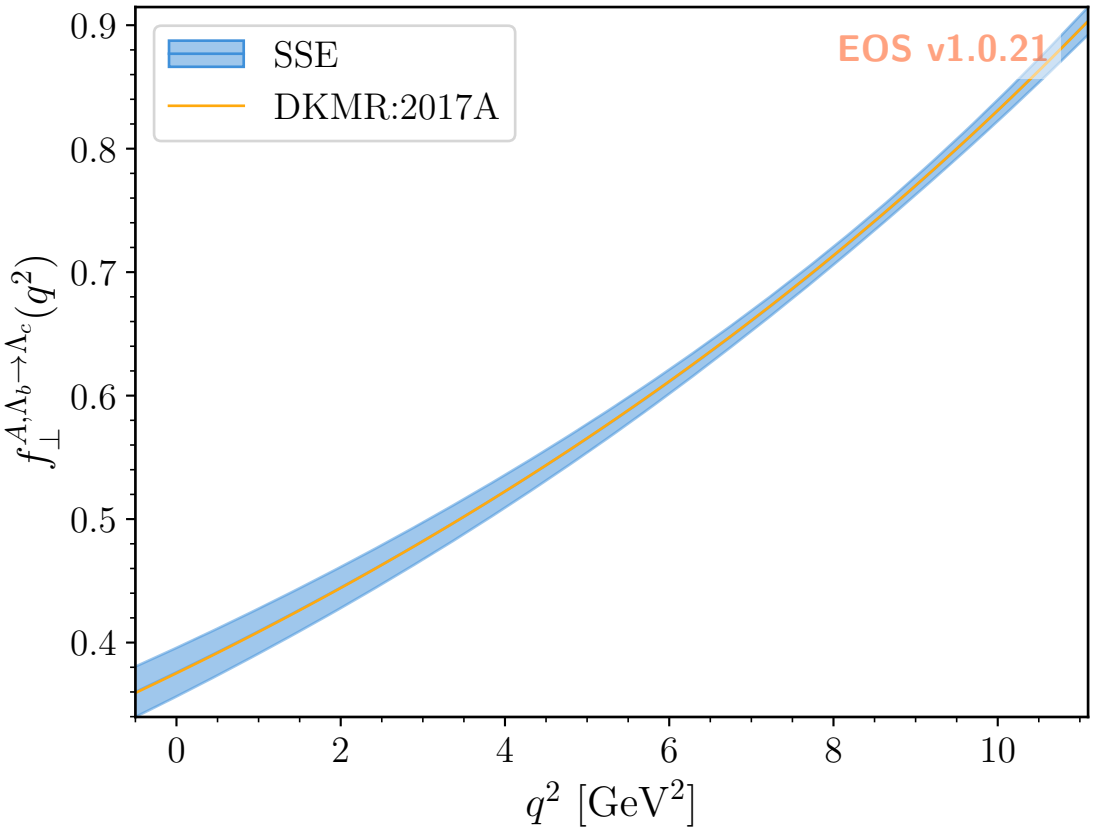

Figure 107: Posterior-predicted form factor for `Lambdab-to-Lambdac` (`Lambdab-to-Lambdac-fperpA`).

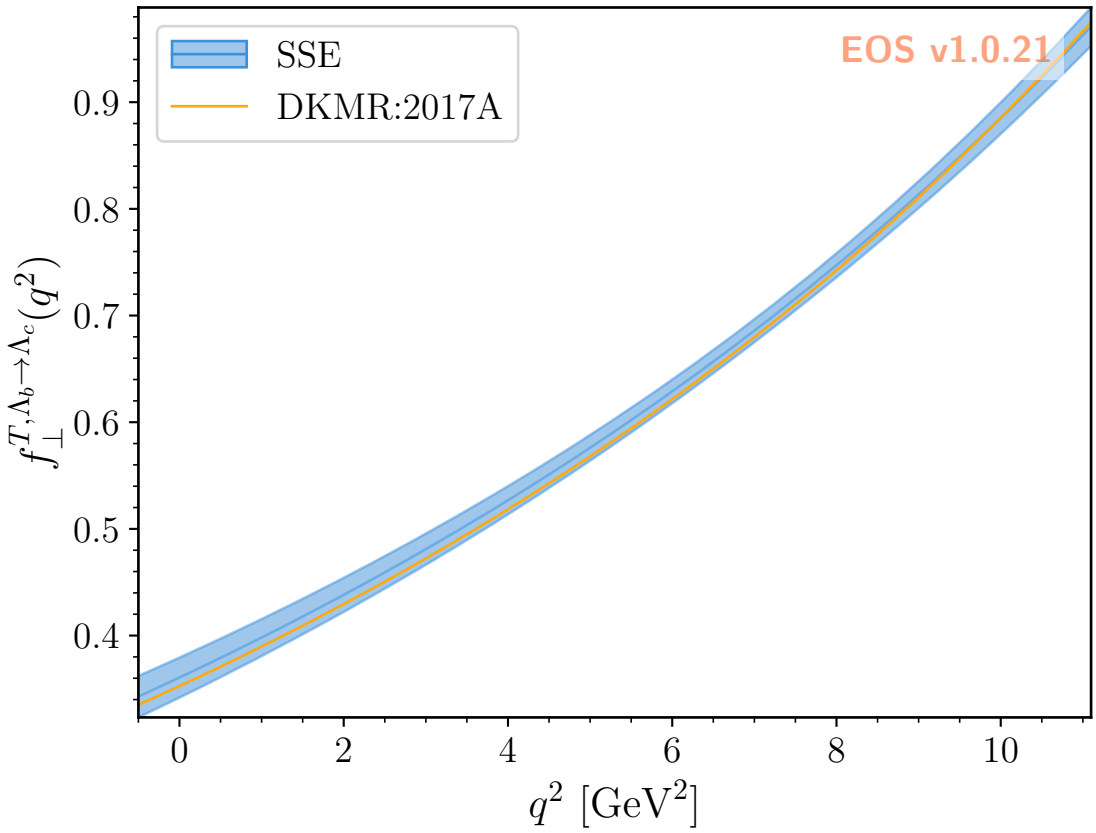

Figure 108: Posterior-predicted form factor for `Lambdab-to-Lambdac` (`Lambdab-to-Lambdac-fperpT`).

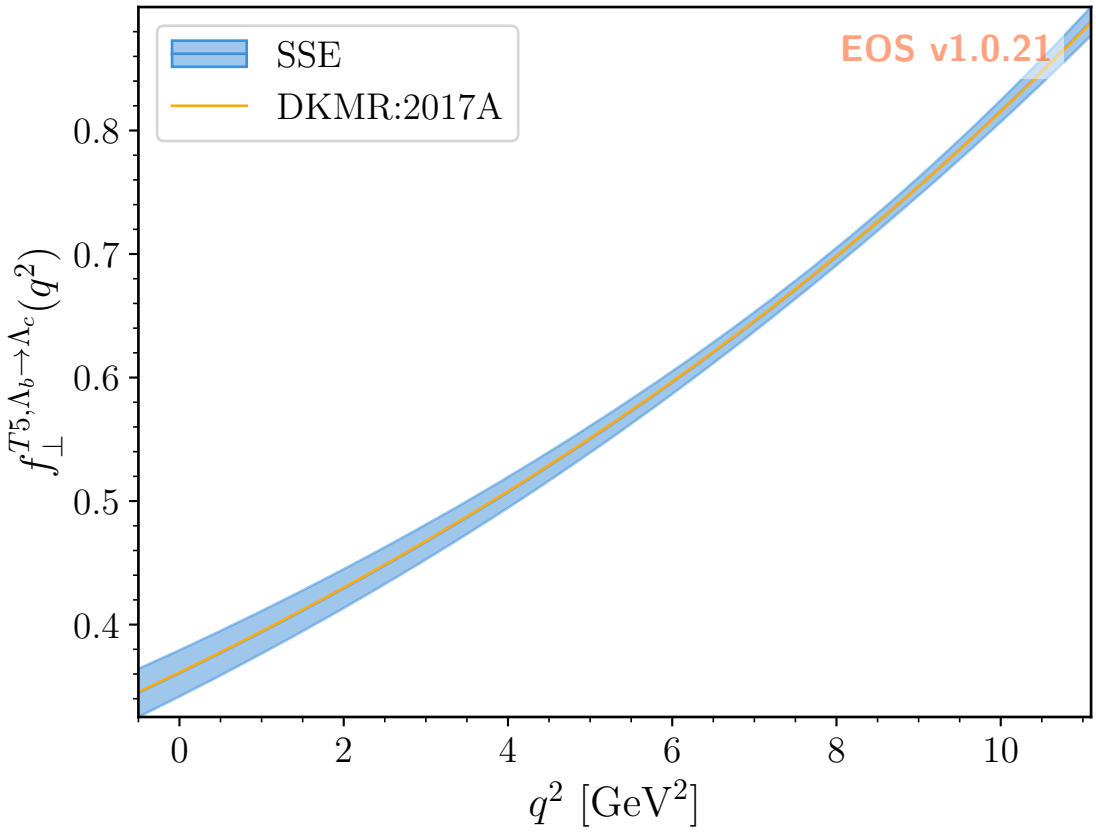

Figure 109: Posterior-predicted form factor for `Lambdab-to-Lambdac` (`Lambdab-to-Lambdac-fperpT5`).

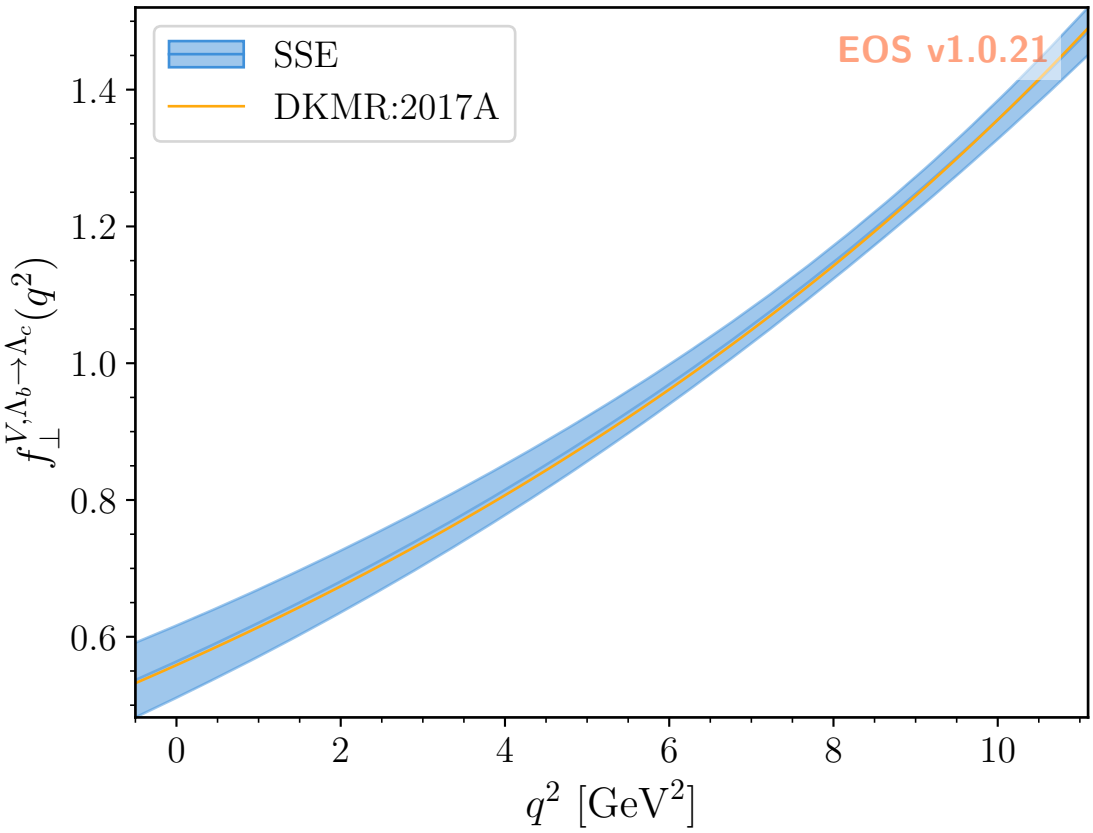

Figure 110: Posterior-predicted form factor for `Lambdab-to-Lambdac` (`Lambdab-to-Lambdac-fperpV`).

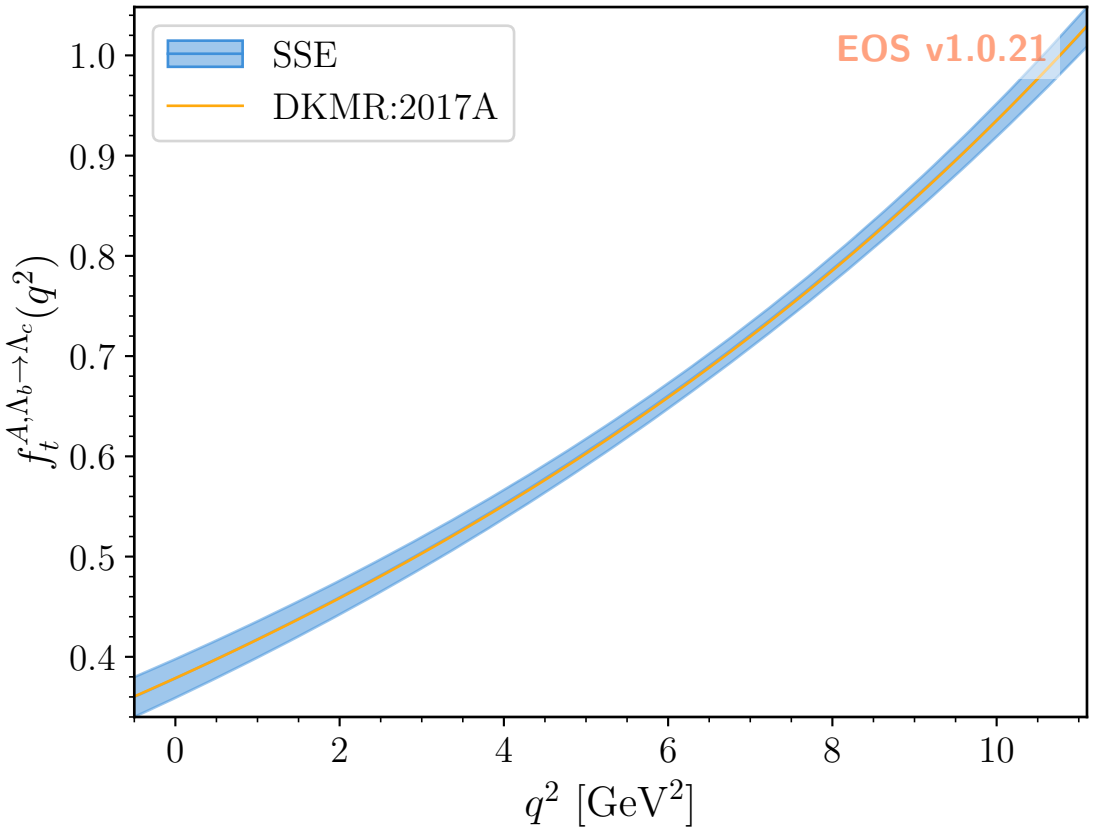

Figure 111: Posterior-predicted form factor for `Lambdab-to-Lambdac` (`Lambdab-to-Lambdac-ftA`).

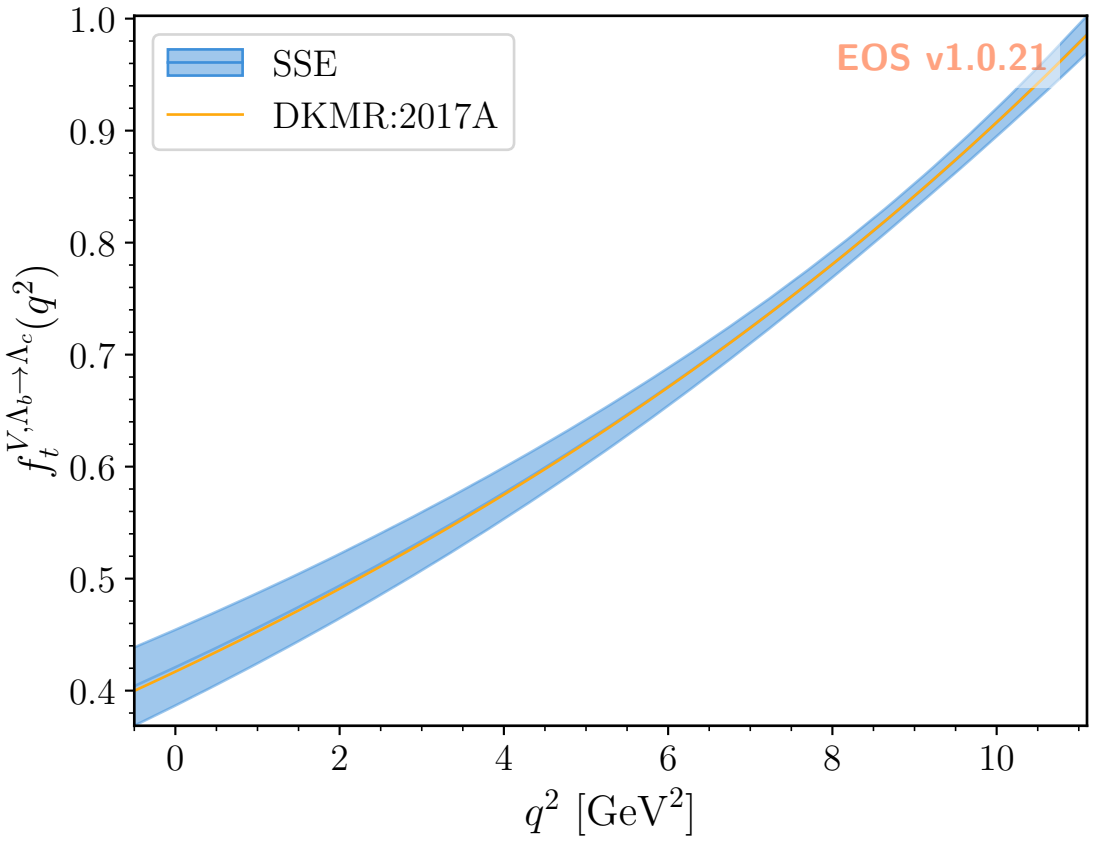

Figure 112: Posterior-predicted form factor for `Lambdab-to-Lambdac` (`Lambdab-to-Lambdac-ftV`).

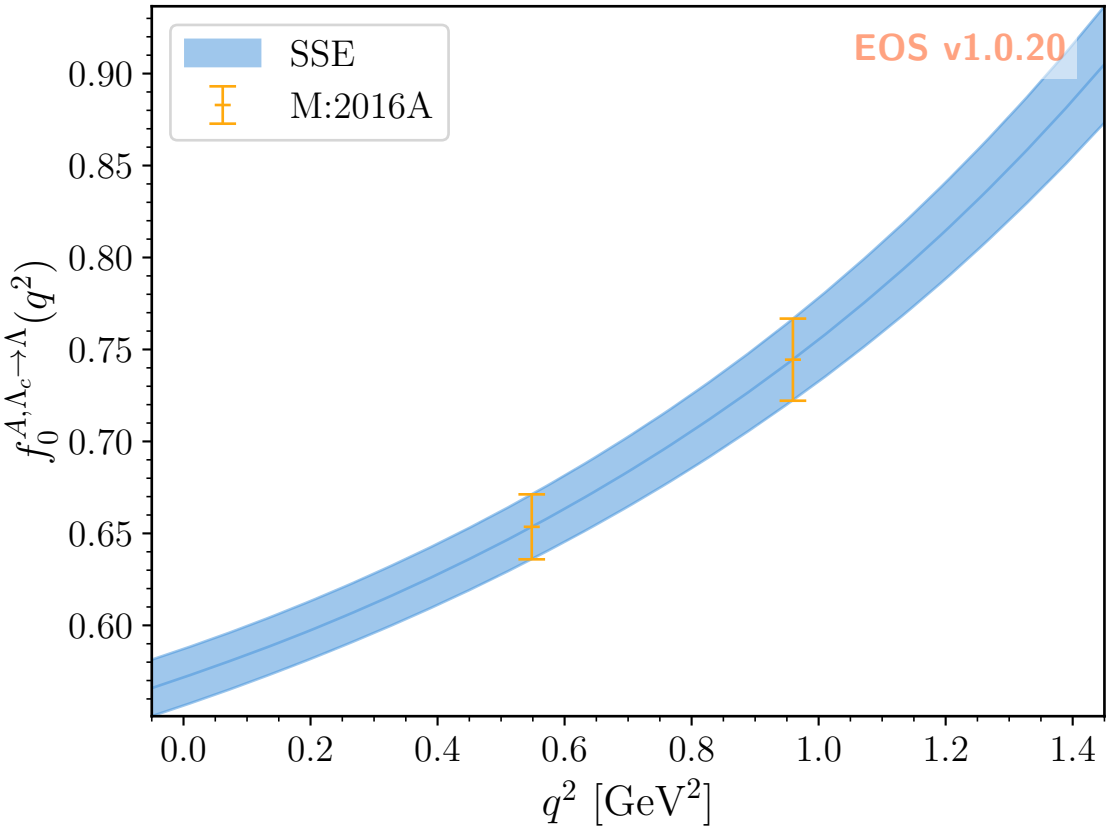

Figure 113: Posterior-predicted form factor for `Lambdac-to-Lambda` (`Lambdac-to-Lambda-f0A`).

Posterior `Lambdac-to-Lambda`

This posterior varies 15 `SSE` coefficients. It is constrained by:

- `Lambda_c->Lambda::f_time+long+perp^V+A@M:2016A`

  Note. No (lattice or LCSR) tensor form-factor data are available for this transition. The tensor coefficients are therefore not varied and retain their database default values; the tensor form factors below are not determined by this fit.

Goodness of fit

The table lists the local p value of each constraint together with the global p value and total χ^2 of the best-fit point.

Constraint	local p value
`Lambda_c->Lambda::f_time+long+perp^V+A@M:2016A`	100.0%
total (χ^2 / d.o.f. = 0.01 / 2)	**99.7%**

Posterior-predicted form factors

The bands show the posterior-predicted form factors (`SSE`); the points show the constraints entering the fit.

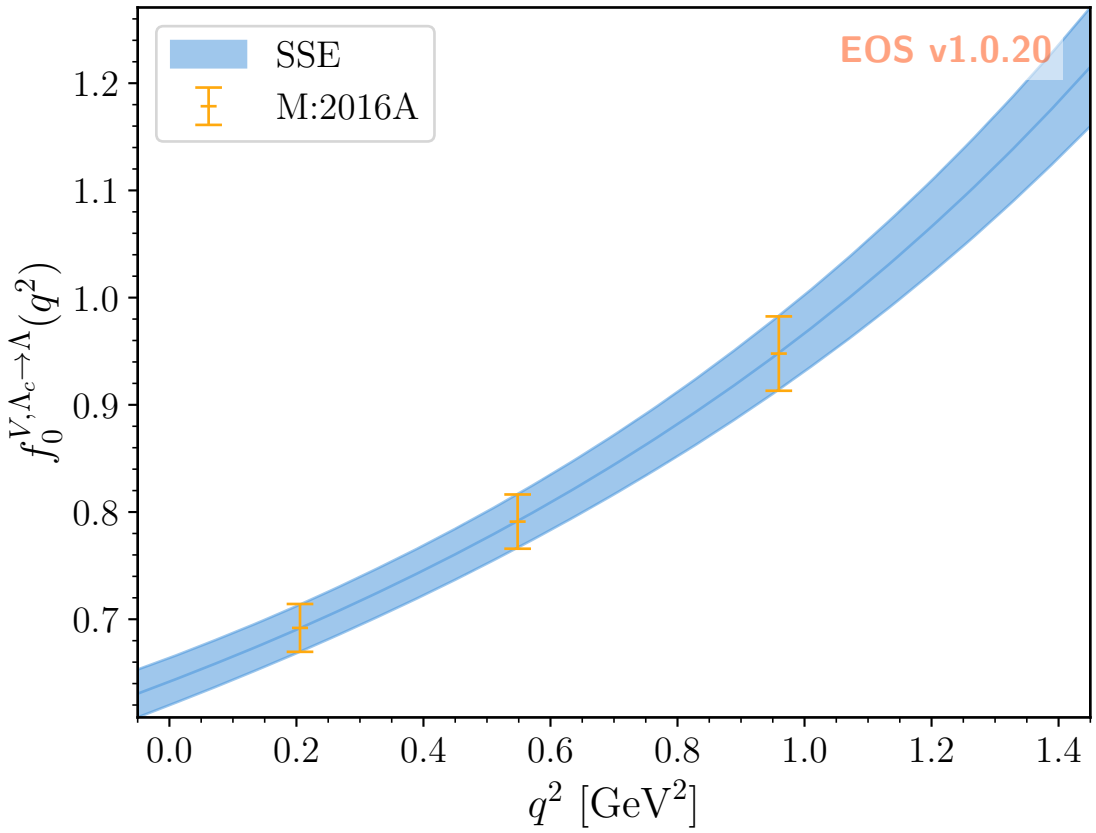

Figure 114: Posterior-predicted form factor for `Lambdac-to-Lambda` (`Lambdac-to-Lambda-f0V`).

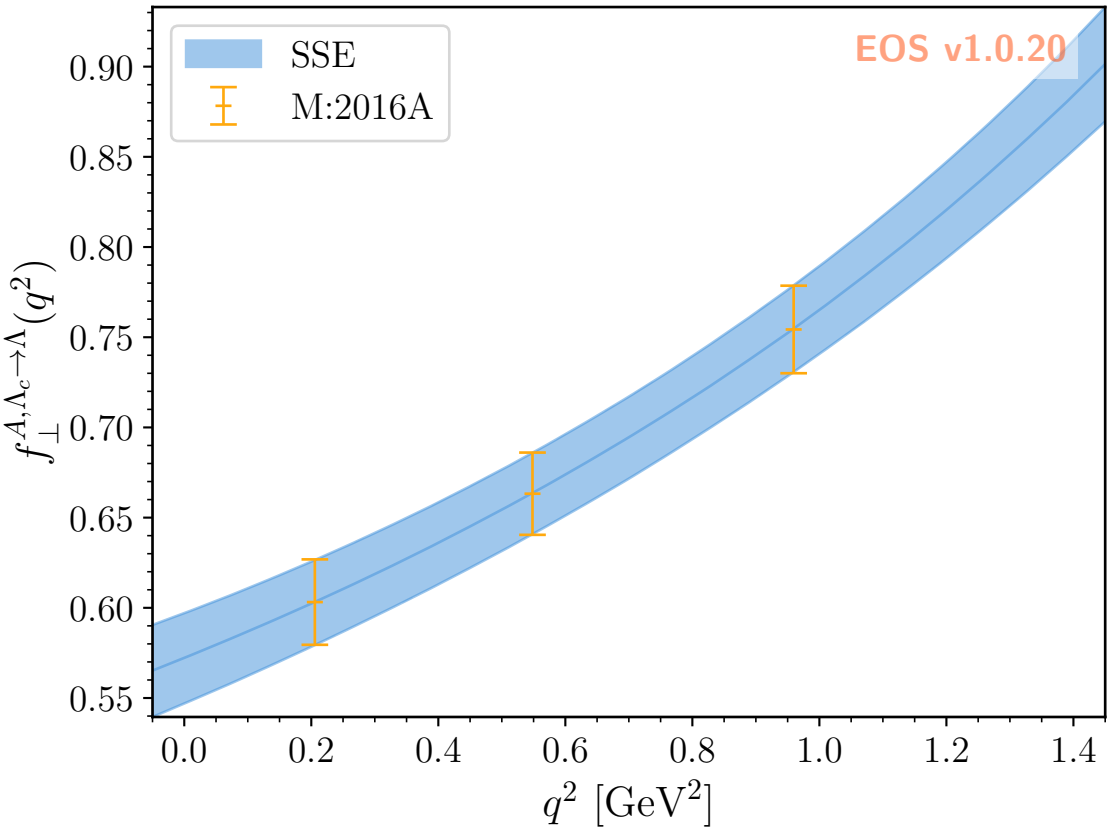

Figure 115: Posterior-predicted form factor for `Lambdac-to-Lambda` (`Lambdac-to-Lambda-fperpA`).

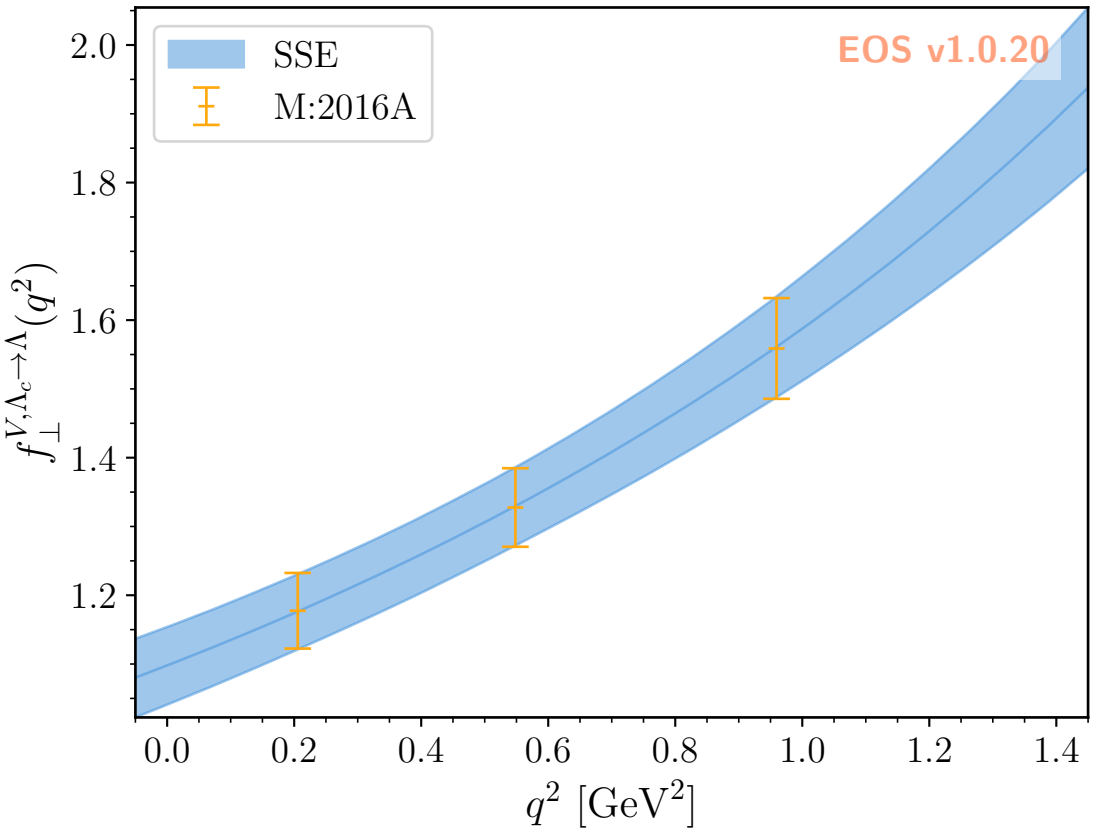

Figure 116: Posterior-predicted form factor for `Lambdac-to-Lambda` (`Lambdac-to-Lambda-fperpV`).

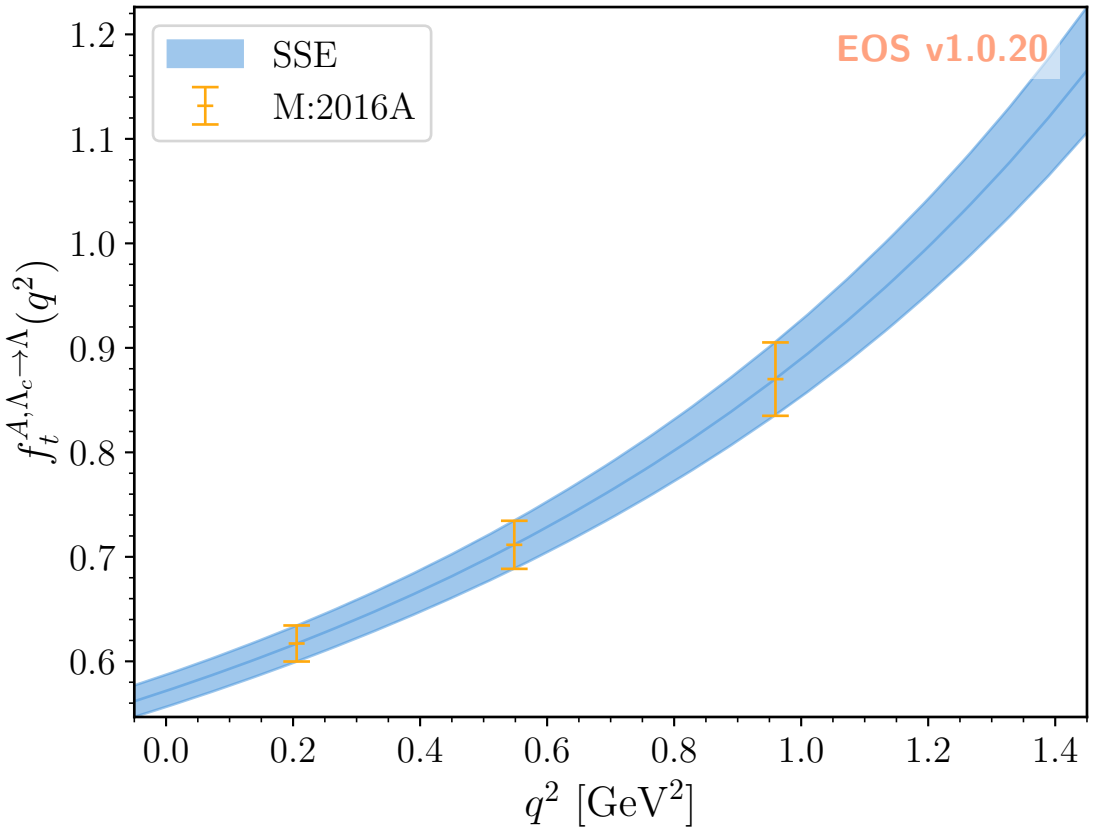

Figure 117: Posterior-predicted form factor for `Lambdac-to-Lambda` (`Lambdac-to-Lambda-ftA`).

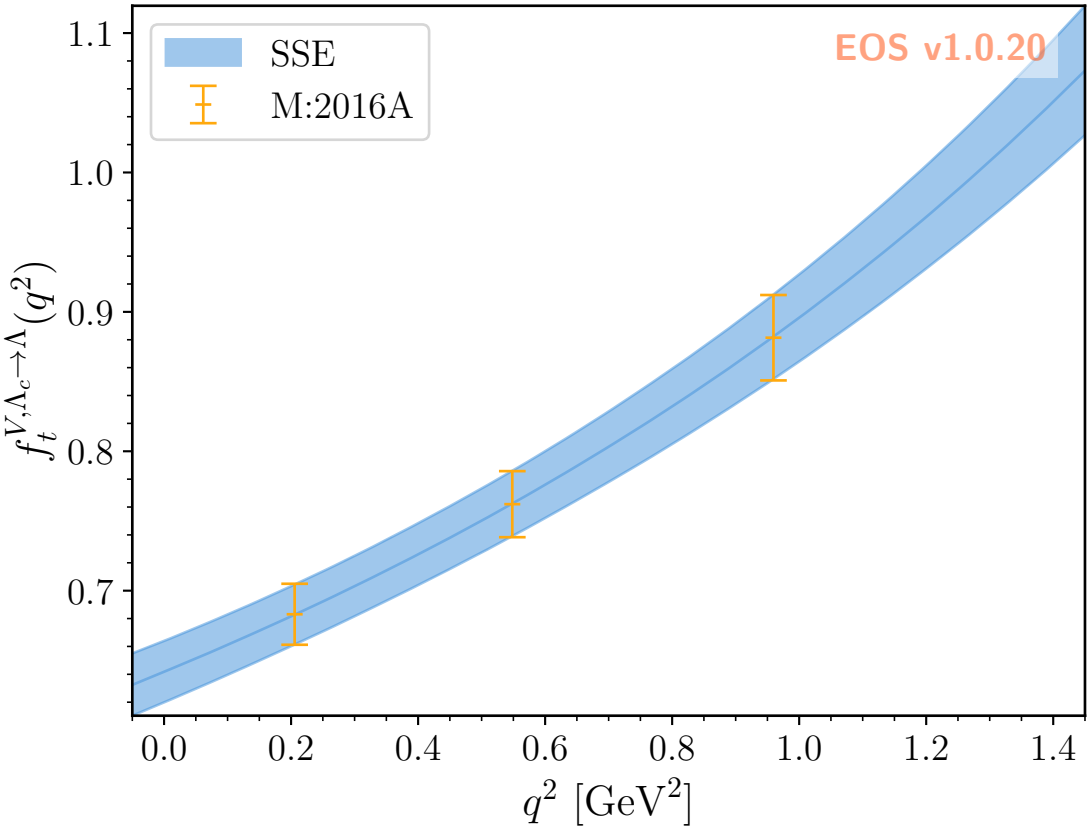

Figure 118: Posterior-predicted form factor for `Lambdac-to-Lambda` (`Lambdac-to-Lambda-ftV`).

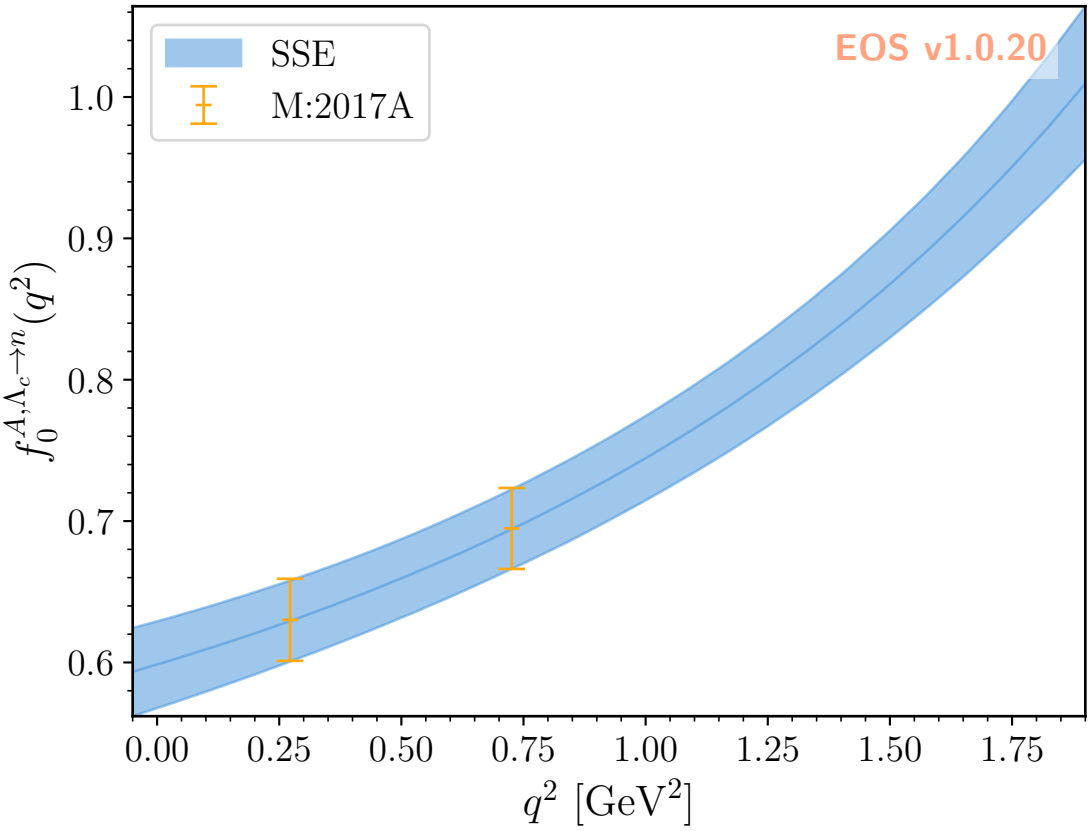

Figure 119: Posterior-predicted form factor for `Lambdac-to-neutron` (`Lambdac-to-neutron-f0A`).

Posterior `Lambdac-to-neutron`

This posterior varies 25 `SSE` coefficients. It is constrained by:

- `Lambda_c->neutron::f_time+long+perp^V+A+T+T5[HO]@M:2017A`

Goodness of fit

The table lists the local p value of each constraint together with the global p value and total χ^2 of the best-fit point.

Constraint	local p value
`Lambda_c->neutron::f_time+long+perp^V+A+T+T5[HO]@M:2017A`	100.0%
total (χ^2 / d.o.f. = 0.07 / 3)	**99.5%**

Posterior-predicted form factors

The bands show the posterior-predicted form factors (`SSE`); the points show the constraints entering the fit.

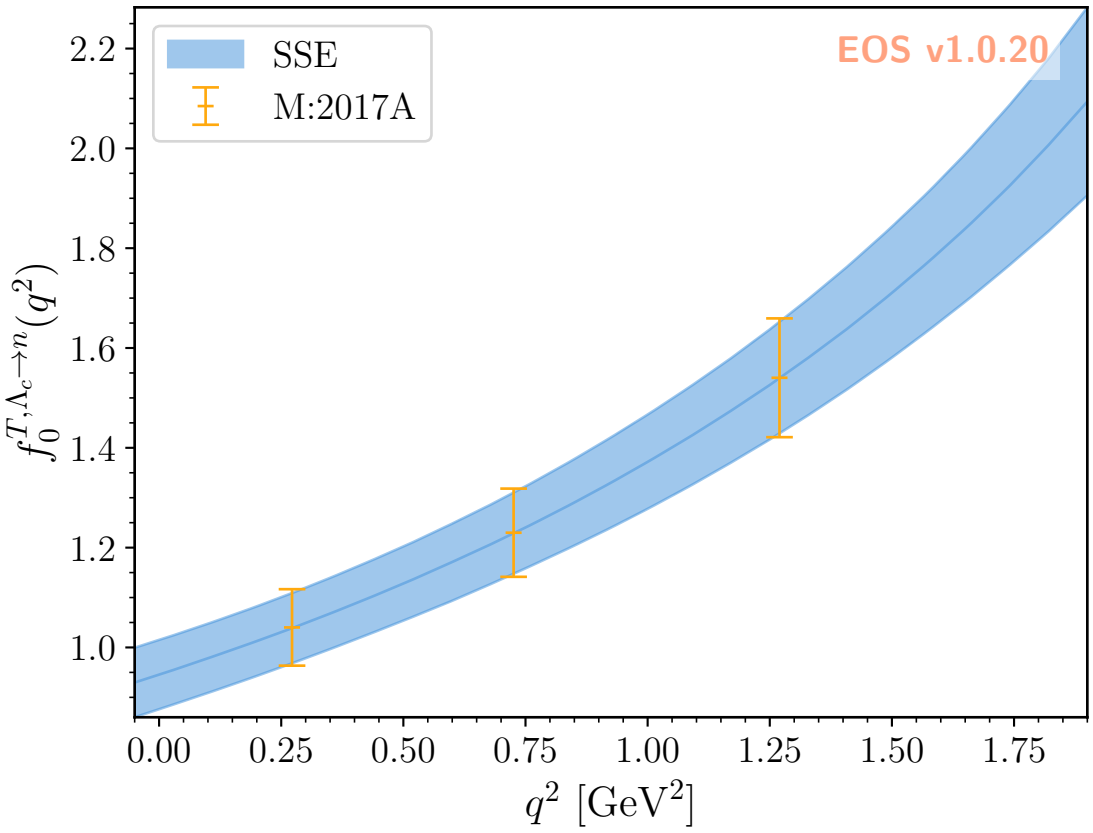

Figure 120: Posterior-predicted form factor for `Lambdac-to-neutron` (`Lambdac-to-neutron-f0T`).

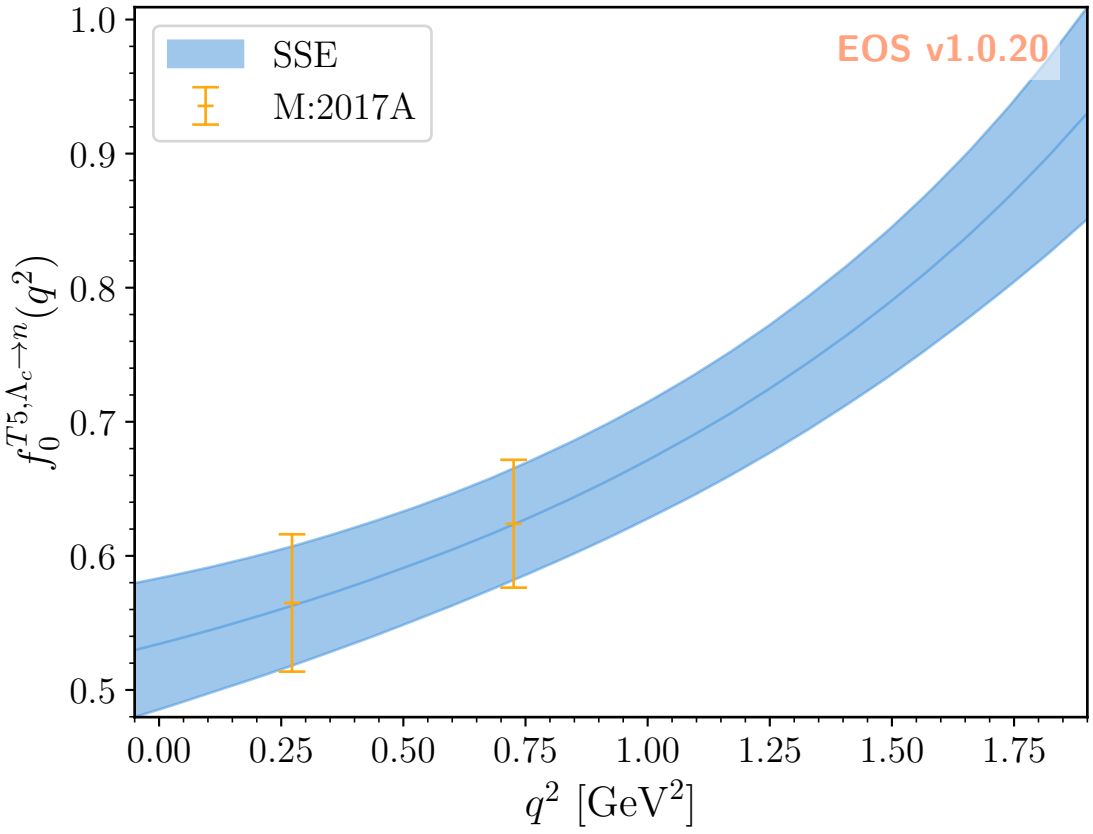

Figure 121: Posterior-predicted form factor for `Lambdac-to-neutron` (`Lambdac-to-neutron-f0T5`).

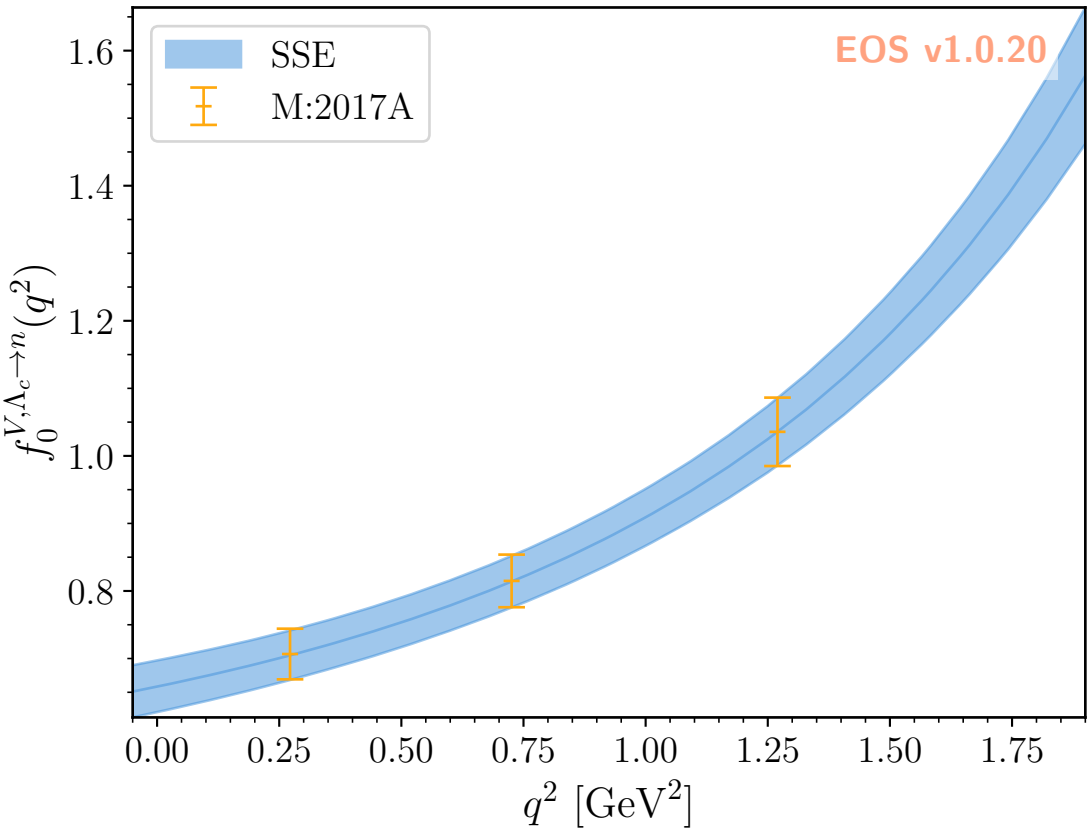

Figure 122: Posterior-predicted form factor for `Lambdac-to-neutron` (`Lambdac-to-neutron-f0V`).

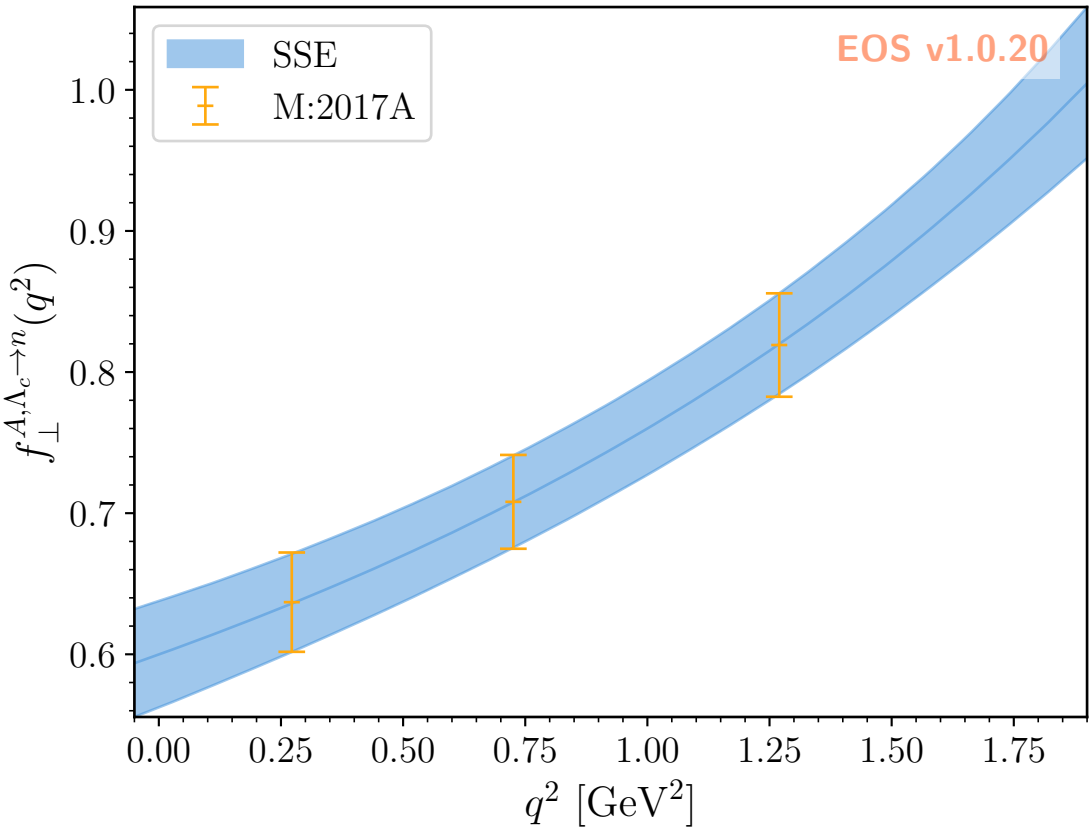

Figure 123: Posterior-predicted form factor for `Lambdac-to-neutron` (`Lambdac-to-neutron-fperpA`).

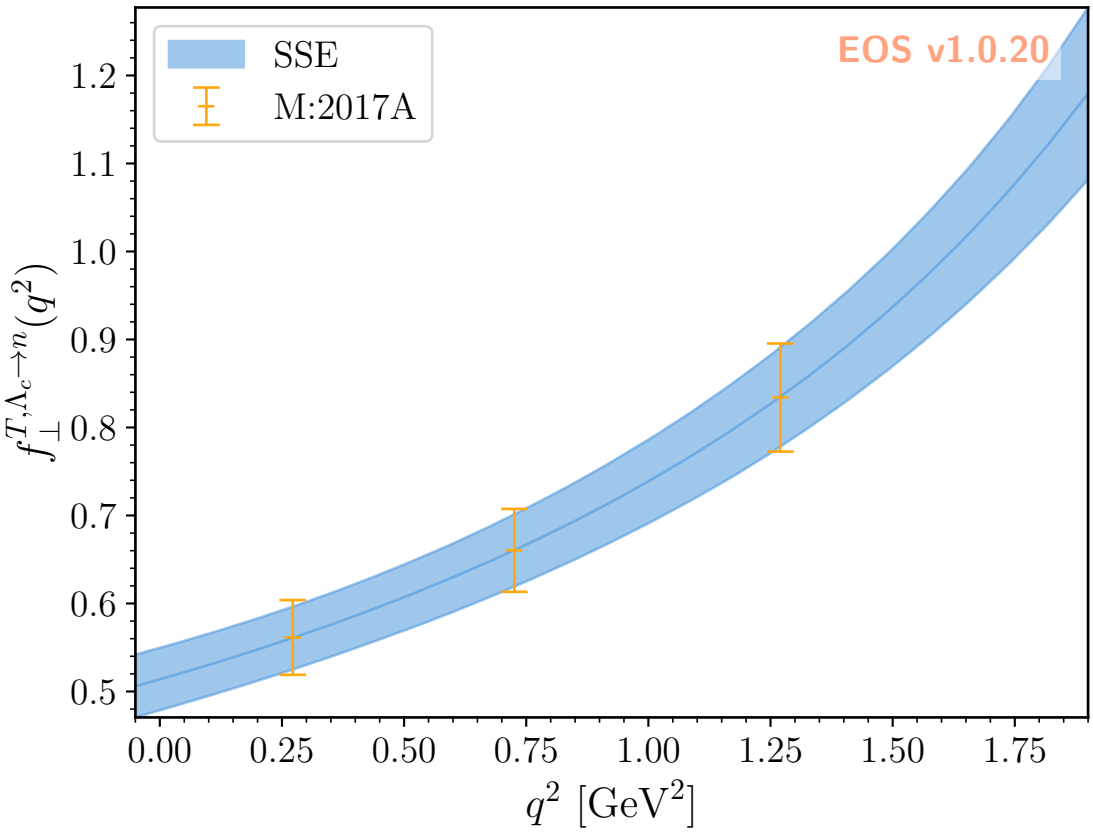

Figure 124: Posterior-predicted form factor for `Lambdac-to-neutron` (`Lambdac-to-neutron-fperpT`).

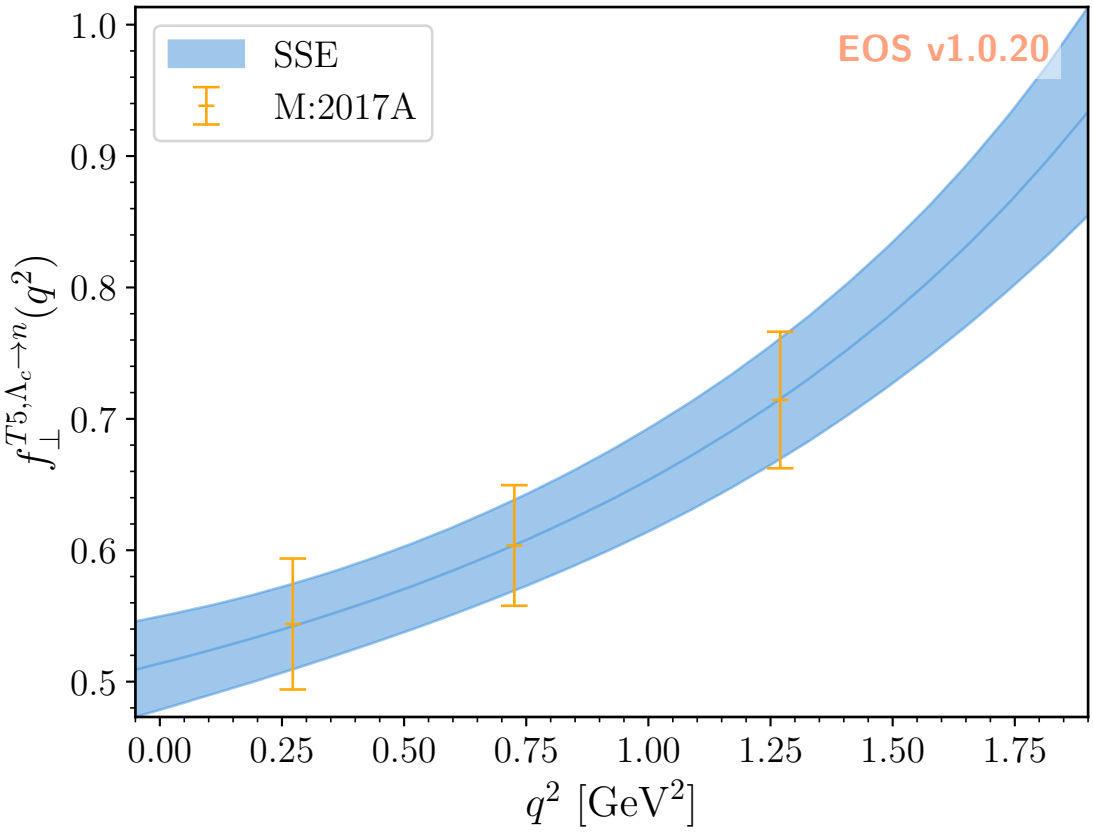

Figure 125: Posterior-predicted form factor for `Lambdac-to-neutron` (`Lambdac-to-neutron-fperpT5`).

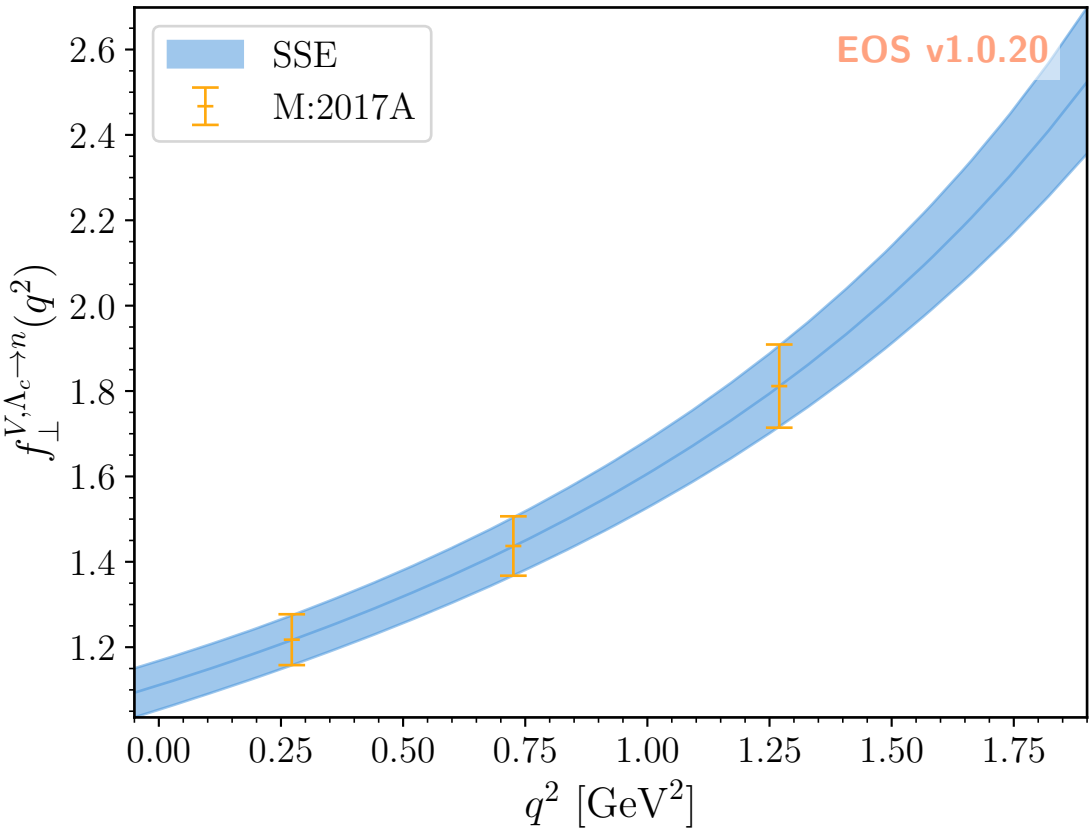

Figure 126: Posterior-predicted form factor for `Lambdac-to-neutron` (`Lambdac-to-neutron-fperpV`).

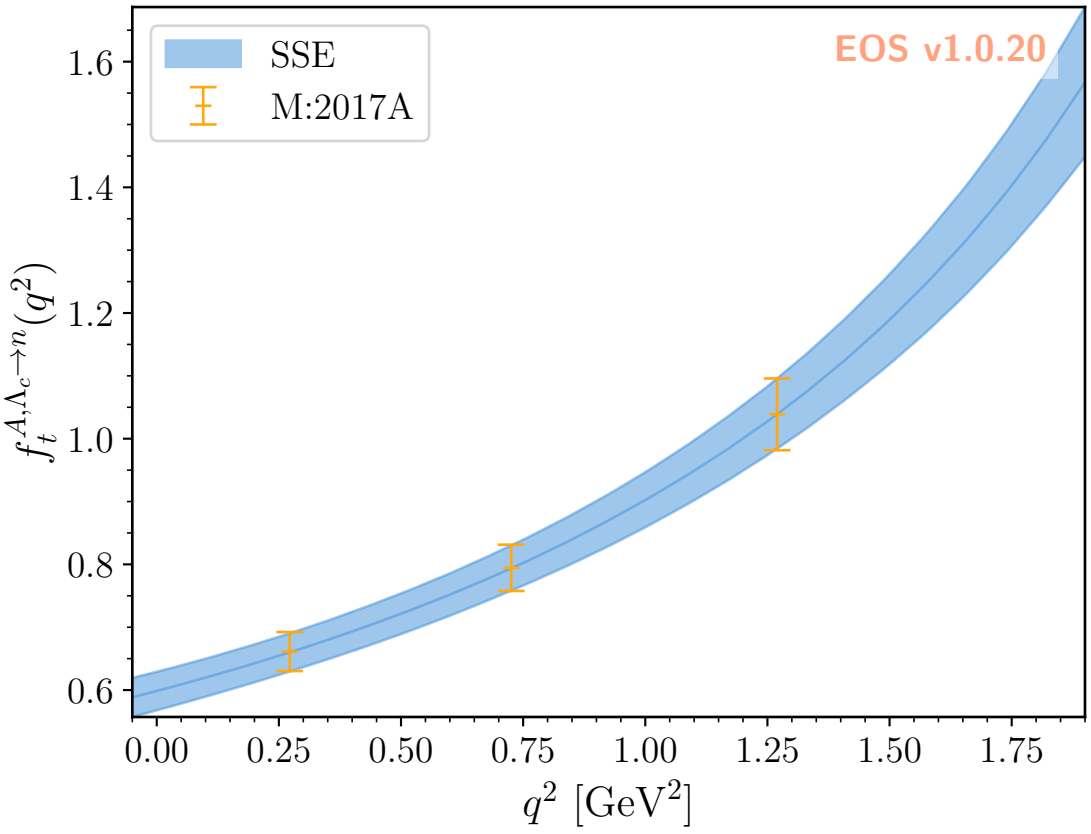

Figure 127: Posterior-predicted form factor for `Lambdac-to-neutron` (`Lambdac-to-neutron-ftA`).

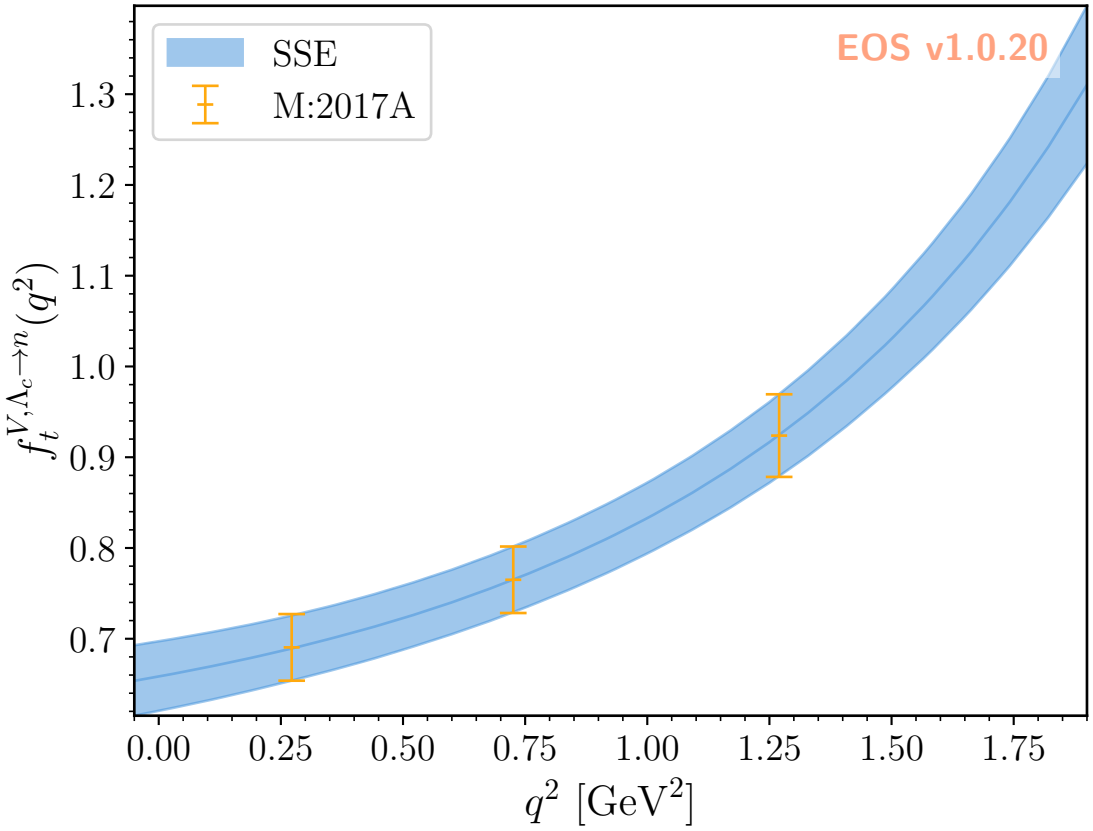

Figure 128: Posterior-predicted form factor for `Lambdac-to-neutron` (`Lambdac-to-neutron-ftV`).

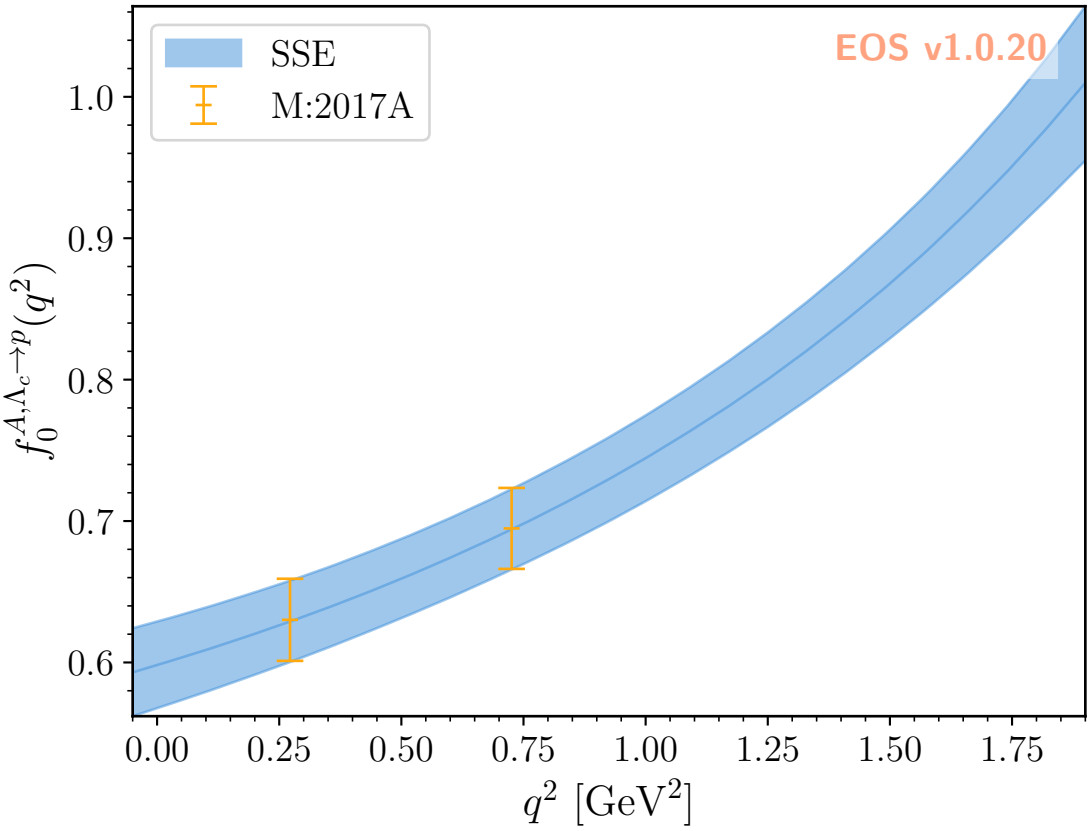

Figure 129: Posterior-predicted form factor for `Lambdac-to-proton` (`Lambdac-to-proton-f0A`).

Posterior `Lambdac-to-proton`

This posterior varies 25 `SSE` coefficients. It is constrained by:

- `Lambda_c->proton::f_time+long+perp^V+A+T+T5[HO]@M:2017A`

Goodness of fit

The table lists the local p value of each constraint together with the global p value and total χ^2 of the best-fit point.

Constraint	local p value
`Lambda_c->proton::f_time+long+perp^V+A+T+T5[HO]@M:2017A`	100.0%
total (χ^2 / d.o.f. = 0.07 / 3)	**99.5%**

Posterior-predicted form factors

The bands show the posterior-predicted form factors (`SSE`); the points show the constraints entering the fit.

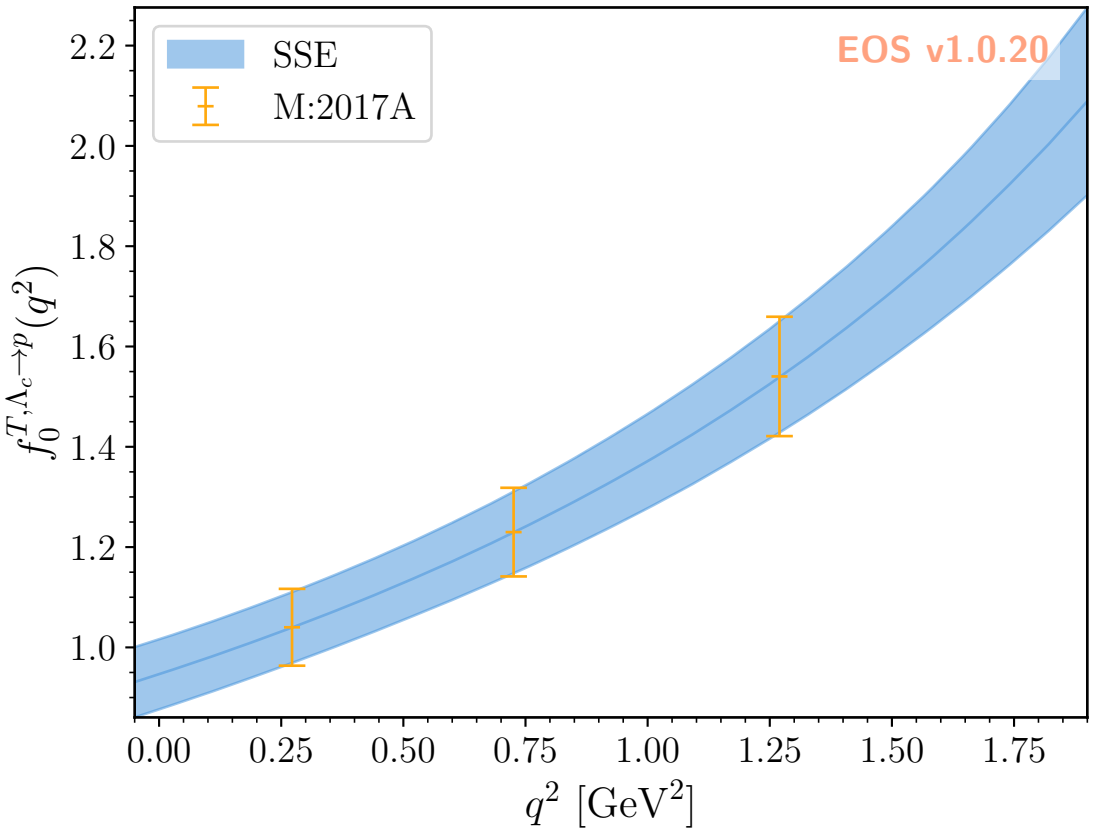

Figure 130: Posterior-predicted form factor for `Lambdac-to-proton` (`Lambdac-to-proton-f0T`).

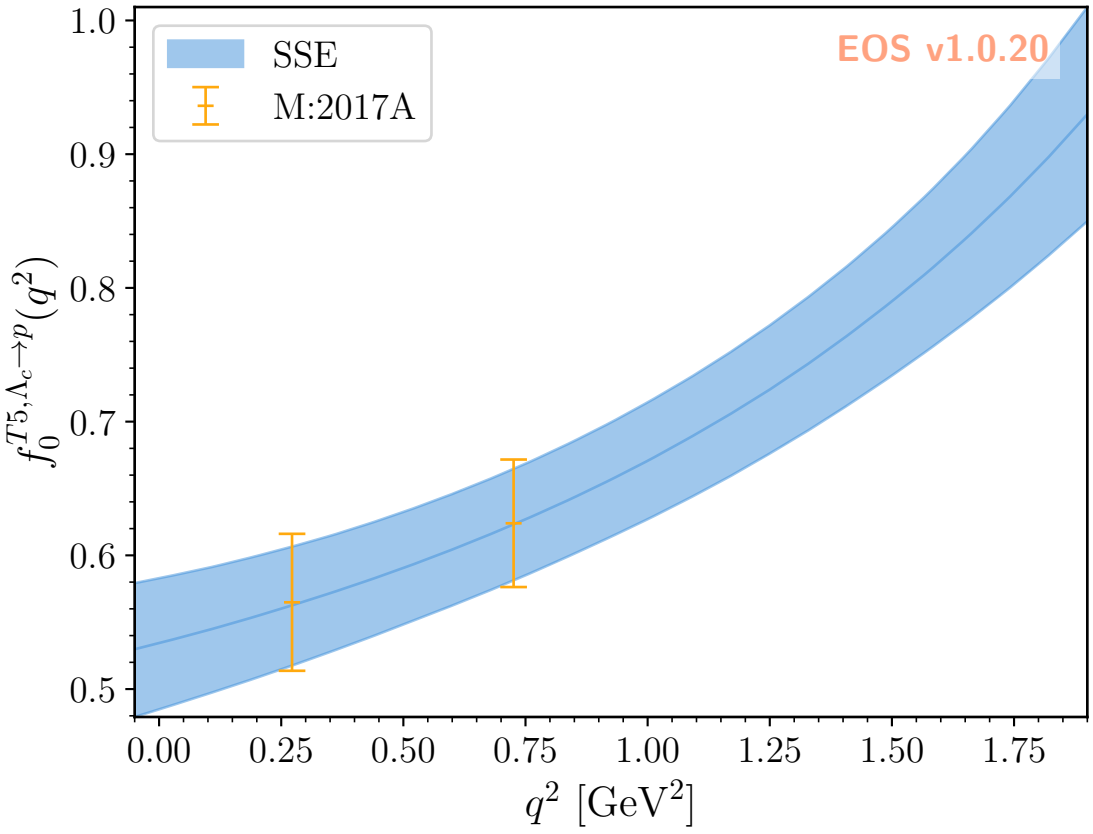

Figure 131: Posterior-predicted form factor for `Lambdac-to-proton` (`Lambdac-to-proton-f0T5`).

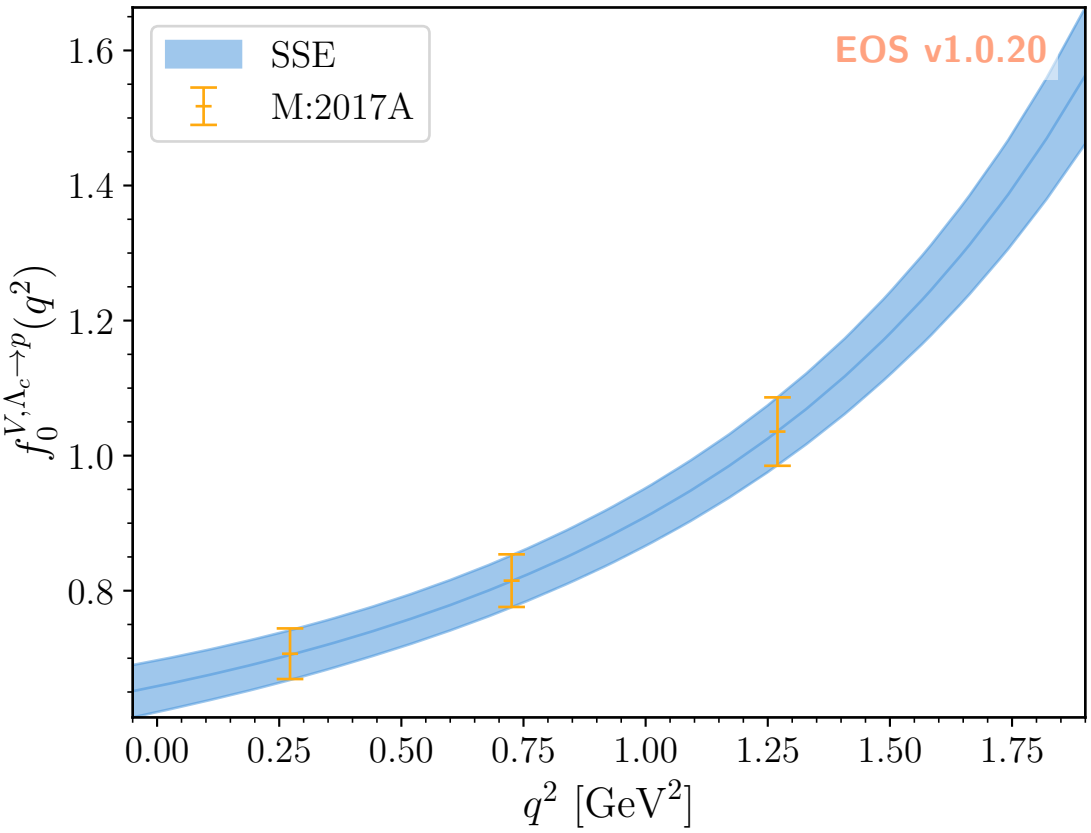

Figure 132: Posterior-predicted form factor for `Lambdac-to-proton` (`Lambdac-to-proton-f0V`).

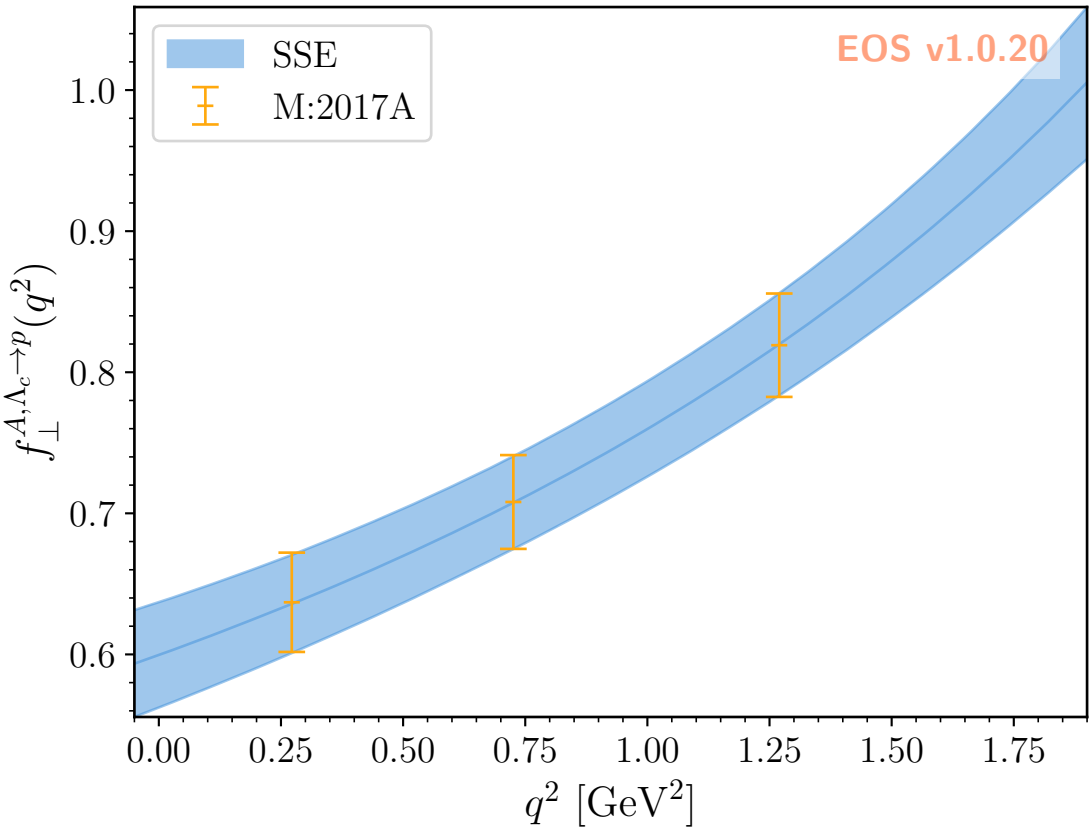

Figure 133: Posterior-predicted form factor for `Lambdac-to-proton` (`Lambdac-to-proton-fperpA`).

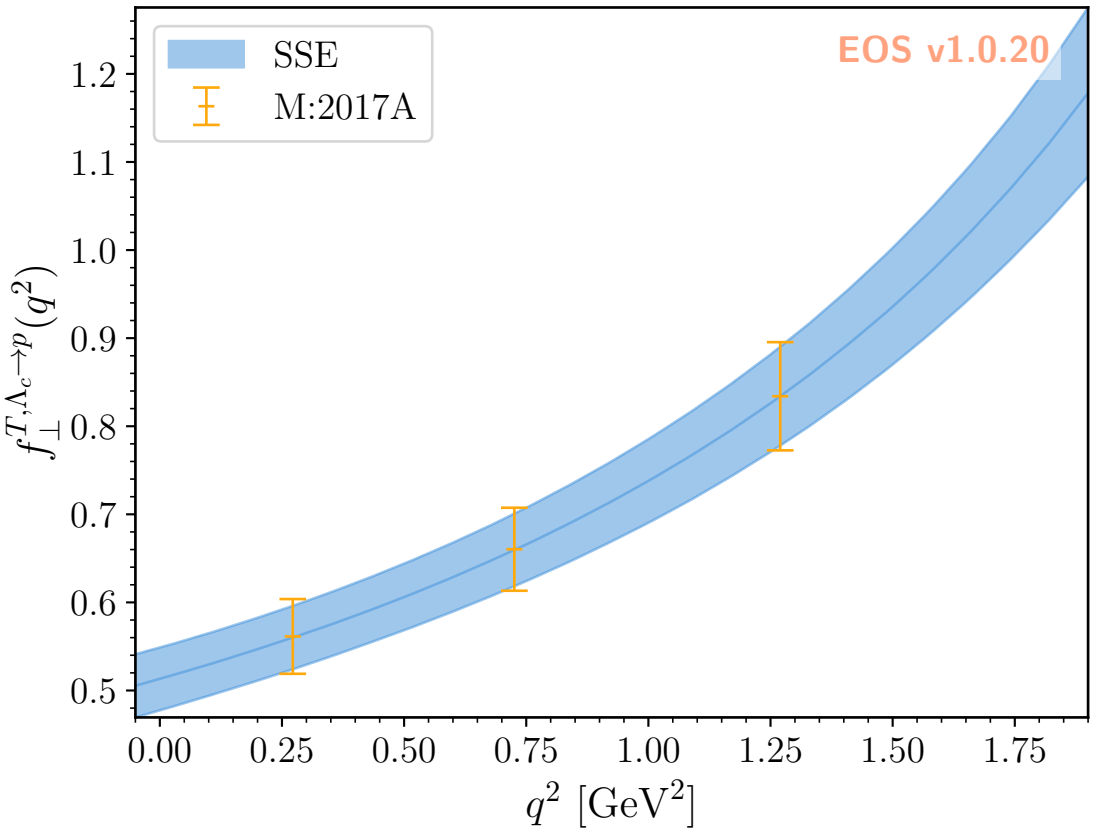

Figure 134: Posterior-predicted form factor for `Lambdac-to-proton` (`Lambdac-to-proton-fperpT`).

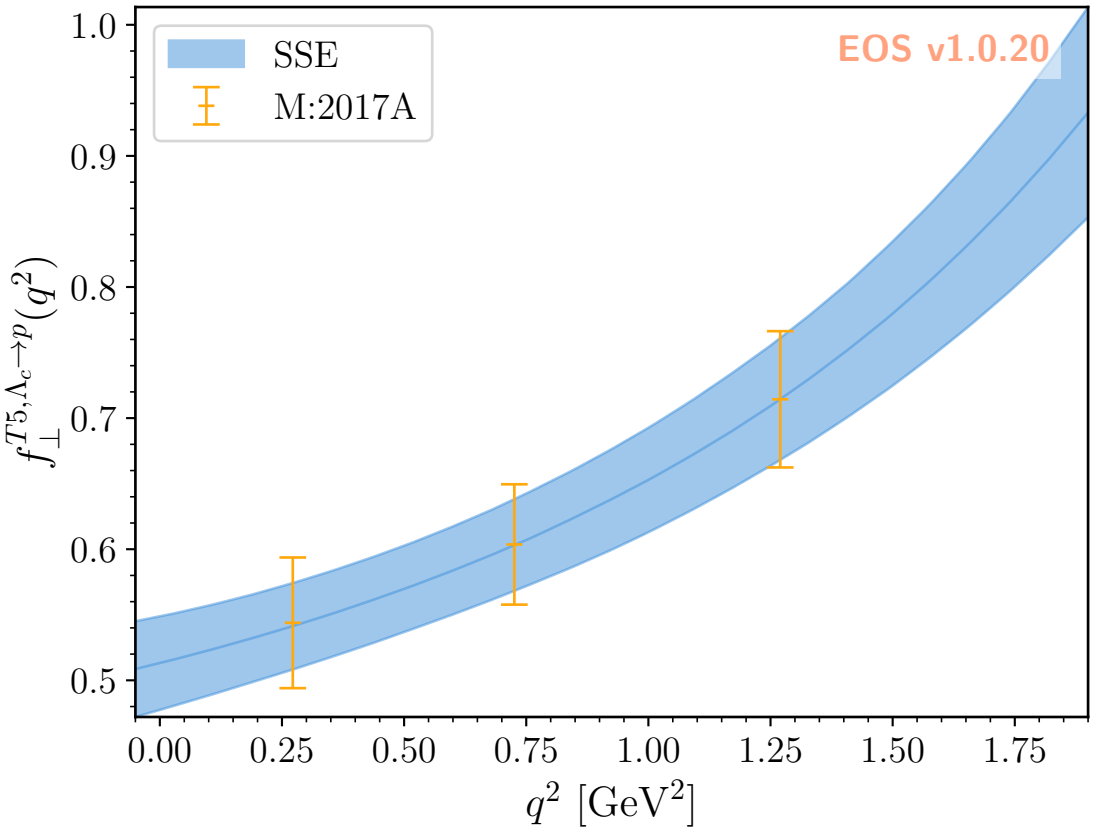

Figure 135: Posterior-predicted form factor for `Lambdac-to-proton` (`Lambdac-to-proton-fperpT5`).

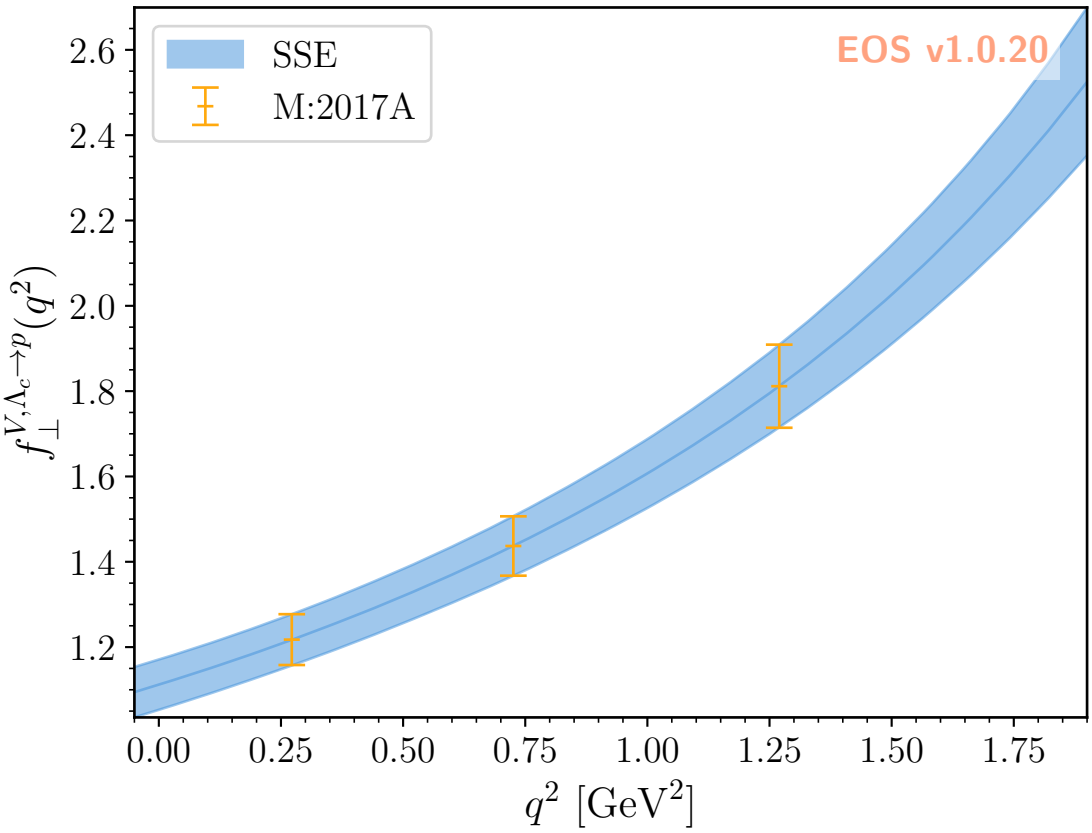

Figure 136: Posterior-predicted form factor for `Lambdac-to-proton` (`Lambdac-to-proton-fperpV`).

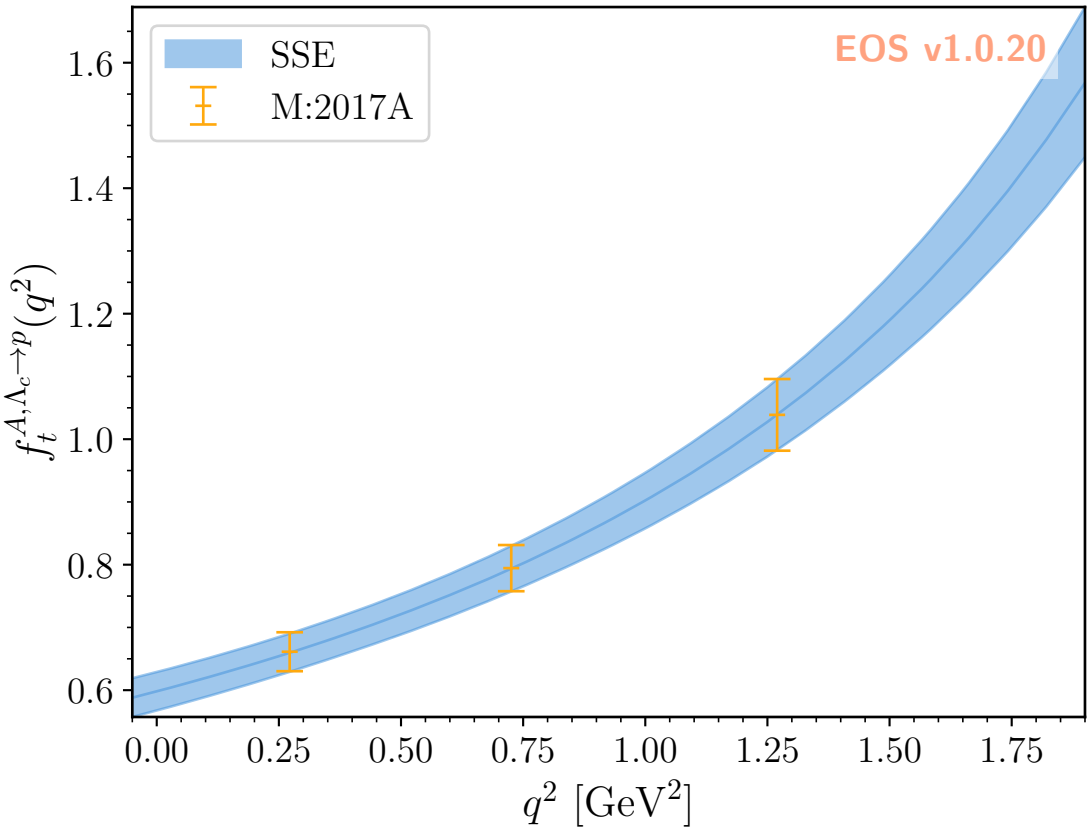

Figure 137: Posterior-predicted form factor for `Lambdac-to-proton` (`Lambdac-to-proton-ftA`).

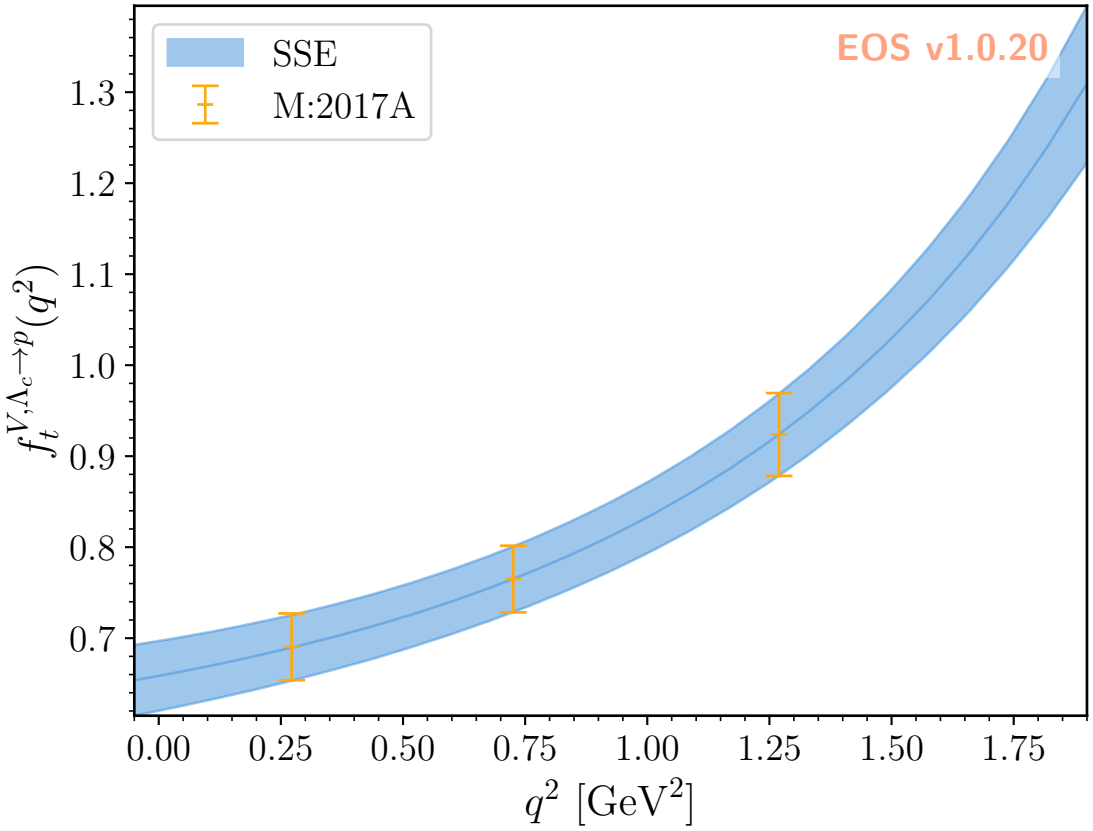

Figure 138: Posterior-predicted form factor for `Lambdac-to-proton` (`Lambdac-to-proton-ftV`).